ON THE INTERNAL HOMOTOPY THEORY OF MOTIVIC CATEGORIES

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Abstract. We develop a general ∞-categorical framework for internal Eilenberg-MacLane objects, internal homotopy groups, and an internal notion of covering spaces, and study their behavior under suitable localizations. Applying this to the ordinary motivic localization, we identify internal n-Eilenberg-MacLane objects with strongly $\mathbb{A}^1$-invariant sheaves of (abelian when $n \geqslant 2$) groups, yielding a formal obstruction to the motivic homotopy category being an ∞-topos. We prove that taking $\mathbb{A}^1$-localizations induces an equivalence between classical $\mathbb{A}^1$-coverings of a Nisnevich local space and the internal motivic coverings of its motivic localization, providing a streamlined proof of a generalized motivic Van Kampen theorem. Along the way, we establish a Nisnevich-local-to-global $\mathbb{A}^1$-connectivity criterion: a k-scheme is $\mathbb{A}^1$-connected if and only if it admits a Nisnevich cover by $\mathbb{A}^1$-connected schemes such that each pairwise intersection has a (k-)point. Finally, we show that over a general Qcqs base, passing to the birational motivic homotopy category recovers certain essential topos-theoretic properties absent in the ordinary $\mathbb{A}^1$-setting. In fact, for a finite-dimensional scheme with finitely many generic points, we show that the birational motivic homotopy category is a Postnikov-complete ∞-topos of cohomological dimension 0.

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1. Introduction

1.1. **Notations and terminologies.** *Categorical.* We shall freely use the language of ∞-categories without referring to any particular choice of a model for the theory. The reader may freely use their preferred model. A standard reference is [Lur09], where the author models these on Kan complexes as models for ∞-groupoids.

We choose and fix Grothendieck universes $\mathcal{G}_0 \in \mathcal{G}_1 \in \mathcal{G}_2$ of small, large, and very large sets, respectively. Unless otherwise specified, by a space we shall always mean a $\mathcal{G}_0$-space. *Spc* will denote the $\mathcal{G}_1$-∞-category of $\mathcal{G}_0$-spaces (or Kan complexes, if the reader prefers). Unless otherwise specified, by an ∞-category we mean a $\mathcal{G}_1$-∞-category. We say that a $\mathcal{G}_1$-∞-category is locally small if it is $\mathcal{G}_0$-*Spc*-enriched. We shall use the term "very large" ∞-categories for locally-$\mathcal{G}_1$-small $\mathcal{G}_2$-∞-categories.

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By Pr^L we shall mean the ∞-category of large presentable ∞-categories, while Cat_∞ will denote the ∞-category of all large ∞-categories. $\mathcal{T}op_\infty$ will stand for the ∞-category of ∞-toposes and geometric morphisms. Thus Cat_∞, Pr^L, etc., are examples of very large ∞-categories.

For an ∞-category $\mathcal{C}$, we use $\mathrm{Map}_{\mathcal{C}}(-,-)$ to denote the Spc-valued hom in $\mathcal{C}$. By $\mathcal{P}(\mathcal{C})$ we mean the category of presheaves of spaces on $\mathcal{C}$. If $\mathcal{C}$ has coproducts, by $\mathcal{P}_\Sigma(\mathcal{C})$ we mean the full subcategory of $\mathcal{P}(\mathcal{C})$ consisting of presheaves that send finite coproducts to products.

Algebraic geometry. Throughout this paper, S will denote a Qcqs scheme, and k will denote a field. We do not assume k is perfect unless otherwise specified. By a morphism of schemes, we shall always mean a quasicompact, separated morphism. In particular, smooth morphisms will be assumed to be of finite presentation.

For a scheme (Qcqs) S, we denote by Sm_S the category of smooth schemes over S. By $\mathcal{P}(S)$ we shall mean the ∞-category of presheaves of spaces on Sm_S and call an object of $\mathcal{P}(S)$ an S-space. Similarly, $\mathcal{P}_\Sigma(S) := \mathcal{P}_\Sigma(Sm_S)$. If σ is a Grothendieck topology on Sm_S, then $\mathcal{P}_\sigma(S)$ stands for the ∞-topos of S-spaces satisfying descent with respect to σ-sieves.

Motivic homotopy. We will use the notation $\mathcal{H}^{mot}(S)$ for the full subcategory of $\mathcal{P}(S)$ consisting of $\mathbb{A}^1$-local, Nisnevich-local objects. The corresponding localization functors will be denoted L_{mot}. Similarly, $\mathcal{H}^{bir}(S)$ and L_{bir} will handle birational motivic localization, i.e., the localization of $\mathcal{H}^{mot}(S)$ at all dense open immersions. When working with the $\mathbb{A}^1$-motivic homotopy category, most of our important results will be sensitive to the dimension of the base scheme and will work best over fields. Thus, we shall mostly be working with $\mathcal{H}^{mot}(k)$, where k is a field. For birational local results, however, we will work over a general Qcqs base scheme. The point is that, due to the main results of [Mai26a], the birational motivic homotopy category admits a simpler construction.

By π_0 (similarly π_n), we shall always mean the presheaf of homotopy groups. This is also the 0th truncation functor for $\mathcal{P}(S)$. π_n^{nis} will denote the homotopy groups internal to the Nisnevich topos, i.e., the Nisnevich sheafification of π_n. $\pi_i^{\mathbb{A}^1}$ will, as usual, stand for $\pi_i^{nis}L_{mot}$. Similarly, π_i^b is defined as $\pi_i^{nis}L_{bir}$, which is known to be equivalent to $\pi_i L_{bir}$ [Mai26a, Corollary 2.1.8]. We will use italic X, Y, Z, R, S, T for schemes, calligraphic letters $\mathcal{C}, \mathcal{D}$, etc., for categories, and $\mathcal{X}, \mathcal{Y}$, etc., for S-spaces.

1.2. **The philosophy of the current article.** Let us return to the starting point of $\mathbb{A}^1$ homotopy theory from the universal homotopical viewpoint. The $\mathbb{A}^1$ homotopy category of [MV99] is obtained from the category of smooth schemes as the universal presentable ∞-category that 'satisfies' Nisnevich descent and '$\mathbb{A}^1$-invariance' [Rob12, Theorem 5.2].

The classical analog of this concept is the following: start with the category of smooth ***manifolds***, then impose open cover descent and $\mathbb{R}^1$-invariance. The beauty of this construction is that it yields an ∞-topos. Indeed, since smooth manifolds have good open covers, the resulting category is the ∞-category Spc of spaces itself [Dug99, Remark 3.4.10], which is an ∞-topos.

It is a well-established property that the $\mathbb{A}^1$-motivic homotopy category does not constitute an ∞-topos, even over fields, which introduces significant complexities to motivic calculations. This structural constraint was initially noted in [SOsr12, Remark 3.5]. While the core logic of their remark is sound, a complete formalization necessitates explicitly identifying the appropriate Eilenberg-MacLane objects within the motivic framework, a technical detail that naturally fell outside the immediate scope of their discussion.

In this work, we will formalize this observation using a categorical framework and then *explicitly identify* the 'n-Eilenberg-MacLane objects' in the $\mathbb{A}^1$ homotopy category with strongly $\mathbb{A}^1$-invariant sheaves of (abelian when $n \geqslant 2$) groups. The same arguments as in *loc. cit.* will then show that the motivic homotopy category is not an ∞-topos.

The real reason for such a failure seems to be that, without a controlled etale localization, smooth schemes have many more 'non-linear' homotopy dimensions. Without imposing such an 'extreme' etale localization, the only way to reduce homotopy dimensions is to identify schemes with their generic points. This is exactly what the birational localization promises to achieve (see §4.0). Throughout this paper, we will see that, over arbitrary base schemes, the birational homotopy category is very close to being an ∞-topos, in that it has many properties similar to those of an ∞-topos. We will finally show that it is indeed an ∞-topos when the base scheme is geometrically unibranch with finitely many generic points (for example, a field) (see Theorem 4.5.2).

1.3. **A Quick summary of the article.** In the second section, we define and study the categorical theory of truncations, homotopy groups, and Eilenberg-MacLane spaces, and discuss their behavior under an accessible (Cartesian) localization. We identify suitable hypotheses on the localization functor to ensure that these notions behave as expected. We recall the definitions here for the convenience of the reader.

Given an ∞-category $\mathcal{C}$, we say that an object $X \in \mathcal{C}$ is n-truncated if the presheaf $\mathrm{Map}_{\mathcal{C}}(-, X) : \mathcal{C}^{op} \to Spc$ is (sectionwise) n-truncated. The full subcategory of n-truncated objects is denoted $\mathcal{C}_{\leqslant n}$ and is closed under small limits [Lur09]. Thus, when $\mathcal{C}$ is presentable, this subcategory is given by a localization $\tau^{\mathcal{C}}_{\leqslant n} : \mathcal{C} \to \mathcal{C}_{\leqslant n}$. At the other extreme lies the iterated loop space construction for pointed objects. Given a pointed object X, one defines the loop space of X, denoted $\Omega_{\mathcal{C}} X$, as the kernel pair of $* \to X$, which is canonically pointed, so this operation can be iterated. One defines the internal homotopy groups $\pi^{\mathcal{C}}_n X$ as the 0th internal truncation of $\Omega^n_{\mathcal{C}} X$. The orthogonal theory of truncated objects is the theory of internal n-connective objects, i.e., pointed objects whose internal homotopy groups vanish below n. Finally, one defines an n-Eilenberg-MacLane object as an n-truncated n-connective object.

To this end, we will discuss the cases of 0- and 1-covering spaces under the names geometric and spatial coverings, respectively. When the presentable category is given by a localization of an ∞-topos, there are two primary ways to define spatial coverings. One is internal to the localized subcategory, while the other is an L-local theory developed inside the base ∞-topos. We will find appropriate conditions so that these agree under the localization.

We study the behavior of each of these constructions under a suitable localization. Let us skip the discussion of the general techniques and see their applications to various motivic categories.

The third section is thus dedicated to the internal homotopy theory of the $\mathbb{A}^1$ motivic homotopy category, $\mathcal{H}^{mot} := L_{\mathbb{A}^1}\mathcal{P}_{nis}$ [MV99]. Since this is a presentable ∞-category, by the general method mentioned above, there is an internal homotopy path component functor providing a let adjoint to the inclusion of $\mathbb{A}^1$-invariant Nisnevich sheaves of sets $Sh^{\mathbb{A}^1}$ into $\mathcal{H}^{mot}$ which we denote by π^{mot}_0. There is a canonical natural transformation $\pi^{\mathbb{A}^1}_0 \to \pi^{mot}_0$. Morel's conjecture [Mor12, Conjecture 1.12], which has been proven to be false by Ayoub [Ayo23], is then read as saying that this transformation is an isomorphism. In the first §3.1, we study some properties of this morphism. To this end, we say that a motivic space $\mathcal{X}$ is motivically connected if $\pi^{mot}_0 \mathcal{X} = *$. The first key result we obtain is:

Theorem 1 (see Lemma 3.1.23)**.** *Over a field k, a motivic space is $\mathbb{A}^1$-connected iff it is motivically connected.*

Using this, we will show that:

Theorem 2 (see Theorem 3.2.13)**.** *Let k be a perfect field. The restriction of the internal motivic homotopy groups functor to motivic Eilenberg-MacLane spaces induces equivalences of categories*

$$\begin{aligned} Sh_k^{\mathbb{A}^1} &\simeq \mathcal{EM}_0(\mathcal{H}^{mot}) : \pi_0^{mot} \\ Grp_k^{\mathbb{A}^1} &\simeq \mathcal{EM}_1(\mathcal{H}^{mot}) : \pi_1^{mot} \\ Ab_k^{\mathbb{A}^1} &\simeq \mathcal{EM}_n(\mathcal{H}^{mot}) : \pi_n^{mot} \text{ for all } n \geqslant 2 \end{aligned}$$

And more generally, that:

Theorem 3 (see Corollary 3.3.6)**.** *Over a perfect field, there is an equivalence of categories*

$$\mathrm{B}^{nis} : \mathcal{G}rp_{mot}(\mathcal{H}^{mot}(k)) \simeq \mathcal{H}_\bullet^{mot}(k)_{\geqslant 1} : \Omega$$

where $\mathcal{G}rp_{mot}(\mathcal{H}^{mot}(S))$ denotes the full subcategory of $Grp(\mathcal{H}^{mot}(k))$ consisting of motivic monoids with strongly L_{mot}-local π_0^{mot}. This provides further evidence that $\mathcal{H}^{mot}$ is not an ∞-topos.

These help us present a complete argument for [SOsr12, Remark 3.5] as a concrete theorem:

Theorem 4 (see Corollary 3.2.14)**.** *Over any perfect field k, the motivic homotopy category $\mathcal{H}^{mot}(k)$ is not an ∞-topos.*

To establish this, we give an explicit counterexample of a motivic sheaf of groups that is not strongly $\mathbb{A}^1$-invariant. A counterexample is already known in the literature (Cho14, Lemma 5.6), but it is abelian. Although this suffices, we provide a non-abelian group example and a non-group monoid example: we show that the free sheaves of groups $F_{gr}\mathbb{G}_m$ and monoids $F_{mon}\mathbb{G}_m$ generated by $\mathbb{G}_m$ are $\mathbb{A}^1$-invariant but not strongly $\mathbb{A}^1$-invariant (see Counterexample 3.2.32 and Remark 3.2.33). Along the way, we develop a general technique for constructing exotic motivic sheaves of sets from existing ones and give several non-standard examples (see Section 3.2).

In the next subsection, we study the geometric and spatial theories of coverings for the motivic homotopy category. The first significant result implied by Lemma 3.1.23 is the following counterpart of the classical result stating that "if X is a topological space with a cover U_α such that each U_α is path connected and $U_{\alpha\beta} \neq \varnothing$, then X is path connected."

Theorem 5 (see Corollary 3.4.3)**.** *A scheme X/k is $\mathbb{A}^1$-connected iff there is a Nisnevich covering $U_i \to X$ such that each U_i is $\mathbb{A}^1$-connected and for every pair i, j the fiber product $U_i \times_X U_j$ has a (k-)point.*

Next, we apply the ∞-categorical notion of covering spaces developed in §2.4 to the $\mathbb{A}^1$-motivic setup. Using the methods from §2.4, we reprove various results from [Mor12, §7]. As the very first step, we have

Theorem 6 (see Corollary 3.4.13)**.** *Suppose $\mathcal{X}$ is a Nisnevich sheaf of spaces. We then have an equivalence of categories*

$$\eta_{mot}^* : Cov_{\mathbb{A}^1}(L_{mot}\mathcal{X}) = Cov_{\mathcal{H}^{mot}}(L_{mot}\mathcal{X}) \leftrightarrows Cov_{\mathbb{A}^1}(\mathcal{X}) : L_{mot}$$

where η_{mot} is the localization unit $1 \to L_{mot}$. The category on the left-hand side is the category of covering spaces of $L_{mot}\mathcal{X}$ internal to the motivic homotopy category, while the right-hand side is the ∞-categorical analogue of Morel's category of $\mathbb{A}^1$-coverings of $\mathcal{X}$, as developed in [Mor12, §7].

Using this, we re-prove Theorem 7.8 of [Mor12] (see Theorem 3.4.15).

The final section aims to show that much of the extra care required in the $\mathbb{A}^1$-local case is unnecessary in the birational-local case, $\mathcal{H}^{bir} := L_{bir}\mathcal{H}^{mot}$ [Mai26a]. and in fact, they work over arbitrary Qcqs base schemes:

Theorem 7 (see Theorem 4.1.3)**.** *Over a general Qcqs scheme S, the birational path component functor $\pi_0^b := \pi_0 L_{bir}$ is the internal homotopy path component of $\mathcal{H}^{bir}$.*

Let us quickly list some of the topos-theoretic properties enjoyed by the birational motivic homotopy category:

Theorem 8 (see Theorem 4.2.6)**.** *Similar to that of an ∞-topos* [Lur09, Lemma 7.2.2.11 (1)]*, there are equivalences of categories:*

$$\begin{aligned} \mathrm{Disc}(\mathcal{H}^{bir})_\bullet &\simeq \mathcal{EM}_0(\mathcal{H}^{bir}) : \pi_0^{bir} \\ Grp(\mathrm{Disc}(\mathcal{H}^{bir})) &\simeq \mathcal{EM}_1(\mathcal{H}^{bir}) : \pi_1^{bir} \\ Ab(\mathrm{Disc}(\mathcal{H}^{bir})) &\simeq \mathcal{EM}_n(\mathcal{H}^{bir}) : \pi_n^{bir} \text{ for all } n \geqslant 2 \end{aligned}$$

over an arbitrary Qcqs bse scheme S.

Theorem 9 (see Proposition 4.3.3)**.** *Over a general Qcqs base scheme S, the functor*

$$\Omega : \mathcal{H}^{bir}_{\geqslant 1} \to Grp(\mathcal{H}^{bir})_\bullet \simeq Grp(\mathcal{H}^{bir})$$

is an equivalence of infinity categories, with inverse given by the internal bar construction $\mathbf{B}_{bir}$.

However, since L_{bir} is neither locally cartesian (see [Mai26a, Counterexample 3.2.13]) nor effective (see [Mai26a, Counterexample 3.5.7]), the theory of birational covering spaces is not as strong as its motivic counterpart. Specifically, we can construct universal birational covers only for $\mathbb{A}^1$-motivically connected spaces, rather than for all birationally connected spaces (Theorem 4.4.16), which follows from:

Theorem 10 (see Theorem 4.4.15)**.** *For a motivically connected space $\mathcal{X}$, we have an equivalence of categories*

$$\eta^*_{bir} : Cov_b(L_{bir}\mathcal{X}) = Cov_{\mathcal{H}^{bir}}(L_{bir}\mathcal{X}) \leftrightarrows Cov_b(\mathcal{X}) : L_{bir}$$

where η_{bir} is the localization unit $1 \to L_{bir}$, $Cov_{\mathcal{H}^{bir}}(-)$ is the category of internal coverings in $\mathcal{H}^{bir}$, while $Cov_b(\mathcal{X})$ is the category of birational coverings of $\mathcal{X}$.

In §4.5, we finally reach the main result of this paper.:

Theorem 11 (see Theorem 4.5.6)**.** *Let S be a Qcqs scheme with finitely many generic points. Then $\mathcal{H}^{bir}(S)$ is an ∞-topos.*

We summarize the remaining properties enjoyed by the birational motivic homotopy category construction as follows:

Theorem 12 (see Corollary 4.5.7)**.** *Let S be a qcqs scheme with finitely many generic points. Then the birational motivic space functor*

$$h^b_S(-) := L_{bir}h_S(-) : Sm_S \to \mathcal{H}^{\mathrm{bir}}(S)$$

exhibits the following structural properties:

(1) It contracts the affine line $\mathbb{A}^1_S$.
(2) It maps Nisnevich and cdh squares to pushout squares.
(3) The target $\mathcal{H}^{\mathrm{bir}}(S)$ is an ∞-topos of cohomological dimension 0. It is postnikov complete when $dim(S) < \infty$.

2. Internal homotopy theory in an ∞-category

In this section, we introduce the internal notions of truncated objects, homotopy groups, and Eilenberg-MacLane objects. We will study their behavior under suitable localizations. For various aspects of this analysis to work as expected, we will require different properties of the localization functor. (One such important condition is the path injectivity condition [Definition 2.2.6], which is key to the most important results of the first three subsections of this section.)

In the fourth and final subsection, we will formalize the theory of covering objects internal to an ∞-category and study this under a localization. (The key condition of this section on localization functors is the locality of the fundamental of local objects and effectivity.)

2.1. **Truncations.** A (-1)-connected space X is called n-truncated if, for all choices of base points $x \in \mathrm{Map}_{Spc}(*, X)$, the set $\pi_0 X$ is a singleton and, for all $i > n$, the groups $\pi_i(X, x)$ are trivial. The following notion from [Lur09, Definition 5.5.6.1] offers an immediate ∞-categorical generalization of this concept:

Definition 2.1.1. *Let $\mathcal{C}$ be an ∞-category. An object $\mathcal{F}$ of $\mathcal{C}$ is said to be n-truncated if, for all $\mathcal{E} \in \mathcal{C}$, the ∞-groupoid $\mathrm{Map}_{\mathcal{C}}(\mathcal{E}, \mathcal{F})$ is n-truncated. In other words, the presheaf $\mathrm{Map}_{\mathcal{C}}(-, \mathcal{F}) : \mathcal{C}^{op} \to Spc$ factors through the ∞-category $Spc_{\leqslant n}$ of n-truncated spaces.*

We denote the category of n-truncated objects in $\mathcal{C}$ by $\mathcal{C}_{\leqslant n}$. In particular, $\mathcal{C}_{\leqslant 0}$ is a discrete category and we denote $\mathrm{Disc}\mathcal{C} := \mathrm{N}(\mathcal{C}_{\leqslant 0})$.

If $\mathcal{C}$ is a presentable ∞-category, the full subcategory of n-truncated objects is closed under limits and is therefore given by a localization $\tau^{\mathcal{C}}_{\leqslant n}$ [Lur09, Proposition 5.5.6.5].

We now analyze the concept of n-truncated objects under a localization.

Lemma 2.1.2. *Let $L\mathcal{C} \subset \mathcal{C}$ be a localizing subcategory. Then the inclusion $\mathcal{C}_{\leqslant n} \bigcap L\mathcal{C} \subset (L\mathcal{C})_{\leqslant n}$ is an equivalence, i.e., we have an equality $(L\mathcal{C})_{\leqslant n} = \mathcal{C}_{\leqslant n} \bigcap L\mathcal{C}$.*

Proof. The containment is obvious. Indeed, if $\mathcal{A}$ belongs to $L\mathcal{C}$ and $\mathcal{C}_{\leqslant n}$, then for any $\mathcal{B} \in L\mathcal{C}$, one has $\mathrm{Map}_{L\mathcal{C}}(\mathcal{B}, \mathcal{A}) = \mathrm{Map}_{\mathcal{C}}(\mathcal{B}, \mathcal{A})$. But this last term is n-truncated since $\mathcal{A} \in \mathcal{C}_{\leqslant n}$.

To see the reverse inclusion, note that since $(L\mathcal{C})_{\leqslant n} \subset L\mathcal{C}$, it suffices to show that $(L\mathcal{C})_{\leqslant n} \subset \mathcal{C}_{\leqslant n}$. So let $\mathcal{X} \in L\mathcal{C}_{\leqslant n}$, and let $\mathcal{Y} \in \mathcal{C}$ be an arbitrary object. Then we have $\mathrm{Map}_{\mathcal{C}}(\mathcal{Y}, \mathcal{X}) \simeq \mathrm{Map}_{L\mathcal{C}}(L\mathcal{Y}, \mathcal{X})$. But this last term is an n-truncated ∞-groupoid because $\mathcal{X} \in (L\mathcal{C})_{\leqslant n}$. This shows that $\mathcal{X} \in \mathcal{C}_{\leqslant n}$ as well. Thus, $(L\mathcal{C})_{\leqslant n} \subset \mathcal{C}_{\leqslant n}$. □

In other words, the following diagram is a pullback square in Cat_∞:

$$\begin{array}{ccc} (L\mathcal{C})_{\leqslant n} & \hookrightarrow & L\mathcal{C} \\ \downarrow & & \downarrow \\ \mathcal{C}_{\leqslant n} & \hookrightarrow & \mathcal{C} \end{array}$$

Proposition 2.1.3. *Let $L\mathcal{C} \subset \mathcal{C}$ be an accessible localization of a presentable ∞-category $\mathcal{C}$. Then $(L\mathcal{C})_{\leqslant n}$ is a localization of $\mathcal{C}_{\leqslant n}$ and of $\mathcal{C}$.*

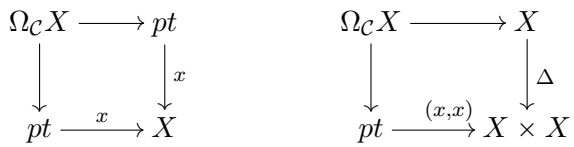


FIGURE 1. loop object

Proof. Since both $\mathcal{C}_{\leqslant n}$ and $L\mathcal{C}$ are closed under limits in $\mathcal{C}$, we conclude immediately from the lemma above that so is $(L\mathcal{C})_{\leqslant n} \subset \mathcal{C}_{\leqslant n}$. □

Let us denote the corresponding localization functor by $L_{\leqslant n} : \mathcal{C}_{\leqslant n} \to (L\mathcal{C})_{\leqslant n}$.

On the other hand, when L is an accessible localization of a presentable ∞-category $\mathcal{C}$, so that $L\mathcal{C}$ is also presentable, we know, by [Lur09, Proposition 5.5.6.5], that $(L\mathcal{C})_{\leqslant n} \subset L\mathcal{C}$ is also a localization. We denote the internal truncation functor for the ∞-category $L\mathcal{C}$ by

$$\tau^{L\mathcal{C}}_{\leqslant n} : L\mathcal{C} \to (L\mathcal{C})_{\leqslant n}.$$

It follows that,

Lemma 2.1.4. $\tau^{L\mathcal{C}}_{\leqslant n} \simeq L_{\leqslant n}\tau_{\leqslant n|L\mathcal{C}}$.

Proof. Suppose $\mathcal{X} \in L\mathcal{C}$ and $\mathcal{Y} \in (L\mathcal{C})_{\leqslant n} = L\mathcal{C} \bigcap \mathcal{C}_{\leqslant n}$. We get

$$\begin{aligned}
\mathrm{Map}_{(L\mathcal{C})_{\leqslant n}}(\tau^{L\mathcal{C}}_{\leqslant n}\mathcal{X}, \mathcal{Y}) &= \mathrm{Map}_{L\mathcal{C}}(\mathcal{X}, \mathcal{Y}) \text{ [by definition of } \tau^{L\mathcal{C}}_{\leqslant n}] && (1)\\
&= \mathrm{Map}_{\mathcal{C}}(\mathcal{X}, \mathcal{Y}) && (2)\\
&= \mathrm{Map}_{\mathcal{C}_{\leqslant n}}(\tau^{\mathcal{C}}_{\leqslant n}\mathcal{X}, \mathcal{Y}) \text{ [since } \mathcal{Y} \in \mathcal{C}_{\leqslant n}] && (3)\\
&= \mathrm{Map}_{(L\mathcal{C})_{\leqslant n}}(L_{\leqslant n}\tau^{\mathcal{C}}_{\leqslant n}\mathcal{X}, \mathcal{Y}) && (4)
\end{aligned}$$

The result now follows from the Yoneda lemma. □

Internal homotopy groups

Let $\mathcal{C}$ be a category with finite limits. The projection map $\Lambda^2_2 \to \Delta^1$ sending $\{0, 1\}$ to 0 and 2 to 1 induces a functor $\mathcal{C}^{\Delta^1} \to \mathcal{C}^{\Lambda^2_2}$. Composing this with the canonical functor $\mathcal{C}_\bullet := \mathcal{C}_{*/} \to \mathcal{C}^{\Delta^1}$, induces another functor $\mathcal{C}_* \to \mathcal{C}^{\Lambda^2_2}$. Finally, taking limits over Λ^2_2 we arrive at the loop space functor $\Omega : \mathcal{C}_\bullet \to \mathcal{C}_\bullet$ as the obvious factorization of the composition:

$$\mathcal{C}_\bullet \to \mathcal{C}^{\Lambda^2_2} \xrightarrow{lim} \mathcal{C}.$$

By iteration, we arrive at the n-fold loop space functor $\Omega^n : \mathcal{C}_\bullet \to \mathcal{C}_\bullet$.

Actually, there are multiple ways to think about this. For example, given a pointing $x : * \to X$, we may define $\Omega_{\mathcal{C}}X$ by either of the following two equivalent pullback diagrams:

Definition 2.1.5. *The 0-th internal homotopy path components object functor of a presentable ∞-category $\mathcal{C}$ is defined as $\pi^{\mathcal{C}}_0 := \tau^{\mathcal{C}}_{\leqslant 0}$. For $n > 0$, the n-th internal homotopy group $\pi^{\mathcal{C}}_n : \mathcal{C}_* \to \mathrm{Disc}(\mathcal{C})_*$ is defined as $\tau^{\mathcal{C}}_{\leqslant 0}\Omega^n_{\mathcal{C}} = \pi^{\mathcal{C}}_0\Omega^n_{\mathcal{C}}$.*

Definition 2.1.6. *We say that the localization is path cartesian if $L_{\leqslant 0}$ is cartesian.*

In the rest of this section, we assume that L is path Cartesian, so that $\tau^{\mathcal{C}}_{\leqslant 0}$ is also cartesian. By formal arguments, it is clear that $\pi^{\mathcal{C}}_n X$ is an n-group object in $\mathrm{Disc}(\mathcal{C})$. In short, if we let $n\text{-}Grp(\mathcal{D})$

stand for n-fold group objects of a 1-category $\mathcal{D}$, i.e., $n\text{-}Grp(\mathcal{D}) = Grp(Grp(...(Grp(\mathcal{D}))))$, it is clear by the Eckmann-Hilton argument that for every $n \geqslant 1$

$$n\text{-}Grp(\mathcal{D}) = \begin{cases} Grp(\mathcal{D}) \text{ for } n = 1 \\ Ab(\mathcal{D}) \text{ for } n \geqslant 2 \end{cases}$$

we have induced functors $\pi_n^{\mathcal{C}} : \mathcal{C}_\bullet \to n\text{-}Grp(\mathrm{Disc}(\mathcal{C}))$.

Lemma 2.1.7. *Let $L\mathcal{C}$ be an accessible localization of a presentable ∞-category $\mathcal{C}$. Then $\pi_0^{L\mathcal{C}} \simeq L_{\leqslant 0}\pi_0^{\mathcal{C}}$.*

Proof. This follows at once from the identification $\tau_{\leqslant 0}^{L\mathcal{C}} = \pi_0^{L\mathcal{C}}$ and Lemma 2.1.4. □

In the case of a localization, the previously obtained truncation formula tells us that

$$\pi_0^{L\mathcal{C}} := \tau_{\leqslant 0}^{L\mathcal{C}} \simeq L_{\leqslant 0}\tau_{\leqslant 0|L\mathcal{C}}^{\mathcal{C}} = L_{\leqslant 0}\pi_{0\,|L\mathcal{C}}^{\mathcal{C}}.$$

In fact,

Corollary 2.1.8. *Let $L\mathcal{C} \subset \mathcal{C}$ be a Bousfield localization of a presentable ∞-category $\mathcal{C}$. Then $\pi_i^{L\mathcal{C}} : L\mathcal{C} \to \mathrm{Disc}(LC)$ is equivalent to $L_{\leqslant 0}\pi_{i\,|L\mathcal{C}}^{\mathcal{C}}$. In fact, this is also equivalent to $L_{\leqslant 0}^{L\mathcal{C}}\pi_{i\,|L\mathcal{C}}^{\mathcal{C}} : L\mathcal{C} \to \mathrm{Disc}(LC)$.*

Proof. First of all, since $L\mathcal{C} \subset \mathcal{C}$ is closed under limits and Ω is constructed from limits, we find that $\Omega_{L\mathcal{C}}^n = (\Omega_{\mathcal{C}}^n)_{|L\mathcal{C}}$.

By Lemma 2.1.2, for any $\mathcal{X} \in L\mathcal{C}$, we have

$$\pi_i^{L\mathcal{C}}\mathcal{X} := \tau_{\leqslant 0}^{L\mathcal{C}}\Omega_{L\mathcal{C}}^n\mathcal{X} \simeq L_{\leqslant 0}\tau_{\leqslant 0}^{\mathcal{C}}\Omega_{\mathcal{C}}^n\mathcal{X} \simeq L_{\leqslant 0}\pi_i^{\mathcal{C}}\mathcal{X}.$$

□

Remark 2.1.9. Note that the above corollary does not say that L takes homotopy groups to homotopy groups. Rather, it says that one must choose a localization at the level of discrete (aka 0-truncated) objects to obtain the correct notion of internal homotopy 'groups'.

Lemma 2.1.10. *Let $\mathcal{C}$ be an ∞-category with finite limits, and let $X \in \mathcal{C}_\bullet$ be an n-truncated object for some n. Then $\Omega_{\mathcal{C}}X$ is $(n-1)$-truncated.*

Proof. Recall that an object X in a presentable ∞-category $\mathcal{C}$ is n-truncated iff the diagonal $X \xrightarrow{\Delta} X \times X$ is $(n-1)$-truncated [Lur09, Lemma 5.5.6.15]. From the pullback diagram on the right of Figure 1, we deduce that $\Omega_{\mathcal{C}}X$ is an $(n-1)$-truncated object. To see this, note that the base change functor $x^* : \mathcal{C}_{/X} \to \mathcal{C}$ is left exact (in fcat, it is a right adjoint). So by [Lur09, Proposition 5.5.6.16], x^* preserves $(n-1)$-truncated objects. In particular, because the right vertical map of the right square of Figure 1 is $(n-1)$-truncated, the pullback condition yields that $\Omega_{\mathcal{C}}X \to *$ is $(n-1)$-truncated. But this is equivalent to saying that $\Omega_{\mathcal{C}}X$ is $(n-1)$-truncated. □

Remark 2.1.11. The slice condition (i.e., $\Omega\tau_{\leqslant n} \simeq \tau_{\leqslant n-1}\Omega$) need not hold unless $\mathcal{C}$ is an ∞-topos. In other words, the canonical map $\tau_{\leqslant n-1}\Omega \to \Omega\tau_{\leqslant n}$ need not be an equivalence in general.

Proposition 2.1.12. *Let $\mathcal{C}$ be a presentable ∞-category and $X \in \mathcal{C}$ be an n-truncated object for some $n \geqslant 0$. Then for every choice of base point $x : * \to X$ and every $i \geqslant n$, $\pi_i^{\mathcal{C}}(X, x) = * \in \mathrm{Disc}(\mathcal{C})$.*

Proof. Using either of the pullback squares of Figure 1, observe that when X is pointed, so is $\Omega_{\mathcal{C}}X$. Fixing a base point of X, we will thus drop the base point from the picture. Using Lemma 2.1.10, it follows by induction that $\Omega^i X$ is $(n-i)$-truncated. So when $i > n+1$, we are done (since (-2)-truncated objects are contractible). It remains to show that $\pi^{\mathcal{C}}_{n+1}(X) = *$. Now, $\Omega^{n+1}X$ is (-1)-truncated. So it suffices to know that a pointed (-1)-truncated object is contractible. This follows from the similar statement in *Spc*, the Yoneda lemma, and the definition of truncated objects. □

Remark 2.1.13. Once again, the converse of the proposition need not be true in general. In fact, this need not even be true in arbitrary ∞-toposes. A minimal hypothesis under which the converse holds is hypercompleteness. Indeed, in any ∞-topos, such a morphism induces an ∞-connective morphism $X \to \tau^{\mathcal{C}}_{\leqslant n}X$. So the converse is true, for example, in $\mathcal{P}(S)$ [Lur09, Example 7.2.1.9], or in $\mathcal{P}_{nis}(S)$ when S is Qcqs of finite Krull dimension (use [CM21, Theorem 3.18] and [Lur09, Corollary 7.2.1.12] along with the fact that the collection of functors $\{p^* : \mathcal{P}_{nis}(Sm_S) \to \mathcal{P}_{nis}(\acute{e}t_X)\}_{p:X\to S\in Sm_S}$ is conservative).

We end this subsection with a few more definitions that we will use in the future:

Definition 2.1.14. *When $L\mathcal{C} \subset \mathcal{C}$ is an accessible localization of presentable ∞-categories, we define $\pi^L_0 : \mathcal{C} \to \mathrm{Disc}(\mathcal{C})$ as the composition $\pi^{\mathcal{C}}_0 \circ L$.*

Definition 2.1.15. *Let $X \in \mathcal{C}$ be an object and let L be a localization of $\mathcal{C}$. We say that X is L-connected if LX is a connected object of $\mathcal{C}$, i.e., $\pi^L_0 X = \pi^{\mathcal{C}}_0 LX \simeq *$. We say that X is $L\mathcal{C}$-connected if $\pi^{L\mathcal{C}}_0 LX \simeq *$.*

Lemma 2.1.16. *An L-connected object $X \in \mathcal{C}$ is $L\mathcal{C}$-connected.*

Proof. This follows at once from Lemma 2.1.7. □

2.2. **Eilenberg Maclane objects.** Throughout this subsection, $\mathcal{C}$ is a presentable ∞-category. In a general presentable ∞-category, there could possibly be many ways to define the notion of n-connective objects. We choose to define it dually to Proposition 2.1.12:

Definition 2.2.1. *We say that a pointed object $x : * \to X$ in a presentable ∞-category $\mathcal{C}$ is*

(1) (0-)connected, or equivalently, 1-connective, if $\pi^{\mathcal{C}}_0 X = \tau^{\mathcal{C}}_{\leqslant 0}X = *$.
(2) globally $(n+1)$-connective (for some $n \geqslant 0$) (or n-connected) if it is connected and the object $\pi^{\mathcal{C}}_i(X, x)$ is the terminal object for all $0 < i \leqslant n$.

Remark 2.2.2. By convention, we say that every object is (-2)-connected, and that an object is (-1)-connected if $\tau^{\mathcal{C}}_{\leqslant -1}X = *$. Similarly, one can define the 0-connectivity condition even when $\mathcal{X}$ is not given to be pointed. When we say that an object X is 0-connected, it is inherently implied that X is (-1)-truncated. This is because $\tau_{\leqslant -1}\tau_{\leqslant 0} \simeq \tau_{\leqslant -1}$. However, the above definition is not a generalization of [Lur09, Definition 6.5.1.10], where the author defines it for an ∞-topos. The problem is that the local homotopy group object $\pi_i X$ cannot always be assembled from the global homotopy groups $\pi_i(X, x)$.

Remark 2.2.3. A version of [Lur09, Proposition 6.5.1.12], i.e., 'an object X is n-connective iff $\tau^{\mathcal{C}}_{\leqslant n}X \simeq *$', might be false unless $\mathcal{C}$ is an ∞-topos.

We denote the full subcategory of $\mathcal{C}$ consisting of n-connective objects by $\mathcal{C}_{\geqslant n}$.

Definition 2.2.4. *An object in $\mathcal{C}$ is called n-Eilenberg-MacLane if it is n-connective and n-truncated in $\mathcal{C}$. We denote by $\mathcal{EM}_n(\mathcal{C})$ the full subcategory of n-EM objects in $\mathcal{C}$.*

Let $\tau^{\mathcal{C}}_{\leqslant 0}(L\mathcal{C})$ denote the image of the composite $L\mathcal{C} \subset \mathcal{C} \xrightarrow{\tau^{\mathcal{C}}_{\leqslant 0}} \mathrm{Disc}\mathcal{C}$.

Lemma 2.2.5. *For any localization $L\mathcal{C} \subset \mathcal{C}$, there are containments*

$$L\mathcal{C} \bigcap \mathcal{C}_{\geqslant n} \subset (L\mathcal{C})_{\geqslant n} \text{ and } L\mathcal{C} \bigcap \mathcal{EM}_n(\mathcal{C}) \subset \mathcal{EM}_n(L\mathcal{C}).$$

Proof. Let $\mathcal{X} \in L\mathcal{C} \bigcap \mathcal{C}_{\geqslant n}$. We shall show that $\mathcal{X} \in (L\mathcal{C})_{\geqslant n}$. That is $\pi_i^{L\mathcal{C}}\mathcal{X} = 0$ for all $0 \leqslant i \leqslant n-1$. But we know from Corollary 2.1.8 that $\pi_i^{L\mathcal{C}}\mathcal{X} = L_{\leqslant 0}\pi_i^{\mathcal{C}}\mathcal{X} = 0$ (since $\mathcal{X} \in \mathcal{C}_{\geqslant n}$).

Using Lemma 2.1.2

$$\begin{aligned} L\mathcal{C} \bigcap \mathcal{EM}_n(\mathcal{C}) &= L\mathcal{C} \bigcap \mathcal{C}_{\leqslant n} \bigcap \mathcal{C}_{\geqslant n} \\ &= (L\mathcal{C})_{\leqslant n} \bigcap L\mathcal{C} \bigcap \mathcal{C}_{\geqslant n} \\ &\subset (L\mathcal{C})_{\leqslant n} \bigcap (L\mathcal{C})_{\geqslant n} = \mathcal{EM}_n(L\mathcal{C}) \end{aligned}$$

□

Definition 2.2.6. *We say that the localization L of the ∞-topos $\mathcal{C}$ is path-injective if the discrete localization functor $\tau^{L\mathcal{C}}_{\leqslant 0} : \tau^{\mathcal{C}}_{\leqslant 0} L\mathcal{C} \to \mathrm{Disc}\mathcal{C}$ detects the terminal object.*

In other words: A localization L is path injective if one has the following: an object $X \in \mathcal{C}$ is L connected iff it is $L\mathcal{C}$ connected.

Proposition 2.2.7. *Let L be a path-injective localization of an ∞-topos $\mathcal{C}$. Then there are equalities*

$$L\mathcal{C} \bigcap \mathcal{C}_{\geqslant n} = (L\mathcal{C})_{\geqslant n} \text{ and } \mathcal{EM}_n(L\mathcal{C}) = L\mathcal{C} \bigcap \mathcal{EM}_n(\mathcal{C}).$$

Proof. Using Lemma 2.1.2,

$$L\mathcal{C} \bigcap \mathcal{EM}_n(\mathcal{C}) = L\mathcal{C} \bigcap \mathcal{C}_{\leqslant n} \bigcap \mathcal{C}_{\geqslant n} = (L\mathcal{C})_{\leqslant n} \bigcap \mathcal{C}_{\geqslant n}.$$

It suffices to show that $\mathcal{C}_{\geqslant n} \bigcap L\mathcal{C} = (L\mathcal{C})_{\geqslant n}$.

By Lemma 2.2.5, it suffices to show that if $\mathcal{X} \in L\mathcal{C}_{\geqslant n}$, then $\mathcal{X} \in \mathcal{C}_{\geqslant n} \bigcap L\mathcal{C}$. This is equivalent to showing that $\pi_i^{\mathcal{C}}\mathcal{X} = 0$ for all $i \leqslant n-1$. Since $\tau^{L\mathcal{C}}_{\leqslant 0}$ is injective on the image of the restriction $\tau^{\mathcal{C}}_{\leqslant 0} : L\mathcal{C} \to \mathrm{Disc}\mathcal{C}$, it suffices to show that $\tau^{L\mathcal{C}}_{\leqslant 0}\pi_i^{\mathcal{C}}\mathcal{X} = 0$. But this is nothing but

$$\tau^{L\mathcal{C}}_{\leqslant 0}\pi_i^{\mathcal{C}}\mathcal{X} = L_{\leqslant 0}\tau^{\mathcal{C}}_{\leqslant 0}\pi_i^{\mathcal{C}}\mathcal{X} = L_{\leqslant 0}\pi_i^{\mathcal{C}}\mathcal{X} = \pi_i^{L\mathcal{C}}\mathcal{X}$$

[thanks to Corollary 2.1.8], which is given to be 0 $\forall i \leqslant n-1$ (by the condition $\mathcal{X} \in (L\mathcal{C})_{\geqslant n}$). □

Remark 2.2.8. A key feature of an ∞-topos (for example, Spc of spaces, $\mathcal{P}(Sm_S)$, $\mathcal{P}_\Sigma(Sm_S)$, $\mathcal{P}_{nis}(Sm_S)$, etc.) is that in an ∞-topos an object is n-connective if and only if its $n-1$-truncation is trivial. In other words, $\mathcal{C}_{\geqslant n}$ coincides with the fiber of $\tau_{\leqslant n-1} : \mathcal{C} \to \mathcal{C}$ at the final object [Lur09, Proposition 6.5.1.12], provided $\mathcal{C}$ is an ∞-topos. Following this, it is not hard to show that in an ∞-topos $\mathcal{C}$, the restriction of the homotopy group functors

$$\pi^{\mathcal{C}}_{n|\mathcal{EM}_n} : \mathcal{EM}_n(\mathcal{C}_*) \to n\text{-}Grp(\mathrm{Disc}(\mathcal{C}))$$

induces equivalences of categories [HTT, Lur09, Proposition 7.2.2.12]. In words, in an ∞-topos, Eilenberg-MacLane objects are nothing more than discrete "group" objects.

Lemma 2.2.9. *Let $\mathcal{C}$ be an ∞-topos and L an accessible localization. Assume that $L_{\leqslant 0}$ is cartesian. Then for every $\mathcal{X} \in L\mathcal{C}$, $\pi_n^{L\mathcal{C}}$ is an n-group object of* $\mathrm{Disc}(L\mathcal{C})$.

Proof. Since $\pi_i^{L\mathcal{C}} = L_{\leqslant 0}\pi_i^{\mathcal{C}}$ (Corollary 2.1.8), this follows from the fact that $\pi_i^{\mathcal{C}}$ has the same properties when $\mathcal{C}$ is an ∞-topos [Lur09, the paragraph after Lemma 6.5.1.2]. □

Given a cartesian ∞-category $\mathcal{C}$, an n-group object is a grouplike E_n-monoid object of $\mathcal{C}$. We denote the full subcategory of $\mathcal{C}$ consisting of n-group objects by $n\text{-}Grp(\mathcal{C})$. When $\mathcal{C}$ is an ordinary 1-category, $0\text{-}Grp(\mathcal{C}) \simeq \mathcal{C}_*$, $1\text{-}Grp(\mathcal{C}) \simeq Grp(\mathcal{C})$, and finally, $n\text{-}Grp(\mathcal{C}) \simeq Ab(\mathcal{C})$ for every $n \geqslant 2$.

Definition 2.2.10. *Let $G \in n\text{-}Grp(\mathrm{Disc}(\mathcal{C}))$. We say that G is n-strictly L-local if the canonical map $\mathrm{B}^n_{\mathcal{C}}G \to \mathrm{B}^n_{L\mathcal{C}}G$ is an equivalence. Denote the full subcategory of $n\text{-}Grp(\mathrm{Disc}(\mathcal{C}))$ consisting of n-strictly L-local objects as $n\text{-}Grp^L$.*

Proposition 2.2.11. *Let L be a path-injective localization of an ∞-topos. Then for every $n \geqslant 0$, there is an equivalence of categories $\mathrm{B}^n_{L\mathcal{C}} : n\text{-}Grp^L \simeq \mathcal{EM}_n(L\mathcal{C}) : \pi_n^{L\mathcal{C}}$. In fact, when restricted to $\mathcal{EM}_n(L\mathcal{C})$, there is an equivalence $\pi_n^{L\mathcal{C}} \simeq \pi_n^{\mathcal{C}}$.*

Proof. Since, by definition,

$$\mathrm{B}^n_{L\mathcal{C}|n\text{-}Grp_L(L\mathcal{C})} \simeq \mathrm{B}^n_{\mathcal{C}},$$

it follows that the functor

$$\mathrm{B}^n_{L\mathcal{C}} : n\text{-}Grp_L(L\mathcal{C}) \to \mathcal{EM}_n(L\mathcal{C})$$

is well-defined. Now the equivalence

$$\mathrm{B}^n_{\mathcal{C}} : n\text{-}Grp(\mathcal{C}) \simeq \mathcal{EM}_n(\mathcal{C})$$

[Lur09, Proposition 7.2.2.12.] guarantees that the above functor is fully faithful. The equivalence, in fact, guarantees that this is an equivalence, by the definition of $n\text{-}Grp^L$. (Observe that the inverse is given by $\pi^{\mathcal{C}}_{n|\mathcal{EM}_n(L\mathcal{C}}$, i.e., $\pi_n^{\mathcal{C}}$ has L-local n-groups as images. Therefore, the inverse is identical to $\pi_n^{L\mathcal{C}}$.) □

Corollary 2.2.12. *Suppose L is a path-injective (and path-cartesian) localization of an ∞-topos such that the inclusion $Grp_L(L\mathcal{C}) \subset Grp(\mathrm{Disc}(L\mathcal{C}))$ is strict. Then $L\mathcal{C}$ is not an ∞-topos.*

Proof. If $L\mathcal{C}$ is an ∞-topos, then by [Lur09, Proposition 7.2.2.12], $\pi_1^{L\mathcal{C}} : \mathcal{EM}_n(L\mathcal{C}) \to Grp(\mathrm{Disc}(L\mathcal{C}))$ should be an equivalence. But by the above Proposition, this should imply $Grp_L(L\mathcal{C}) = Grp(\mathrm{Disc}(L\mathcal{C}))$. □

2.3. **Truncation and group objects.** Let us briefly recall the bar construction more generally for monoids, for the convenience of the reader. Roughly, a monoid object in a presentable ∞-category $\mathcal{C}$ is a Segal object in the monoidal category $\mathcal{C}^{\times}$, i.e., a simplicial object $M_\bullet$ equipped with a compatible family of equivalences $M_n \simeq M_1 \times_{M_0} \cdots \times_{M_0} M_1$. The bar construction $\mathrm{B}M_\bullet$ is defined as the colimit of the simplicial object $M_\bullet$ regarded as a simplicial diagram in $\mathcal{C}$. Because monoids are canonically pointed by the unit morphism, the resulting colimit is also pointed. Thus, the bar construction is given by a functor $\mathrm{B}_{\mathcal{C}} : \mathrm{Mon}(\mathcal{C}) \to \mathcal{C}_\bullet$.

Now, let us return to the internal loop space construction $\Omega_{\mathcal{C}} : \mathcal{C}_\bullet \to \mathcal{C}$. Given a pointed object $* \xrightarrow{x} X$ in $\mathcal{C}$, we define the loop object of x as the following pullback diagram:

$$\begin{array}{ccc} \Omega_{\mathcal{C}} X & \xrightarrow{p_1} & pt \\ {\scriptstyle p_2}\downarrow & & \downarrow \\ pt & \longrightarrow & X \end{array}$$

In the category of spaces, it is well known that the loop space of any pointed space has a monoid structure. In general, the loop construction in an arbitrary presentable ∞-category can be promoted to a functor $\Omega : \mathcal{C}_\bullet \to \mathrm{Mon}(\mathcal{C})$. Indeed, given a pointed space $x : * \to \mathcal{X}$, one can define the monoid structure on $\Omega\mathcal{X}$ by identifying it with the 1-simplex of the Čech nerve $\check{C}x$. But $\check{C}x$ is actually an augmented object, i.e., it belongs to $\mathcal{P}(\Delta_+, \mathcal{C})$ (we shall denote this category by $Aug(\mathcal{C})$). To identify the monoid structure of $\Omega_{\mathcal{C}} X$, we take the unaugmented part of $\check{C}x$, i.e., $(\Omega_{\mathcal{C}} X)_\bullet = (\check{C}x)_{\geqslant 0}$.

Proposition 2.3.1. *Let $\mathcal{C}$ be a presentable ∞-category. Then there exists an adjunction:*

$$\mathrm{B}_{\mathcal{C}} : \mathrm{Mon}(\mathcal{C}) \leftrightarrows \mathcal{C}_\bullet : \Omega$$

Proof. For a monoid object $M_\bullet$ and a pointed object $x : * \to X$, we have,

$$\begin{aligned} \mathrm{Map}_{\mathrm{Mon}(\mathcal{C})}(M_\bullet, \Omega_{\mathcal{C}} X) &\simeq \mathrm{Map}_{\mathrm{Mon}(\mathcal{C})}(M_\bullet, \check{C}x_{\geqslant 0}) \\ &\simeq \mathrm{Map}_{Aug(\mathcal{C})}(M_\bullet \to |M_\bullet|, \check{C}x) \\ &\simeq \mathrm{Map}_{\mathcal{P}(\Delta_+^{\leqslant 0}, \mathcal{C})}(* \xrightarrow{1} |M_\bullet|, * \xrightarrow{x} X) \text{ [Lur09, Proposition 6.1.2.11]} \\ &= \mathrm{Map}_{\mathcal{C}_\bullet}(|M_\bullet|, X) \end{aligned}$$

□

It is worth recalling that in the category of spaces, the loop space construction yields a grouplike space. One way to see this is to note that the path component of the loop space of a space is a group (namely the fundamental group at the given base point), with the inverse of a based loop γ given by $t \mapsto \gamma(-t)$.

In general, the loop construction in an arbitrary presentable ∞-category is also a grouplike monoid in $\mathcal{C}$. To see this, it suffices to note that the switching of p_1 and p_2 (defining $\Omega_{\mathcal{C}} X$) induces an inverse operation $\iota : \Omega_{\mathcal{C}} X \to \Omega_{\mathcal{C}} X$ for the aforementioned monoid structure of $\Omega_{\mathcal{C}} X$. Therefore, the adjunction of Proposition 2.3.1 restricts to an adjunction:

$$\mathrm{B}_{\mathcal{C}} : \mathrm{Grp}(\mathcal{C}) \leftrightarrows \mathcal{C}_\bullet : \Omega$$

Remark 2.3.2. If $\mathscr{C}$ is an ∞-topos, then the adjunction $\mathrm{B}_{\mathcal{C}} \dashv \Omega$ is an equivalence of ∞-categories [Lur09, Lemma 7.2.2.11 (1)].

When we are in the situation of an accessible localization $L\mathcal{C} \subset \mathcal{C}$, the equivalence from the remark above does not immediately restrict to a similar equivalence for the presentable ∞-category $L\mathcal{C}$. Of course, the first issue is that $(L\mathcal{C})_{\geqslant 1}$ is not always identical to $\mathcal{C}_{\geqslant 1} \bigcap L\mathcal{C}$. Under the hypothesis of Proposition 2.2.7, we can get rid of this issue. However, there is another issue, which we have already introduced in the previous section, namely, that $\mathrm{B}_{\mathcal{C}}$ need not take local objects to local objects, and therefore $\mathrm{B}_{L\mathcal{C}} \neq \mathrm{B}_{\mathcal{C}|L\mathcal{C}}$.

Lemma 2.3.3. *Suppose L is a cartesian localization of an ∞-topos. Then $\mathrm{B}_{L\mathcal{C}} L \simeq L\mathrm{B}_{\mathcal{C}}$.*

Proof. This is obvious because the bar construction is given by finite products and geometric realizations, and $L : \mathcal{C} \to L\mathcal{C}$ commutes with both. □

Remark 2.3.4. Clearly, $\mathrm{B}_{L\mathcal{C}} \simeq L\mathrm{B}_{\mathcal{C}|L\mathcal{C}}$. So the adjunction of Proposition 2.3.1 can be deduced for $L\mathcal{C}$ using the adjunction for $\mathcal{C}$. In particular, since every presentable ∞-category arises as a localization of a presheaf ∞-topos, the adjunction of Proposition 2.3.1 can be deduced even more easily, since the adjunction is well known for presheaf ∞-toposes.

Let $Grp_L L\mathcal{C}$ denote the full subcategory of $GrpL\mathcal{C}$ consisting of objects $\mathcal{X}$ such that $\mathrm{B}_{\mathcal{C}}\mathcal{X} \in L\mathcal{C}$, i.e., $\mathrm{B}_{L\mathcal{C}}X \simeq \mathrm{B}_{\mathcal{C}}X$. Following [EHK$^+$21, §3.4], we will call such objects strongly L-local monoids.

Lemma 2.3.5. *For a localization $L\mathcal{C} \subset \mathcal{C}$, there is a canonical equivalence $\Omega_{L\mathcal{C}} \simeq \Omega_{\mathcal{C}|L\mathcal{C}}$.*

Proof. This is clear because $L\mathcal{C} \subset \mathcal{C}$ is closed under limits and, in particular, under fiber products. □

Theorem 2.3.6. *Suppose L is a path-injective localization of an ∞-topos $\mathcal{C}$. Then the adjunction*

$$\mathrm{B}_{L\mathcal{C}} : Grp(L\mathscr{C}) \leftrightarrows L\mathcal{C}_\bullet : \Omega_{L\mathcal{C}} \simeq \Omega_{\mathcal{C}}$$

restricts to an equivalence

$$\mathrm{B}_{L\mathcal{C}} : Grp_L(L\mathscr{C}) \simeq (L\mathcal{C})_{\geqslant 1} : \Omega_{L\mathcal{C}} \simeq \Omega_{\mathcal{C}}.$$

Proof. Under the hypothesis, we know that $(L\mathscr{C})_{\geqslant 1} = L\mathcal{C} \bigcap \mathscr{C}_{\geqslant 1}$ (Proposition 2.2.7). So it suffices to show that if $\mathcal{X} \in L\mathcal{C} \bigcap \mathscr{C}_{\geqslant 1}$, then $\Omega^{\mathcal{C}}\mathcal{X} \in Grp_L(L\mathscr{C})$. But we know that $\mathrm{B}_{\mathcal{C}}\Omega^{\mathcal{C}}\mathcal{X} \simeq \mathcal{X} \in L\mathcal{C}$ (the first equivalence holds because $\mathcal{X} \in \mathcal{C}_{\geqslant 1}$, [Lur09, Theorem 5.2.6.10]), and we are done. □

2.4. **Covering theory and localizations. Geometric coverings: (a version of (-1)-truncated morphisms)**

We first standardize the notion of an effective epimorphism from [Lur09, the paragraph following Corollary 6.2.3.5]. Recall that this notion is defined in *loc.cit.* only for semi toposes. In this text, we adopt the following definition for arbitrary ∞-categories:

Definition 2.4.1 ([nLa26, effective epimorphism])**.** *We say that a morphism in an ∞-category is an effective epimorphism if its Čech nerve is a colimit diagram.*

Remark 2.4.2. This, however, should be contrasted with the notion of an (-1)-truncated morphism in $\mathcal{C}$. Unless $\mathcal{C}$ is a semi-topos, there is no good reason for these two notions to coincide, as in [Lur09, Corollary 6.2.3.5].

Definition 2.4.3 (Geometric covering)**.** *Let $\mathcal{C}$ be an ∞-category and $L\mathcal{C}$ a localization. We say that a collection of morphisms $\{p_i : U_i \to X\}_{i\in I}$ in $\mathcal{C}$ is:*

(1) A geometric $\mathcal{C}$ cover of X if they are jointly effective epimorphic, i.e., $p : \bigsqcup U_i \to X$ is an effective epimorphism, or equivalently, if the Čech nerve of p is a colimit diagram in $\mathcal{C}$.
(2) A geometric L-cover of X if $L\check{C}_\bullet p$ is a colimit diagram in $L\mathcal{C}$.

Lemma 2.4.4. *A geometric $\mathcal{C}$-cover is a geometric L-cover.*

Proof. This is obvious since L preserves colimits. □

Lemma 2.4.5. *An L geometric covering of L-local objects is an $L\mathcal{C}$-geometric covering.*

Proof. This is clear since $L\mathcal{C}$ is closed under fiber product in $\mathcal{C}$. □

Lemma 2.4.6 (L-connectivity)**.** *Suppose L is a path-injective localization of an ∞-topos $\mathcal{C}$. Let $\{p_i : U_i \to X\}_{i\in I}$ be a geometric L-covering such that each U_i is L-connected and each $U_i \times_X U_j$ is a globally nontrivial L-local path component (i.e.,* $\mathrm{Disc}(L\mathcal{C})(*, \pi_0^L(U_i \times_X U_j)) \neq \emptyset$*; (for example, when these fiber products are pointed). Then X is L-connected.*

Proof. First, note that the functor $\tau_{\leqslant 0}^{L\mathcal{C}} : L\mathcal{C} \to \mathrm{Disc}(L\mathcal{C})$, being a localization, preserves colimits. So by applying $\tau_{\leqslant 0}^{L\mathcal{C}}$ to the colimit diagram $L\check{C}_\bullet p$ in $L\mathcal{C}$ (by definition of geometric L-covering), we see that $\tau_{\leqslant 0}^{L\mathcal{C}} LX$ is the colimit of the path components of $L\check{C}_\bullet p$ in $L\mathcal{C}$, computed in $\mathrm{Disc}(L\mathcal{C})$. As $\mathrm{Disc}(L\mathcal{C})$ is an ordinary category, this colimit can actually be computed as the colimit of the 1-skeleton of the simplicial diagram. In other words, $\tau_{\leqslant 0}^{L\mathcal{C}} LX$ is the coequalizer of the following diagram in $\mathrm{Disc}(L\mathcal{C})$:

$$\coprod_{i,j} \pi_0^{L\mathcal{C}}(U_i \times_X U_j) \rightrightarrows \coprod_i \pi_0^{L\mathcal{C}}(U_i)$$

But by formal arguments, this colimit is equal to the colimit of the pairwise pushouts

$$* \simeq \pi_0^{L\mathcal{C}} U_i \leftarrow \pi_0^{L\mathcal{C}} U_i \times_X U_j \to \pi_0^{L\mathcal{C}} U_j \simeq *$$

in $L\mathcal{C}$, say denoted E_{ij}. Since every leg $\pi_0^{L\mathcal{C}} U_i \times_X U_j \to \pi_0^{L\mathcal{C}U_i} \simeq *$ is identical and is an epimorphism (by the given condition of the existence of a global section), it follows that the pushout maps $\pi_0^{L\mathcal{C}} U_j \to E_{ij}$ are all isomorphisms. So $E_{ij} \simeq *$. Thus, we have shown that the colimit of each of the pairwise diagrams in $\mathrm{Disc}(L\mathcal{C})$:

$$* \simeq \pi_0^{L\mathcal{C}} U_i \leftarrow \pi_0^{L\mathcal{C}} U_i \times_X U_j \to \pi_0^{L\mathcal{C}} U_j \simeq *$$

yields the terminal object. Hence, the final colimit in $\mathrm{Disc}(L\mathcal{C})$, which is $\pi_0^{L\mathcal{C}} LX$, is equivalent to the colimit of the diagram $* \simeq \pi_0^{L\mathcal{C}} LU_i \to *$ in $\mathrm{Disc}(L\mathcal{C})$ and is thus equivalent to $*$ itself. Since L is path injective, it follows that $\pi_0^L X \simeq *$. □

Corollary 2.4.7. *Let $\{p_i : U_i \to X\}_{i\in I}$ be a geometric $\mathcal{C}$ covering such that each U_i is L-locally 0-connected and each $U_i \times_X U_j$ is globally L-connected (e.g., pointed). Then X is L-locally 0-connected.*

Theorem 2.4.8 (L-local Van Kampen theorem)**.** *Let L be a path-injective localization of an ∞-topos $\mathcal{C}$. Let $\{p_i : U_i \to X\}_{i\in I}$ be a well-pointed $\mathcal{C}$ (or, more generally L) geometric covering in $\mathcal{C}$. Assume that $U_i \times_X U_j \times_X U_k$ is L-connected for every $i, j, k \in \{I\} \cup I$ (where we denote $U_I := X$) (i.e., 1,2-, 2-, and 3-fold intersections are connected). Then the L-local Van Kampen theorem states that there is a coequalizer diagram in Grp^L:*

$$\underset{i,j}{*} \pi_1^{L\mathcal{C}}(U_i \times_X U_j) \rightrightarrows \underset{i}{*} \pi_1^{L\mathcal{C}}(U_i) \to \pi_1^{L\mathcal{C}}(X) \to *$$

Proof. Let us construct a diagram $I^{\Delta^{inj}}$ as follows. Its objects are pairs $(n, i_1 \cdots i_n)$ with $n \geqslant 0$ and $i_1 \cdots i_n \in I^n$, and its morphisms are projection maps. Consider the functor $C : (I^{\Delta^{inj}})^{op} \to \mathcal{C}$ given by

$$(n, i_1 \cdots i_n) \mapsto U_{i_1} \times_X \cdots \times_X U_{i_n}.$$

By the effectivity condition, we have a colimit diagram in $\mathcal{C}$:

$$\operatorname*{colim}_{x \in I^{\Delta^{inj}, op}} \Pi x \simeq X.$$

Applying the left adjoint, we obtain a colimit diagram in $L\mathcal{C}$:

$$\operatorname*{colim}_{x \in I^{\Delta^{inj}, op}} L\Pi x \simeq LX.$$

Applying the truncation $\tau^{L\mathcal{C}}_{\leqslant 1}$ (which is a left adjoint), we find the colimit,

$$\operatorname*{colim}_{x\in I^{\Delta^{inj},op}} \tau^{L\mathcal{C}}_{\leqslant 1} L\Pi x \simeq \tau^{L\mathcal{C}}_{\leqslant 1} LX$$

in $(L\mathcal{C})_{\leqslant 1}$. Moreover, because $I^{\Delta^{inj},op}$ is a semisimplicial diagram and $(L\mathcal{C})_{\leqslant 1}$ is a $(2,1)$-category, the colimit can be computed using its 2-skeleton. Since each object in this diagram is connected in $L\mathcal{C}$ and so is the colimit LX (by Theorem 2.4.6), it follows that

$$\operatorname*{colim}_{x\in (I^{\Delta^{inj}}_{\leqslant 2})^{op}} \tau^{L\mathcal{C}}_{\leqslant 1} L\Pi x \simeq \tau^{L\mathcal{C}}_{\leqslant 1} LX$$

in $\mathcal{EM}_1(L\mathcal{C})$. For $x \in I^{\Delta^{inj},op}_{\leqslant 2}$, $L\Pi x$ is given to be 0-connected in $L\mathcal{C}$, hence so is $\tau^{L\mathcal{C}}_{\leqslant 1} L\Pi x$. Thus,

$$\tau^{L\mathcal{C}}_{\leqslant 1} L\Pi x \in \mathcal{EM}_1(L\mathcal{C}).$$

Since the colimit in $(L\mathcal{C})_{\leqslant 1}$ is connected, it is computed in $\mathcal{EM}_1(L\mathcal{C})$. By Proposition 2.2.11, the functor

$$\pi_1^{L\mathcal{C}} : \mathcal{EM}_1(L\mathcal{C}) \to Grp^L$$

preserves colimits. Therefore, we have a diagram in Grp^L (for $(i,j,k) \in I^3$:

$$\pi_1^{L\mathcal{C}}(U_i \times_X U_j \times U_k) \to \pi_1^{L\mathcal{C}}(U_i \times_X U_j) \to \pi_1^{L\mathcal{C}}(U_i)$$

whose colimit is $\pi_1^{L\mathcal{C}} X$. Since Grp^L is a discrete category, this colimit can also be computed as the colimit of the subdiagram (for $(i,j) \in I^2$)

$$\pi_1^{L\mathcal{C}}(U_j) \leftarrow \pi_1^{L\mathcal{C}}(U_i \times_X U_j) \to \pi_1^{L\mathcal{C}}(U_i).$$

The colimit of this last diagram can be obtained from the diagram in question, namely the coequalizer in Grp^L of the parallel arrows:

$$\underset{i,j}{*} \pi_1^{L\mathcal{C}}(U_i \times_X U_j) \rightrightarrows \underset{i}{*} \pi_1^{L\mathcal{C}}(U_i)$$

□

Spatial Covering (0-truncated morphisms):

Definition 2.4.9. *Let $\mathcal{C}$ be a presentable ∞-category. A morphism $f : Y \to X$ in $\mathcal{C}$ is said to be a $\mathcal{C}$-covering if $f \in (\mathcal{C}_{/X})_{\leqslant 0}$, i.e., $f \to \tau^{\mathcal{C}_{/X}}_{\leqslant 0} f$ is an equivalence. The category of coverings of X is the full subcategory of $\mathcal{C}_{/X}$ consisting of coverings of X. We denote this category by $\mathcal{C}ov_{\mathcal{C}}(X) := (\mathcal{C}_{/X})_{\leqslant 0}$.*

Proposition 2.4.10. *When $L\mathcal{C} \subset \mathcal{C}$ is a localization, a morphism $X \to Y$ in $L\mathcal{C}$ is a covering in $L\mathcal{C}$ if and only if it is a covering in $\mathcal{C}$.*

Proof. When $X \in L\mathcal{C}$, the inclusion $L\mathcal{C}_{/X} \subset \mathcal{C}_{/X}$ is closed under limits. Indeed, if $D : \mathscr{I} \to L\mathcal{C}_{/X}$ is a diagram, then the limit of the composite $\mathscr{I} \to L\mathcal{C}_{/X} \subset \mathcal{C}_{/X}$ is the limit of the extended cone $D^{\triangleright} : \mathscr{I}^{\triangleright} \to \mathcal{C}$ (mapping the terminal object to X), with structure map to X given by the leg to the terminal object X. But since $L\mathcal{C} \subset \mathcal{C}$ is closed under limits, this limit belongs to $L\mathcal{C}_{/X}$. Since both $\mathcal{C}_{/X}$ and $L\mathcal{C}_{/X}$ are presentable, it follows that $L\mathcal{C}_{/X} \subset \mathcal{C}_{/X}$ is a localization. Therefore, Lemma 2.1.2 applies, and it follows that

$$L\mathcal{C}_{/X_{\leqslant 0}} = (\mathcal{C}_{/X})_{\leqslant 0} \bigcap L\mathcal{C}_{/X}.$$

□

Remark 2.4.11. For convenience, we say that a localization is effective if, for every object X, the localization unit $X \to LX$ is an effective epimorphism. An example of an effective localization is the localization of a sheaf topos at an interval object, in the sense of [MV99]. Indeed, when a sheaf topos $\mathcal{T}$ is localized at an interval object I, there is an epimorphism $\tau^{\mathcal{T}}_{\leqslant 0}\mathcal{X} \to \tau^{\mathcal{T}}_{\leqslant 0} L_I \mathcal{X}$

[MV99, Corollary 2.3]. By [Lur09, Proposition 7.2.1.14], it follows that $\mathcal{X} \to L_I\mathcal{X}$ is an effective epimorphism. More generally, one can prove that a cartesian localization at any split object (objects admitting global sections) is effective (see [Mai26b]).

Lemma 2.4.12. *An effective localization $L : \mathcal{C} \to \mathcal{C}$ preserves effective epimorphisms.*

Proof. Since $\mathcal{C}$ is an ∞-topos, by [Lur09, Proposition 7.2.1.14], this follows from the fact that the property of being an epimorphism in an ordinary category has right cancellation. □

Lemma 2.4.13. *Let $\mathcal{C}$ be an ∞-topos and L a localization of $\mathcal{C}$. Then*

(1) A morphism of $L\mathcal{C}$-coverings is itself an $L\mathcal{C}$-covering.
(2) Composition of $L\mathcal{C}$ coverings is an $L\mathcal{C}$ covering.
(3) Let $f : Y \to X$ be a morphism in $\mathcal{C}$. Then the pull back functor $\mathcal{C}/LX \to \mathcal{C}/LY$ takes $L\mathcal{C}$ coverings to $L\mathcal{C}$ coverings.
(4) If L is effective, then the canonical map $\mathcal{C}_{/LX} \to \mathcal{C}_{/X}$ (induced by pullback along the localization unit $X \to LX$) preserves and detects coverings.
(5) If L is effective, then a morphism $Z \to LX$ is an $L\mathcal{C}$-covering if and only if Z is L-local and $Z \times_X LX \to X$ is a $\mathcal{C}$-covering.

Proof. (1,2). Using the proposition above, for (1) and (2), it suffices to assume $L = id_{\mathcal{C}}$. Now, suppose we have a commutative triangle:

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ & {\scriptstyle h}\searrow \quad \swarrow{\scriptstyle g} & \\ & Z & \end{array}$$

thought of as a morphism $f : h \to g$ in the slice category $\mathcal{C}_Z$. By [Lur09, Lemma 5.5.6.14], it follows that when g is 0-truncated, h is 0-truncated if and only if f is.

(3, 4). When L is effective, i.e., $X \to LX$ is an effective epimorphism for every $X \in \mathcal{C}$, the result [Lur09, Proposition 6.2.3.17] applies.

(5). Follows immediately from (4) and the definition of $L\mathcal{C}$ coverings. □

Remark 2.4.14. Under the conditions of statement 5 of the above proposition, that is, when the localization L is effective, it follows that there is a pullback square:

$$\begin{array}{ccc} \mathcal{C}ov_{L\mathcal{C}}(LX) & \hookrightarrow & L\mathcal{C}_{/LX} \\ \downarrow & & \downarrow \\ \mathcal{C}ov_{\mathcal{C}}(X) & \hookrightarrow & \mathcal{C}_{/X} \end{array}$$

Lemma 2.4.15. *Suppose X is a pointed, connected object. Then a morphism $f : Z \to X$ in $\mathcal{C}$ is a covering if and only if the fiber of f is a discrete object of $\mathcal{C}$.*

Proof. When X is connected, the pointing map $* \to X$ is an effective epimorphism. As before [Lur09, Proposition 6.2.3.17], the result applies again. The result follows from the fact that a morphism $Z \to *$ is 0-truncated iff Z is 0-truncated. □

Corollary 2.4.16. *Suppose L is closed under extensions over LX. Then a morphism $f : Z \to LX$ with LX pointed and connected is an $L\mathcal{C}$ covering if and only if $fib(f) \in Disc(L\mathcal{C})$.*

Proof. It follows that Z is L-local. So it suffices to know that f is a $\mathcal{C}$-covering (see Proposition 2.4.10). This is immediate from the Lemma above. □

Definition 2.4.17. *Let $\mathcal{C}$ be a presentable ∞-category and L an accessible localization. Let $X \in \mathcal{C}$ be an object. The category of L-fibrations over X is the full subcategory of $\mathcal{C}_{/X}$ consisting of objects over X that are local with respect to L-local equivalences in $\mathcal{C}_{/X}$. We denote it by $L\mathcal{F}ib_X$. An object $Y \in L\mathcal{F}ib_X$ is said to be an L-fibration over X. The category of L-covering fibrations of X, denoted $Cov_L(X)$, is the truncation $\tau_{\leqslant 0} L\mathcal{F}ib_X$.*

Remark 2.4.18. Since the forgetful functor $\mathcal{C}_{/X} \to \mathcal{C}$ preserves and creates colimits, the class of L-local equivalences in $\mathcal{C}_{/X}$ is again saturated. Hence the inclusion $L\mathcal{F}ib_X \subset \mathcal{C}_{/X}$ is an accessible localization, say given by L^{fib}. Therefore, using Lemma 2.1.2, we find:

$$(L\mathcal{F}ib_X)_{\leqslant 0} = (\mathcal{C}_{/X})_{\leqslant 0} \bigcap L\mathcal{F}ib_X.$$

Lemma 2.4.19. *A morphism between L-local objects is always an L-fibration.*

Proof. Given a morphism $f : Y \to X$, let us consider the L^{fib} localization defined by the following triangle

$$\begin{array}{ccc} X & \xrightarrow{e} & dom(L^{fib} f) \\ & \searrow^{f} & \downarrow{\scriptstyle L^{fib} f} \\ & & Y \end{array}$$

Because Y is L-local and $L^{fib} f$ is an L-fibration, it follows that $dom(L^{fib} f)$ is L-local. Since e is an L-equivalence by definition, and both X and $dom(L^{fib} f)$ are L-local, it follows that e is an equivalence, and therefore f is equivalent to the L-fibration $L^{fib} f$. □

Lemma 2.4.20. *(1) The pullback of an L-fibration (resp. covering) is a fibration (resp. covering).*
(2) The composition of L fibrations (resp. coverings) is a fibration (resp. covering).

Proof. (1). Since the left adjoint $f : \mathcal{C}_{/Y} \to \mathcal{C}_{/X}$ given by precomposition with f clearly preserves L-equivalences, the right adjoint f^* preserves the local objects, i.e., L-fibrations. On the other hand, the pullback functor $f^* : \mathcal{C}_{/X} \to \mathcal{C}_{/Y}$ is left exact. By [Lur09, Proposition 5.5.6.16], it preserves 0-truncated objects.

(2). The fact that the composition of L-fibrations is an L-fibration follows easily. That the composition of 0-truncated morphisms is 0-truncated is well known. In fact, this is true in general for n-truncated morphisms. For example, if $f : X \to Y$ and $g : Y \to Z$ are n-truncated, the claim follows from applying [Lur09, Lemma 5.5.6.14] to the morphism f thought of as a morphism from gf to g in the slice category $\mathcal{C}_{/Z}$. □

Proposition 2.4.21. *Let $X \in L\mathcal{C}$ be an L-local object. Then $Cov_L(X) = Cov_{L\mathcal{C}}(X)$.*

Proof. The point is that a morphism $f : Y \to X$ over an L-local object X is an L-fibration iff $Y \in L\mathcal{C}$ (see Lemma 2.4.19). The claim now follows easily from Lemma 2.4.20. □

For the rest of this section, assume that $\mathcal{C}$ is an ∞-topos and that L is an accessible localization (not necessarily left exact).

Corollary 2.4.22. *Let $G \in (Grp^L(\mathcal{C}))$ and let $Y \to X$ be a G-torsor [NSS15 §3]. Then $Y \to X$ is an L-covering.*

Proof. Note that for any $G \in Grp(\mathrm{Disc}(\mathcal{C}))$, $* \to \mathrm{B}_{\mathcal{C}}G$ is a $\mathcal{C}$ covering by [Corollary 2.4.16]. Thus, when $G \in Grp^L\mathcal{C}$ so that $\mathrm{B}_{\mathcal{C}}G$ is L local, the morphism is an $L\mathcal{C}$ covering. But then, by the above proposition Proposition 2.4.21, it is also an L covering. Because $Y \to X$ is pulled back from $* \to \mathrm{B}_{\mathcal{C}}G$ along some map $X \to \mathrm{B}_{\mathcal{C}}G$, Lemma 2.4.20 (1) yields the desired result. □

For an $X \in \mathcal{C}$, let $\eta : X \to LX$ be the corresponding localization unit. Pulling back along this unit morphism yields a canonical map

$$\eta^* : Cov_{L\mathcal{C}}LX = Cov_L(LX) \to Cov_L(X).$$

This map fits into the following pullback square (even when L is not effective):

$$\begin{array}{ccc} \mathcal{C}ov_{L\mathcal{C}}(LX) & \hookrightarrow & L\mathcal{C}_{/LX} \\ \downarrow{\scriptstyle \eta^*} & & \downarrow{\scriptstyle \eta^*} \\ \mathcal{C}ov_L(X) & \hookrightarrow & \mathcal{C}_{/X} \end{array}$$

Proposition 2.4.23. *If L is locally cartesian and effective on X, then the functor*

$$\eta_X^* : Cov_{L\mathcal{C}}LX = Cov_L(LX) \to Cov_L(X)$$

given by pull back along the localization $\eta_X : X \to LX$ is an equivalence of categories, with inverse given by L.

Proof. Given an L-covering $f : Y \to X$, consider the L-fibration $Lf : LY \to LX$ (see Lemma 2.4.19) and the pullback

$$\begin{array}{ccccc} Y & \longrightarrow & LY \times_{LX} X & \longrightarrow & X \\ & \searrow & \downarrow & & \downarrow \\ & & LY & \xrightarrow{Lf} & LX \end{array}$$

Because L is locally cartesian on X, the morphism $Y \to LY \times_{LX} X$ is an L-equivalence. Since Lf (being a morphism of L-local objects) is an L-fibration (by Lemma 2.4.19), so is the base change $LY \times_{LX} X \to X$ (use Lemma 2.4.20). Since f (being an L-covering) is also an L-fibration, the L-equivalence $Y \to LY \times_{LX} X$ is, in fact, an equivalence (being a morphism between L-fibrations). It follows that the following is a pullback square:

$$\begin{array}{ccc} Y & \longrightarrow & X \\ \downarrow & & \downarrow \\ LY & \longrightarrow & LX \end{array}$$

Since $Y \to X$ is a covering and $X \to LX$ is an effective epimorphism, it follows from [Lur09, Proposition 6.2.3.17] that $LY \to LX$ is also a covering.

From the proof, it is clear that the functor $L : Cov_L(X) \to L\mathcal{C}_{/LX}$ factors through $Cov_{L\mathcal{C}}(LX)$ (which we again denote by $L : Cov_L(X) \to Cov_{L\mathcal{C}}(LX)$) and that η^*L is equivalent to the identity. On the other hand, local cartesianity guarantees that the composition $L\eta^*$ is also equivalent to the identity. □

Remark 2.4.24. Suppose $\mathcal{C}$ is an ∞-topos with an initial object $\varnothing$. If $X \in \mathcal{C}$, the unique map $\varnothing \to X$ is always 0-truncated and hence forms an (uninteresting) initial object of the category $Cov_{\mathcal{C}}(X)$. When coverings are pointed (so the empty covering is discarded), $Cov_{\mathcal{C}}(X)_*$ has an

interesting initial object. To see this, let $X \in \mathcal{C}_*$. Consider the zero truncation $\tau_{\leqslant 0}^{/X}$ (which is a left adjoint) of the pointing map $* \to X$ and denote it by $\tau_{>1}^{\mathcal{C}} X \to X$. Since $* \to X$ is an initial object of $\mathcal{C}_{/X,*}$, and the truncation functor

$$\tau_{\leqslant 0}^{/X} : \mathcal{C}_{/X,*} \to Cov_{\mathcal{C}}(X)_*$$

preserves colimits, it follows that $\tau_{>1}^{\mathcal{C}} X \to X$ is an initial object of $Cov_{\mathcal{C}}(X)_*$. In fact, by computing homotopy groups, one can show that this is part of a fiber sequence

$$\tau_{>1}^{\mathcal{C}} X \to X \to \tau_{\leqslant 1}^{\mathcal{C}} X.$$

Proposition 2.4.25. *Suppose $\mathcal{C}$ is an ∞-topos with an accessible localization L. Suppose moreover that $X \in \mathcal{C}_*$ is such that $\tau_{\leqslant 1} LX \simeq \tau_{\leqslant 1}^{L\mathcal{C}} LX$. Then the category $\mathcal{C}ov_{L\mathcal{C}}(LX)_*$ has an initial object $\tilde{X} := \tau_{>1} LX$.*

Proof. Clearly, $\mathcal{C}ov_{L\mathcal{C}}(LX) \subset \mathcal{C}ov_{\mathcal{C}}(LX)$. Therefore, it suffices to show that the initial object of $\mathcal{C}ov_{\mathcal{C}}(LX)$ is L-local. This initial object is the 1-connected covering $\tau_{>1}^{\mathcal{C}} LX \to LX$. In the fiber sequence

$$\tau_{>1}^{\mathcal{C}} LX \to LX \to \tau_{\leqslant 1}^{\mathcal{C}} LX,$$

the term $\tau_{\leqslant 1}^{\mathcal{C}} LX$ is L-local. Since $L\mathcal{C} \subset \mathcal{C}$ is closed under limits, we find that $\tau_{>1}^{\mathcal{C}} LX$ is L-local and hence

$$(\tau_{>1}^{\mathcal{C}} LX \to LX) \in \mathcal{C}ov_{L\mathcal{C}}(LX).$$

□

We will call $\tau_{\leqslant 1}^{\mathcal{C}} Y$ the fundamental groupoid of Y and denote it by $\Pi_1^{\mathcal{C}}(Y)$. We will use the following notation:

$$\Pi^L Y := \tau_{\leqslant 1}^{\mathcal{C}} LY \text{ and } \Pi^{L\mathcal{C}} Y := \tau_{\leqslant 1}^{L\mathcal{C}} LY \simeq L_{\leqslant 1} \tau_{\leqslant 1}^{\mathcal{C}} LY \simeq L_{\leqslant 1} \Pi^L Y.$$

Example 2.4.26. For a space $\mathcal{X} \in \mathcal{P}(Sm_S)$ and a Grothendieck topology σ, the ordinary σ-local fundamental groupoid will be denoted by $\Pi_1^{\sigma} \mathcal{X}$. Similarly, when $L = L_{mot}$ and $L = L_{bir}$, the corresponding fundamental groupoids will be denoted by $\Pi_1^{\mathbb{A}^1} \mathcal{X}$ and $\Pi_1^{b} \mathcal{X}$, respectively.

The condition in the above proposition is then read as $\Pi^L X \simeq \Pi^{L\mathcal{C}} X$. While it might be possible to deduce more general statements without assuming this equivalence, at this point we will work with this condition.

Definition 2.4.27. *We say that a localization L is fundamental on X if $\Pi^L X \simeq \Pi^{L\mathcal{C}} X$. We say that L is fundamental if it is fundamental on every object.*

Lemma 2.4.28. *Suppose L is fundamental on a pointed object $y : * \to Y$. Then*

$$\pi_1^L(Y, y) = \pi_1^{\mathcal{C}}(LY, Ly) \simeq \pi_1^{L\mathcal{C}}(LY, Ly).$$

Proof. This follows from the fact that

$$\pi_1^{\mathcal{C}}(LY, Ly) \simeq \Omega \tau_{\leqslant 1}^{\mathcal{C}} LY \simeq \Omega \tau_{\leqslant 1}^{L\mathcal{C}} LY,$$

and that the loop space of a local object is local. □

Suppose $Y \in \mathcal{C}$ admits a global section $y : * \to Y$. Then, by Lemma 2.4.13 (3), pullback along y induces a functor

$$y^* : Cov_{L\mathcal{C}}(LY) \to Cov_L(*) = Disc(LC).$$

Lemma 2.4.29. *Let $Z \to Y$ be a covering in $\mathcal{C}$. Then the fiber y^*Z has an action by $\pi_1(Y,y)$. When $\Pi^L X \simeq \Pi^{L\mathcal{C}} X$, y^* of an $L\mathcal{C}$ covering is a $\pi_1^L(Y,y) = \pi_1^{\mathcal{C}}(LY, Ly) \simeq \pi_1^{L\mathcal{C}}(LY, Ly)$ module.*

Proof. Again, we may reduce to $L = id$. By applying $Map(W, -)$ and using the equivalence of the Yoneda functor $\mathcal{C} \to \infty\text{-}Top(\mathcal{C}^{op}, Spc)$, we reduce this to the case of spaces, where it is well known that ΩY acts coherently on y^*Z, inducing the required action on π_0. □

Thus, there is a canonical functor

$$\Gamma_y^L : Cov_{L\mathcal{C}}(LY) \longrightarrow \pi_1^L(Y,y)\text{-}Disc(L\mathcal{C})$$

provided L is fundamental on Y.

Proposition 2.4.30. *Suppose $L\mathcal{C}$ is closed under extensions over LY. When LY is connected, the functor*

$$\Gamma_y^L : Cov_{L\mathcal{C}}(LY) \longrightarrow \pi_1^L(Y,y)\text{-}Disc(L\mathcal{C})$$

is an equivalence.

Proof. When $L = id$, the inverse is given by

$$F \mapsto \big(\mathrm{B}_{\mathcal{C}}(F, \Omega Y, *) \to \mathrm{B}_{\mathcal{C}}(*, \Omega Y, *) \simeq Y\big).$$

Since the morphism $\mathrm{B}_{\mathcal{C}}(F, \Omega Y, *) \to \mathrm{B}_{\mathcal{C}}(*, \Omega Y, *) \simeq Y$ has a canonical fiber equal to the discrete space F, connectivity of Y implies that it is a covering (see Lemma 2.4.15).

When L is nontrivial, it suffices to know the L-local counterpart, namely Corollary 2.4.16. □

Before we state the next result, it is worth noting that when X is an n-truncated object in a presentable ∞-category $\mathcal{D}$, then for any object Y, the object $\underline{Map}_{\mathcal{C}}(Y, X)$ is also n-truncated. Hence, the following makes sense, where $\underline{End}_{\mathcal{D}}(X) := \underline{Map}_{\mathcal{D}}(X)$.

Lemma 2.4.31. *Let L be a localization of an ∞-topos $\mathcal{C}$. Suppose L is locally cartesian and fundamental on a pointed object X. Then the canonical map $\pi_1^L X \to \underline{End}_{L\mathcal{C}_{/LX}}(\tilde{X})$ in $\mathcal{C}$ is an isomorphism. It follows that this induces isomorphisms in $\mathcal{C}$*

$$\pi_1^{L\mathcal{C}} LX \simeq \pi_1^L X \cong \underline{Aut}_{L\mathcal{C}_{/LX}}(\tilde{X}) \cong \underline{End}_{L\mathcal{C}_{/LX}}(\tilde{X}).$$

Proof. Again, it suffices to prove this for $L = id_{\mathcal{C}}$, since

$$\underline{End}^{L\mathcal{C}}_{LX}(\tilde{X}) \simeq \underline{End}^{\mathcal{C}}_{LX}(\tilde{X}) \text{ and } \pi_1^L = \pi_1^{\mathcal{C}} L.$$

Indeed, for any $Z \in \mathcal{C}_{/X}$,

$$\begin{aligned}
\mathrm{Map}_{\mathcal{C}_{/LX}}(Z, \underline{End}_{L\mathcal{C}_{/LX}}(\tilde{X})) &\simeq \mathrm{Map}_{L\mathcal{C}_{/LX}}(LZ, \underline{End}_{L\mathcal{C}_{/LX}}(\tilde{X})) \\
&\simeq \mathrm{Map}_{L\mathcal{C}_{/LX}}(LZ \times_{LX} \tilde{X}, \tilde{X}) \\
&\simeq \mathrm{Map}_{L\mathcal{C}_{/LX}}(L(Z \times_{LX} \tilde{X}), \tilde{X}) \text{ [since } L \text{ is locally cartesian]} \\
&\simeq \mathrm{Map}_{\mathcal{C}_{/LX}}(Z \times_{LX} \tilde{X}, \tilde{X}) \\
&\simeq \mathrm{Map}_{\mathcal{C}_{/LX}}(Z, \underline{End}_{\mathcal{C}_{/LX}}(\tilde{X})
\end{aligned}$$

So we are reduced to proving that

$$Aut_{\mathcal{C}_{/Y}}(\tau_{>1}Y) \simeq \underline{End}_{\mathcal{C}_{/Y}}(\tau_{>1}Y) \cong \pi_1^{\mathcal{C}} Y$$

for an object $Y \in \mathcal{C}$. Let $u : Y \to \tau_{\leqslant 1}Y$ be the canonical truncation with basepoint $f : * \to \tau_{\leqslant 1}Y$. We wish to compute the absolute mapping object $End_{\mathcal{C}_{/Y}}(\tau_{>1}Y) \in \mathcal{C}$.

Let $i : \tau_{>1}Y \to Y$ denote the canonical inclusion. By definition of morphisms in the slice category $\mathcal{C}_{/Y}$, we obtain a fiber sequence in $\mathcal{C}$:

$$\underline{End}_{\mathcal{C}_{/Y}}(\tau_{>1}Y) \to \underline{Map}_{\mathcal{C}}(\tau_{>1}Y, \tau_{>1}Y) \xrightarrow{i\circ -} \underline{Map}_{\mathcal{C}}(\tau_{>1}Y, Y)$$

taken over the structure map i.

Next, we map the object $\tau_{>1}Y$ into the defining fiber sequence of the connected cover:

$$\tau_{>1}Y \xrightarrow{i} Y \xrightarrow{u} \tau_{\leqslant 1}Y.$$

Because mapping objects in an ∞-topos preserve limits, this induces another fiber sequence in $\mathcal{C}$:

$$\underline{Map}_{\mathcal{C}}(\tau_{>1}Y, \tau_{>1}Y) \xrightarrow{i\circ -} \underline{Map}_{\mathcal{C}}(\tau_{>1}Y, Y) \xrightarrow{u\circ -} \underline{Map}_{\mathcal{C}}(\tau_{>1}Y, \tau_{\leqslant 1}Y).$$

By the adjunction properties of truncations, $\tau_{>1}Y$ is > 1-connected and $\tau_{\leqslant 1}Y$ is 1-truncated. Thus, any map between them factors uniquely through the terminal object. $*$:

$$\underline{Map}_{\mathcal{C}}(\tau_{>1}Y, \tau_{\leqslant 1}Y) \simeq \underline{Map}_{\mathcal{C}}(*, \tau_{\leqslant 1}Y) \simeq \tau_{\leqslant 1}Y.$$

The composition $u \circ i$ is canonically identified with the basepoint $f : * \to \tau_{\leqslant 1}Y$. Therefore, the fiber of the map $u \circ -$ is precisely the loop space of $\tau_{\leqslant 1}Y$ at the basepoint f.

By comparing the fiber sequences, we see that the object $\underline{End}_{\mathcal{C}_{/Y}}(\tau_{>1}Y)$ is exactly the fiber of $u \circ -$ over f. Hence:

$$\underline{End}_{\mathcal{C}_{/Y}}(\tau_{>1}Y) \simeq \Omega(\tau_{\leqslant 1}Y, f) \simeq \pi_1^{\mathcal{C}}(Y).$$

Finally, because $\tau_{>1}Y$ is an object of $\mathcal{C}_{/Y}$, its endomorphism object $\underline{End}_{\mathcal{C}_{/Y}}(\tau_{>1}Y)$ naturally inherits a monoid structure in $\mathcal{C}$. Since $\pi_1^{\mathcal{C}}(Y)$ is inherently group-like, the inclusion of the maximal subgroupoid is an equivalence. This yields the desired isomorphism:

$$\underline{Aut}_{\mathcal{C}_{/Y}}(\tau_{>1}Y) \simeq \underline{End}_{\mathcal{C}_{/Y}}(\tau_{>1}Y) \simeq \pi_1^{\mathcal{C}}(Y).$$

□

The L-local Galois correspondence

Theorem 2.4.32. *Let L be a localization of an ∞-topos that is fundamental on connected objects and is path injective on all objects. Suppose we are given a pointed L- connected object $X \in \mathcal{C}$ such that L is locally cartesian, effective, and extensions closed over X. Then there is an equivalence of categories between the 1-category $Cov_L(X)^{\geqslant 1}$ of L-coverings of X with $L\mathcal{C}$-connected total space and the strongly L-local subgroups of $\pi_1^{L\mathcal{C}}(X)$:*

$$Sub_{Grp_L(\mathrm{Disc}L\mathcal{C})}(\pi_1^{L\mathcal{C}}(X)) \simeq Cov_{L\mathcal{C}}(LX) \times_{L\mathcal{C}_{/LX}} L\mathcal{C}_{>0/LX}.$$

Proof. To set the stage, let us first consider the well-known case $L = id$. In this situation, the equivalence is comes from the following construction:

$$_*\pi_1^{\mathcal{C}} : Cov_{\mathcal{C}}(X) \to Sub_{Grp(\mathrm{Disc}\mathcal{C})}(\pi_1^{\mathcal{C}}(X)).$$

Given a $\mathcal{C}$-covering $p : Y \to X$ of X define $_*\pi_1^{\mathcal{C}}(p) := p_*(\pi_1^{\mathcal{C}}(Y))$. The construction giving the converse is as follows:

$$p_- : Sub_{Grp(\mathrm{Disc}\mathcal{C})}(\pi_1^{\mathcal{C}}(X)) \to Cov_{\mathcal{C}}(X).$$

Given a subgroup $i : H \leqslant \pi_1^{\mathcal{C}}(X)$ the covering space p_H is given by the following pull back:

$$\begin{array}{ccc} E_H & \longrightarrow & \mathrm{B}_{\mathcal{C}}H \\ {\scriptstyle p_H}\downarrow & & \downarrow{\scriptstyle \mathrm{B}_{\mathcal{C}}i} \\ X & \xrightarrow[q]{} & \mathrm{B}_{\mathcal{C}}\pi_1^{\mathcal{C}}(X) \end{array}$$

It is clear, using a long exact sequence argument of $\mathcal{C}$-homotopy groups, that E_H is connected and $\pi_1^{\mathcal{C}} E_H = H$. Since for every covering p the morphism

$$p_* : \pi_1^{\mathcal{C}} dom(p) \to \pi_1^{\mathcal{C}} cod(p)$$

is injective (since its fiber of p is discrete) we have $p_{H*}\pi_1^{\mathcal{C}} E_H \simeq H$. For the other equivalence, note that given a covering $p : Y \to X$ there is a morphism of coverings $Y \to E_{p_*\pi_1^{\mathcal{C}} Y}$. By a long exact sequence of homotopy groups, we know that both coverings have identical fibers with identical action of $\pi_1^{\mathcal{C}} X$ (this is what uses the connectivity assumption on Y). By Proposition 2.4.30 these dta yield the equivalence:

$$Sub_{Grp(\mathrm{Disc}\mathcal{C})}(\pi_1^{\mathcal{C}}(X)) \simeq Cov_{\mathcal{C}}(X) \times_{\mathcal{C}_{/X}} \mathcal{C}^{\geqslant 1}{}_{/X}.$$

Now we come to the L-local part. It suffices to identify both sides of the equivalence. First of all, by the assumption on L we know that for any connected Y $\pi_1^{L\mathcal{C}} Y$ is strongly L-local. So the morphism ${}_*\pi_1^{\mathcal{C}}$ restricts to

$${}_*\pi_1^{L\mathcal{C}} : Cov_{L\mathcal{C}}(LX) \to Sub_{Grp_L(\mathrm{Disc}L\mathcal{C})}(\pi_1^{L\mathcal{C}}(LX)).$$

On the other hand, because the fiber of p_H is identical to the fiber of $\mathrm{B}_{\mathcal{C}} i$, by the extension colsedness property of L over X, Y is L local whenever $\mathrm{B}_{\mathcal{C}} H$ is. Thus p_- restricts to

$$p_- : Sub_{Grp_L(\mathrm{Disc}L\mathcal{C})}(\pi_1^{L\mathcal{C}}(X)) \to Cov_{L\mathcal{C}}(LX).$$

So the equivalence established above restricts to the equivalence:

$$Sub_{Grp_L(\mathrm{Disc}L\mathcal{C})}(\pi_1^{L\mathcal{C}}(X)) \simeq Cov_{L\mathcal{C}}(LX) \times_{\mathcal{C}_{/LX}} \mathcal{C}^{\geqslant 1}{}_{/LX}.$$

The claimed equivalence is then a consequence of path injectivity. □

Theorem 2.4.33. *Let L be a localization of an ∞-topos that is fundamental on connected objects and is path injective on all objects. Suppose we are given a pointed $L\mathcal{C}$- connected object $LX \in L\mathcal{C}$ such that L is extension-closed over X. Then there is an equivalence of categories between the 1-category $Cov_L(X)^{>_L 0}$ of L-coverings of X with L-connected total space and the strongly L-local subgroups of $\pi_1^L(X)$.*

Proof. Composing the above equivalence with the equivalence of Proposition 2.4.23 (this part requires effectivity and locally cartesian ness of L over X). □

3. The internal homotopy theory of $\mathcal{H}^{mot}$

In this section, we apply the definitions and results from the previous section to the motivic localization functor $L_{mot} : \mathcal{P}_{nis}(S) \to \mathcal{H}^{mot}(S)$. Before we do that, let us have a quick recap of how this localization is constructed.

Let S be a Qcqs scheme. By an S-space, we mean a presheaf of spaces on Sm_S. We denote by $\mathcal{P}(S)$ the ∞-category of S-spaces. We say that an S-space is Nisnevich-local if it takes Nisnevich cd squares to pullback squares of spaces (and the empty scheme to the contractible space). Recall that a Nisnevich cd square is a pullback square:

$$Q: \quad \begin{array}{ccc} V & \hookrightarrow & Y \\ \downarrow & & \downarrow p \\ U & \underset{j}{\hookrightarrow} & X \end{array}$$

of schemes smooth over S, where p is etale and j is an open immersion such that the complements induce an isomorphism with the reduced scheme structure. We denote the full subcategory of $\mathcal{P}(S)$ consisting of Nisnevich local spaces by $\mathcal{P}_{nis}(S)$. It is easy to see that the Nisnevich locality

condition is equivalent to the condition of being local with respect to the following morphism of k-spaces:

$$\{h_S U \bigsqcup_{h_S V} h_S Y \to h_S X \mid Q \text{ is a Nisnevich square}\} \bigcup \{\varnothing \to h_S \varnothing\}$$

Since the collection above is small, it follows that $\mathcal{P}_{nis}(S) \subset \mathcal{P}(S)$ is an accessible localization [Lur09, Proposition 5.5.1.14]. We denote the corresponding localization functor by L_{nis}. It turns out that L_{nis} is left exact, and thus $\mathcal{P}_{nis}(S)$ is an ∞-topos [Hoy17, Corollary 3.9].

We say that an S-space $\mathcal{X}$ is $\mathbb{A}^1$-local if, for every $X \in Sm_S$, the restriction map $\mathcal{X}(X) \xrightarrow{p^*} \mathcal{X}(\mathbb{A}^1 \times X)$, induced by the projection $\mathbb{A}^1 \times X \xrightarrow{p} X$, is an equivalence of spaces. Again, since the collection of $\mathbb{A}^1$-projections is a small set, it follows that the full subcategory of $\mathcal{P}(S)$ consisting of $\mathbb{A}^1$-local objects is an accessible localization, which we denote by $L_{\mathbb{A}^1}$.

Finally, an S-space is called motivic if it is both Nisnevich local and $\mathbb{A}^1$-local. We denote the full subcategory of $\mathcal{P}(S)$ consisting of motivic S-spaces by $\mathcal{H}^{mot}(S)$. Because both conditions are given by small sets, it follows that the subcategory $\mathcal{H}^{mot}(S) \subset \mathcal{P}(S)$ is an accessible localization. This localization is denoted by L_{mot}. It is a cartesian, locally cartesian, and accessible localization.

Morel defines the $\mathbb{A}^1$-homotopy group of a space as the Nisnevich homotopy group of its motivic localization, i.e.,

$$\pi_i^{\mathbb{A}^1} := \pi_i^{nis} L_{mot} := L_{nis} \pi_i L_{mot}.$$

It is evident that, in terms of Definition 2.1.14, this is precisely $\pi_i^{L_{mot}}$. Nonetheless, it is evident that Morel's $\mathbb{A}^1$-homotopy groups of a motivic space are simply its Nisnevich homotopy group sheaves.

Unfortunately, the $\mathbb{A}^1$-homotopy path component construction $\pi_0^{\mathbb{A}^1}$ is not as well behaved categorically. In particular, it fails to be the internal 0-truncation functor for the motivic homotopy category, as it is in an ∞-topos. Consequently, it does not preserve (motivic) colimits. The agenda of this section is to elaborate on this issue and correct it by introducing the internal notions of motivic truncations and motivic homotopy groups.

3.1. **Motivic truncations and Motivic homotopy groups.** For the convenience of the motivic readers (those who are reading this section directly without reading §2), we shall first restate the definitions of §2.1 for the motivic localization $L_{mot} : \mathcal{P}(S) \to \mathcal{H}^{mot}(S)$. To simplify notation, we shall drop the scheme S, assuming it to be fixed. This assumption shall continue until Remark 3.1.15, after which we shall assume that the base is a field (though not necessarily perfect, unless otherwise specified).

Motivic truncations:

Definition 3.1.1. *We say that a motivic space $\mathcal{X}$ is motivically n-truncated if, for every motivic space $\mathcal{Y}$, the space $\mathrm{Map}_{\mathcal{H}^{mot}}(\mathcal{Y}, \mathcal{X})$ is n-truncated. We denote the full subcategory of n-truncated motivic spaces by $\mathcal{H}^{mot}_{\leqslant n}$ or, equivalently, by $\tau^{mot}_{\leqslant n}\mathcal{H}^{mot}$.*

From Lemma 2.1.2, it follows that:

Proposition 3.1.2. *A motivic space is motivically n-truncated if and only if it is Nisnevich locally n-truncated, i.e.,*

$$\mathcal{H}^{mot}_{\leqslant n} = \mathcal{H}^{mot} \bigcap \mathcal{P}^{nis}_{\leqslant n}.$$

Remark 3.1.3. Recall that, when $n \geqslant 0$ and S has finite Krull dimension, by Remark 2.1.13 this last condition is equivalent to the requirement that for all $i > n$ the Nisnevich-local space has trivial π_i^{nis}.

Since $\mathcal{P}$ is an ∞-topos, Proposition 2.1.3 implies that $\mathcal{H}^{mot}_{\leqslant n} \subset \mathcal{P}$ is also a localization. We let

$$L^{mot}_{\leqslant n} : \mathcal{P} \to \mathcal{H}^{mot}_{\leqslant n}$$

stand for the induced motivic truncation functors. In particular, these restrict to the internal motivic truncation functors

$$\tau^{mot}_{\leqslant n} : \mathcal{H}^{mot} \to \mathcal{H}^{mot}_{\leqslant n},$$

i.e., $\tau^{mot}_{\leqslant n}$ is the localization functor for $\mathcal{H}^{mot}_{\leqslant n} \subset \mathcal{H}^{mot}$.

Construction 3.1.4. To set this in concrete terms, let us discuss how one constructs a formula for $L^{mot}_{\leqslant n}$.

We keep in mind that, from Lemma 2.1.2,

$$\mathcal{H}^{mot}_{\leqslant n} = \mathcal{H}^{mot} \bigcap \mathcal{P}_{\leqslant n} = L_{\mathbb{A}^1}\mathcal{P} \bigcap \mathcal{P}_{nis} \bigcap \mathcal{P}_{\leqslant n}.$$

The idea is then standard: define

$$L^{mot}_{\leqslant n} := \operatorname*{colim}_m \left(L_{nis}\tau_{\leqslant n} L_{\mathbb{A}^1} \right)^m$$

where $L_{nis} : \mathcal{P} \to \mathcal{P}^{nis}$ is the Nisnevich ∞-sheafification, and $\tau_{\leqslant n} : \mathcal{P} \to \mathcal{P}_{\leqslant n}$ is the ordinary Postnikov truncation of the presheaf topos (which can, for example, be modeled sectionwise by any simplicial truncation model, such as the Moore-Postnikov model P_n) and $L_{\mathbb{A}^1} : \mathcal{P} \to L_{\mathbb{A}^1}\mathcal{P}$ is the (presheaf) $\mathbb{A}^1$-localization functor (which can be modeled by taking the singular Suslin-type construction). In fact, since $\tau^{nis}_{\leqslant n} = L_{nis}\tau_{\leqslant n}$, this formula reads $L^{mot}_{\leqslant n} = \operatorname{colim}_m (\tau^{nis}_{\leqslant n} L_{\mathbb{A}^1})^m$.

Theorem 3.1.5. *The construction above provides the desired localization functor*

$$L^{mot}_{\leqslant n} : \mathcal{P} \to \mathcal{H}^{mot}_{\leqslant n}.$$

Proof. To see why this works, first observe that all three subcategories in question are closed under filtered colimits. Indeed, $\mathcal{P}_{nis} \subset \mathcal{P}$ is closed under filtered colimits because the Nisnevich topology is a cd topology; $L_{\mathbb{A}^1}\mathcal{P} \subset \mathcal{P}$ is closed under all small colimits, as can be checked on sections; and $\mathcal{P}_{\leqslant n} \subset \mathcal{P}$ is closed under filtered colimits, which can be proven by induction: the base case $n = -2$ follows from [Lur09, Definition 5.5.6.1], while the inductive step follows from [Proposition 5.5.6.5, Lur09] and the commutation of pullbacks and filtered colimits in toposes. A cofinality argument then shows that

$$\varinjlim_m (L_{nis}\tau_{\leqslant n} L_{\mathbb{A}^1}\mathcal{X})^m \in \mathcal{P}^{nis} \bigcap L_{\mathbb{A}^1}\mathcal{P} \bigcap \mathcal{P}_{\leqslant n} = \mathcal{H}^{mot} \bigcap \mathcal{P}_{\leqslant n} = \mathcal{H}^{mot}_{\leqslant n}.$$

That the natural transformation

$$1 \to \operatorname*{colim}_m (L_{nis}\tau_{\leqslant n} L_{\mathbb{A}^1})^m$$

is an $L^{mot}_{\leqslant n}$ equivalence follows from the definition of associated saturated classes. □

Remark 3.1.6. It follows that $\tau^{mot}_{\leqslant n} \simeq L^{mot}_{\leqslant n\,|\mathcal{H}^{mot}}$.

Remark 3.1.7. From the arguments of the above proof, it is not hard to see that the same truncation formula can equivalently be given by:

$$L^{mot}_{\leqslant n} = \operatorname*{colim}_m (\tau^{nis}_{\leqslant n} L_{mot})^m = \operatorname*{colim}_m (\tau_{\leqslant n} L_{mot})^m.$$

The following is not generally true for arbitrary localizing categories. In fact, it is a property of an ∞-topos [Lur09, Lemma 6.5.1.2]:

Proposition 3.1.8. *The localization $L^{mot}_{\leqslant n}$ is effective, i.e., for every S-space $\mathcal{X}$, the map $\mathcal{X} \to L^{mot}_{\leqslant n}\mathcal{X}$ is an effective epimorphism.*

Proof. By [Lur09, Proposition 7.2.1.14], it suffices to prove that the morphism is a surjection of Nisnevich sheaves under π_0^{nis}. But this morphism of Nisnevich sheaves is identified with $\pi_0^{nis}\mathcal{X} \to \mathrm{colim}_m \pi_0^{\mathbb{A}^1}\mathcal{X}$. The claim thus follows from [MV99, Corollary 1.2.3]. □

Remark 3.1.9. It is clear from the above proof that:

$$\tau_{\leqslant 0}^{mot} = \operatorname*{colim}_m (\pi_0^{\mathbb{A}^1})^{\circ m}.$$

Proposition 3.1.10. *$L_{\leqslant n}^{mot}$ preserves products.*

Proof. We use the formula constructed above. Since filtered colimits commute with finite products, it suffices to observe that each component functor in the formula has the desired property. The case of L_{nis} is standard (it is, in fact, left exact); the case of $L_{\mathbb{A}^1}$ is well known (using the fact that $\mathbb{A}^1$-projections are stable under arbitrary base change); for $\tau_{\leqslant n}$, use [Lur09, Lemma 6.5.1.2]. □

Corollary 3.1.11. *$\tau_{\leqslant n}^{mot}$ preserves products.*

Corollary 3.1.12. *L_{mot} is a path cartesian localization (see Definition 2.1.6).*

In the particular case of discrete motivic spaces, one finds

$$\mathrm{Disc}(\mathcal{H}^{mot}) = \mathcal{H}^{mot} \bigcap disc(\mathcal{P}_{nis}) = \mathcal{H}^{mot} \bigcap Sh_{nis}(Sm_S, Sets) = Sh_{nis}^{\mathbb{A}^1}$$

(here $Sh_{nis}^{\mathbb{A}^1}$ is the category of $\mathbb{A}^1$-invariant Nisnevich sheaves of sets on Sm_S). Following the above machinery, a localization for the inclusion $Sh_{nis}^{\mathbb{A}^1} \subset \mathcal{P}$ can be given by the formula:

$$L_{\leqslant 0}^{mot} := \operatorname*{colim}_m (L_{nis}\tau_{\leqslant 0}L_{\mathbb{A}^1})^m.$$

A well-developed version of this was obtained for sheaves of sets in [BHS15], where it was denoted by $\mathcal{L}$.

Corollary 3.1.13. *For motivic spaces $\mathcal{X}$, one has the sheaf isomorphism $\mathcal{L}\pi_0^{nis}\mathcal{X} \cong \pi_0^{mot}\mathcal{X}$.*

Proof. By taking the $\mathbb{A}^1$-singular formulae $L_{\mathbb{A}^1} := |(-)^{\mathbb{A}^\bullet}|$, one finds that

$$\tau_{\leqslant 0}|(-)^{\mathbb{A}^\bullet}| \simeq |(\tau_{\leqslant 0}(-))^{\mathbb{A}^\bullet}| \cong coeq\big((-)^{\mathbb{A}^1} \rightrightarrows (-)\big).$$

But this is precisely the formulae obtained in [BHS15] for presheaves of sets, which they called $\mathcal{S}^{pre}$. Consequently,

$$L_{nis}\tau_{\leqslant 0}L_{\mathbb{A}^1} = L_{nis}\pi_0^{ch} = \mathcal{S}.$$

Thus their formulae $\mathcal{L} := \mathrm{colim}_m \mathcal{S}^m \equiv \mathrm{colim}_m (L_{nis}\tau_{\leqslant 0}L_{\mathbb{A}^1})^m$ is indeed the motivic truncation functor

$$L_{\leqslant 0}^{mot} : Sh_{nis} \to disc(\mathcal{H}^{mot}) = Sh_{nis}^{\mathbb{A}^1}$$

as constructed in Theorem 3.1.4. □

The actual motivic truncation is thus given by $\tau_{\leqslant 0}^{mot} := L_{\leqslant 0}^{mot}{}_{|\mathcal{H}^{mot}} \simeq \mathcal{L}(\pi_0^{nis}{}_{|\mathcal{H}^{mot}})$.

Motivic homotopy groups:

Let π_i^{mot} be the internal homotopy groups of $\mathcal{H}^{mot}(S)$, following Definition 2.1.5. By definition, it is the internal motivic 0-truncation of the i-fold loop space:

$$\pi_i^{mot} := \tau_{\leqslant 0}^{mot}\Omega^i_{\mathcal{H}^{mot}(k)} \simeq \tau_{\leqslant 0}^{mot}\Omega^i.$$

Corollary 3.1.14. *For all $i \geqslant 0$ and for $\mathcal{X}$ a motivic space, we have*

$$\pi_i^{mot}\mathcal{X} = \tau_{\leqslant 0}^{mot}\pi_i^{nis}\mathcal{X} \simeq \mathcal{L}\pi_i^{nis}\mathcal{X}.$$

where $\mathcal{L}$ is a localization for the inclusion $Sh_{nis}^{\mathbb{A}^1} \subset Sh_{nis}$.

Proof. This is Corollary 2.1.8 applied to L_{mot}, together with the observation $L_{\leqslant 0}^{mot} \simeq \mathcal{L}(\pi_0^{nis})$. □

(For the rest of this section, we shall work over fields. Thus, we let $S = Speck$ for a field k.)

Remark 3.1.15. In [Mor12], Morel proved that, at least over perfect fields, all higher Nisnevich homotopy sheaves of a motivic local space are $\mathbb{A}^1$-invariant (in fact, strongly/strictly!), rendering a further discrete motivic localization unnecessary. Following this, he conjectured that the same would hold even for π_0. Since $\mathcal{L}$ is a localization for the inclusion $Sh_{nis}^{\mathbb{A}^1} \subset Sh_{nis}$, this conjecture is equivalent to the statement that

$$\pi_0^{mot} \simeq \mathcal{L}\pi_0^{nis}{}_{|\mathcal{H}^{mot}} \simeq \pi_0^{nis}{}_{|\mathcal{H}^{mot}}$$

[Mor12, Conjecture 1.12]. Thus, in the early stages of motivic homotopy theory, no importance was given to the internal truncation functor $L_{\leqslant 0}^{mot}$ and internal homotopy groups.

Though it was known that the above conjecture is very close to being true:

Lemma 3.1.16. *If $\mathcal{X}$ is a k-motivic space over a perfect field k, then $\pi_0^{nis}\mathcal{X}$ is $\mathbb{A}^1$-invariant if and only if it is weakly unramified if and only if, for any irreducible $X \in Sm_k$, the map $\pi_0^{nis}\mathcal{X}(X) \to \pi_0^{nis}\mathcal{X}(k(X))$ is injective.*

Proof. If $\pi_0^{nis}\mathcal{X}$ is $\mathbb{A}^1$-invariant, then it is weakly unramified by [Bac24, Corollary 2.4].

Conversely, if $\pi_0^{nis}\mathcal{X}$ is weakly unramified, then its $\mathbb{A}^1$-invariance follows from [Cho14, Corollary 3.2 and Lemma 4.14]. □

It was known that $\pi_0^{nis}\mathcal{X}(X) \to \pi_0^{nis}\mathcal{X}(k(X))$ has a trivial fiber for any pointed motivic space $\mathcal{X}$. Building on this observation and using the lemma above, Choudhury proved in [Cho14] that, for motivic spaces with a minimal amount of group structure, the Nisnevich homotopy path component π_0^{nis} is indeed $\mathbb{A}^1$-invariant:

Proposition 3.1.17. *Let $\mathcal{X}$ be a k-motivic H-group space (i.e., a group object in the homotopy 1-category $h\mathcal{H}^{\mathbb{A}^1}(k)$) over a perfect field k. Then $\pi_0^{mot}\mathcal{X} \simeq \pi_0^{nis}\mathcal{X}$.*

Proof. This is [Cho14, Theorem 4.18] using Corollary 3.1.14. □

Unfortunately, despite all that, it is now known, thanks to Ayoub, that there are motivic spaces $\mathcal{X}$ such that $\pi_0^{nis}\mathcal{X}$ is not $\mathbb{A}^1$-invariant [Ayo23]. Thus motivic truncation functors $L_{\leqslant n}^{mot}$ are unavoidable. At best, what is known is:

Proposition 3.1.18. *Let the base be a field k. Then for a k-motivic space $\mathcal{X}$, the canonical map*

$$\pi_0^{\mathbb{A}^1}\mathcal{X} \simeq \pi_0^{nis}\mathcal{X} \to \pi_0^{mot}\mathcal{X}$$

is an isomorphism on $\mathcal{F}_k$ and on $\mathbb{A}^1_{\mathcal{F}_k}$.

Proof. This is a restatement of [CR24, Corollary 2.15]. □

Remark 3.1.19. Since $\tau^{mot}_{\leqslant 0}$ preserves products by Corollary 3.1.11, it follows that π^{mot}_1 is a group object in $Sh^{\mathbb{A}^1}_k$ and that π^{mot}_i is an abelian group object in $Sh^{\mathbb{A}^1}_k$. Although, these observations are immediate when k is perfect:

Proposition 3.1.20. *Let k be a perfect field. Then for all $i \geqslant 1$ and for every k-motivic space $\mathcal{X}$, there are equivalences*

$$\pi^{mot}_i \mathcal{X} \simeq \pi^{nis}_i \mathcal{X} \simeq \pi^{\mathbb{A}^i}_i \mathcal{X}.$$

Proof. By the corollary above, this is an immediate consequence of the fact that for every $i \geqslant 1$ the sheaf $\pi^{\mathbb{A}^1}_i \mathcal{X}$ is $\mathbb{A}^1$-invariant [Mor12, Corollary 6.3]. □

Proposition 3.1.21. *Let $\mathcal{X} \in \mathcal{P}_{nis}(k)_{\geqslant 1}$ be pointed. Then the following conditions are equivalent:*

(1) $\mathcal{X} \in \mathcal{H}^{mot}(k)$.
(2) $\mathrm{K}_{nis}(\pi^{nis}_i \mathcal{X}, i) \in \mathcal{H}^{mot}(k)$ for all i.
(3) $\tau^{nis}_{\leqslant i} \mathcal{X} \in \mathcal{H}^{mot}(k)$ for all i.

In this situation, $\tau^{nis}_{\leqslant i} \mathcal{X} \simeq \tau^{mot}_{\leqslant i} \mathcal{X}$ for all $i \geqslant 0$.

Proof. The equivalence of the three conditions is well known (see [Corollary 6.3 in Mor12]); we merely repeat it to set the stage.

(1) $\implies$ (2) follows from [Corollary 6.2, Mor12].

Suppose (2) holds, i.e., for all i we have $\mathrm{K}_{nis}(\pi^{nis}_i \mathcal{X}, i) \in \mathcal{H}^{mot}(k)$. Since $\mathcal{X}$ is Nisnevich locally connected, we have $\tau^{nis}_{\leqslant 0} \mathcal{X} \simeq * \in \mathcal{H}^{mot}(k)$. This proves the base case of (3). Suppose inductively that $\tau^{nis}_{\leqslant i} \mathcal{X} \in \mathcal{H}^{mot}(k)$ for some i. Consider the fiber sequence

$$\mathrm{K}^{nis}(\pi^{nis}_{i+1}(\mathcal{X}), i+1) \to \tau^{nis}_{\leqslant i+1} \mathcal{X} \to \tau^{nis}_{\leqslant i} \mathcal{X}$$

Because of (2), we know that the fiber $\mathrm{K}^{nis}(\pi^{nis}_{i+1}(\mathcal{X}), i+1)$ lies in $\mathcal{H}^{mot}(k)$. By [Mor12, Lemma 6.51], it then follows that $\tau^{nis}_{\leqslant i+1} \mathcal{X}$ lies in $\mathcal{H}^{mot}(k)$.

Finally, if (3) holds, i.e., $\tau^{nis}_{\leqslant n} \mathcal{X} \in \mathcal{H}^{mot}(k)$. By the Postnikov convergence of $\mathcal{P}_{nis}(Sm_k)$, we have

$$\mathcal{X} \simeq \varprojlim_n \tau^{nis}_{\leqslant n} \mathcal{X}.$$

Since the inclusion $\mathcal{H}^{mot}(k) \subset \mathcal{P}_{nis}(Sm_k)$ is closed under limits, it follows that $\varprojlim_n \tau^{nis}_{\leqslant n} \mathcal{X} \in \mathcal{H}^{mot}(k)$, and consequently, $\mathcal{X} \in \mathcal{H}^{mot}(k)$, i.e., (1) holds.

Now we come to the final statement. From Theorem 3.1.5 and Theorem 3.1.4 we know that $\tau^{mot}_{\leqslant i} \simeq \mathrm{colim}_m (\tau^{nis}_{\leqslant n} L_{\mathbb{A}^1})^m$. Since $\mathcal{X}$ is motivic local, it follows that $\tau^{nis}_{\leqslant i} L_{\mathbb{A}^1} \mathcal{X} \simeq \tau^{nis}_{\leqslant i} \mathcal{X}$, which from the second condition is motivic local as well. Hence for all n we have $(\tau^{nis}_{\leqslant n} L_{\mathbb{A}^1})^m \mathcal{X} \simeq \tau^{nis}_{\leqslant n} \mathcal{X}$. Hence,

$$\tau^{mot}_{\leqslant i} \mathcal{X} \simeq \varinjlim_m (\tau^{nis}_{\leqslant n} L_{\mathbb{A}^1})^m \mathcal{X} \simeq \varinjlim_m \tau^{nis}_{\leqslant n} \mathcal{X} \simeq \tau^{nis}_{\leqslant i} \mathcal{X}$$

as claimed. □

Remark 3.1.22 (On the postnicov convergence of the motivic hhomotopy category)**.** To the best of my knowledge, the Postnikov convergence of the motivic homotopy category is not yet known. However, it is known that the presentable subcategory of Nisnevich-connected motivic spaces $\tau^{nis}_{\geqslant 1} \mathcal{H}^{mot}$ is Postnikov convergent, at least over base fields. Indeed, the higher motivic truncations $\tau^{mot}_{\leqslant n}$ for objects in $\tau^{mot}_{\geqslant 1} \mathcal{H}^{nis}$ agree with their Nisnevich truncation $\tau^{nis}_{\leqslant n}$ in this case

(see Proposition 3.1.21). Since, by closedness under limits and Remark 2.9 of *loc.cit.* (again), $\tau^{nis}_{\geqslant 1}\mathcal{H}^{mot}$ is Postnikov convergent, we conclude that so is $\tau^{mot}_{\geqslant 1}\mathcal{H}^{mot}$.

We now come to one of the most important observations of this paper, namely that the motivic localization functor is path-injective (see Definition 2.2.6). As we have seen in the categorical section, §2, this will have far-reaching consequences.

Lemma 3.1.23. *Let $\tau^{nis}_{\leqslant 0}\mathcal{H}^{mot}$ denote the image of $\mathcal{H}^{mot}$ under $\tau^{nis}_{\leqslant 0}$ (we may call it the category of $\mathbb{A}^1$-homotopy path components). Then the restriction functor*

$$L^{mot}_{\leqslant 0} \simeq \mathcal{L} : \tau^{nis}_{\leqslant 0}\mathcal{H}^{mot} \to Sh^{\mathbb{A}^1}_{nis}$$

is injective. In other words, the motivic localization is path injective (Definition 2.2.6).

Proof. Using the isomorphism on $\mathcal{F}_k$ as in Proposition 3.1.18, this follows immediately from [Corollary 2.10, Bac24]. □

Remark 3.1.24. Here is a complete argument for the benefit of the reader: First, since $\pi_0\mathcal{X}(K) \simeq \pi^{nis}_0\mathcal{X}(\mathcal{K})$ for every $K \in \mathcal{F}_k$, we want to show that for any $K \in \mathcal{F}_k$ any two $t, t' \in \mathcal{X}_0(K)$ are connected by a simplicial homotopy in $\mathcal{X}(K)$. Since $\mathcal{L}\pi^{nis}_0\mathcal{X}(K) = *$, there is an n-ghost homotopy

$$H = (\pi^{nis}_0\mathcal{X} \xleftarrow{h} V \xrightarrow{nis} \mathbb{A}^1_K, h^W : W \underset{hb}{\overset{ha}{\rightrightarrows}} \pi^{nis}_0\mathcal{X}, \tilde{\sigma}_i),$$

where the last two maps a and b are the following composites (in which the first map is a Nisnevich cover):

$$a, b : W \xrightarrow{nis} V \times_{\mathbb{A}^1_K} V \underset{pr_2}{\overset{pr_1}{\rightrightarrows}} V.$$

Note that since V is a 1-dimensional scheme, it has homotopy dimension 1, and $\mathcal{X}$ has Nisnevich descent. By [Bac24, Lemma 2.17], we have a surjection $[V, \mathcal{X}] \to [V, \pi^{nis}_0\mathcal{X}]$. Therefore, up to homotopy, there is a map $u : V \to \mathcal{X}$ descending to h.

Now, we claim that for every $K \in \mathcal{F}_k$ and any two sections t, t', as above, are simplicially homotopic. We prove this by induction on n. As in [BRS22, Theorem 2.2], we may assume that V is an elementary Nisnevich cover $V = V_1 \sqcup V_2$ and let $h_i = h_{|V_i}$ and $u_i = u_{|V_i}$, and denote $W' = V_1 \times_{\mathbb{A}^1_K} V_2$. Let $W' = \cup W'_j$ be the irreducible components of W', and let η_j be the corresponding generic points. We deduce that H'_{η_j} is an $(n-1)$ ghost homotopy of $(u_1)_{\eta_j}$ and $(u_2)_{\eta_j}$. By the induction hypothesis, this implies that $(u_1)_{\eta_j}$ and $(u_2)_{\eta_j}$ are simplicially homotopic in $\mathcal{X}$. By definition, these simplicial homotopies are defined over an open subset $W'' \subset W'$. Consequently, $(u_1)_{|W''}$ and $(u_2)_{|W''}$ are simplicially homotopic in $\mathcal{X}$. Moreover, by further shrinking V_2, we may assume that actually $W'' = V_1 \times_{\mathbb{A}^1_K} V_2$.

Therefore, we may assume that $u_1 : V_1 \to \mathcal{X}$ and $u_2 : V_2 \to \mathcal{X}$ agree on $V_1 \times_{\mathbb{A}^1_k} V_2$ up to a simplicial homotopy. Since $V = V_1 \sqcup V_2 \to \mathbb{A}^1_K$ is a Nisnevich cover and $\mathcal{X}$ is a Nisnevich ∞-sheaf, we may therefore find a lift of $u : V \to \mathcal{X}$ to $u' : \mathbb{A}^1_K \to \mathcal{X}$. But since $\mathcal{X}$ is $\mathbb{A}^1$-local, this yields a simplicial homotopy between the endpoints, namely t and t'.

The base case is easy. Indeed, a 0-ghost homotopy from t to t' in $\pi^{nis}_0\mathcal{X}$ is given by an explicit map $\mathbb{A}^1_K \to \pi^{nis}_0\mathcal{X}$, which, by appeal to the same homotopy dimension argument [Bac24, Lemma 2.17], comes from $[\mathbb{A}^1_K, \mathcal{X}]$ and, by the $\mathbb{A}^1$-locality of $\mathcal{X}$, is nothing but a simplicial homotopy.

3.2. **Motivic Eilenberg Maclane spaces.** Let $L_{\mathbb{A}^1}Grp_k$ and $L_{\mathbb{A}^1}Ab_k$ denote the full subcategories of Sh_k^{nis} consisting of $\mathbb{A}^1$-invariant sheaves of groups and abelian groups, respectively. Both are full subcategories of $Sh_k^{\mathbb{A}^1}$. As introduced by Morel in his textbook [Mor12], $Grp_k^{\mathbb{A}^1}$ denotes the full subcategory of $L_{\mathbb{A}^1}Grp_k$ consisting of strongly $\mathbb{A}^1$-invariant sheaves of groups, while $Ab_k^{\mathbb{A}^1}$ is the full subcategory consisting of strictly $\mathbb{A}^1$-invariant sheaves of abelian groups. In the rest of section §3, we will require the base to be a field k that we don't assume to be perfect unless otherwise specified.

As we did in the previous subsection, we shall restate the definitions of §3.2 for the motivic localization.

Definition 3.2.1. *A motivic space $\mathcal{X}$ is motivically 1-connective (equivalently, 0-connected) if $\pi_0^{mot}\mathcal{X}$ is trivial. We say that a pointed motivic space $x : * \to \mathcal{X}$ is motivically $(n+1)$-connective (or n-connected) for $n > 0$ if, in addition, $\pi_i^{mot}(\mathcal{X}, x) \simeq *$ for all $i \leqslant n$. We denote the full subcategory of n-connective objects of $\mathcal{H}^{mot}(k)$ by $\tau_{\geqslant n}^{mot}\mathcal{H}^{mot}(k)$.*

Theorem 3.2.2. *For every $n \geqslant 1$ and k a field (assumed perfect when $n \geqslant 2$),*

$$\tau_{\geqslant n}^{mot}\mathcal{H}^{mot}(k) \simeq \tau_{\geqslant n}^{nis}\mathcal{H}^{mot}(k).$$

Proof. When $n = 1$, this is Lemma 3.1.23. With this, the $n \geqslant 2$, the cases follow immediately from Proposition 3.1.20. □

Definition 3.2.3. *Let $n \geqslant 1$. A Nisnevich sheaf of n-groups (i.e., of groups when $n = 1$ and of abelian groups when $n \geqslant 2$) G is said to be n-strictly $\mathbb{A}^1$-invariant if $\mathrm{B}_{nis}^n G$ is $\mathbb{A}^1$-local. We denote the full subcategory of $n - Grp_k^{nis}$ consisting of n-strictly $\mathbb{A}^1$-invariant sheaves by n-$Grp^{L_{mot}}$.*

Remark 3.2.4. Since Ω^n preserves local objects, it follows from the equivalence $G \simeq \Omega^n \mathrm{B}_{nis}^n G$ that n-strictly $\mathbb{A}^1$-invariant sheaves are $\mathbb{A}^1$-invariant.

Remark 3.2.5. This is equivalent to saying that for all $i \leqslant n$, the presheaves $H_{nis}^i(-, G)$ are $\mathbb{A}^1$-local.

Remark 3.2.6. Clearly, 1-strictly $\mathbb{A}^1$-invariant sheaves are precisely the strongly $\mathbb{A}^1$-invariant sheaves in the sense of [Mor12, Definition 7 (2)]. In other words, $1\text{-}Grp^{L_{mot}} = Grp_k^{\mathbb{A}^1}$. Whereas G is strictly $\mathbb{A}^1$-invariant in the sense of [Mor12, Definition 7 (3)] if and only if it is n-strictly $\mathbb{A}^1$-invariant for all $n \geqslant 1$.

Lemma 3.2.7. *Let k be a perfect field. Then, for all $n \geqslant 2$*

$$n\text{-}Grp_{L_{mot}} = Ab_k^{\mathbb{A}^1}.$$

Proof. First, observe that for every $n \geqslant 2$, an object of $n\text{-}Grp^{L_{mot}}(\mathcal{H}^{mot}(k))$ is, in particular, a strongly $\mathbb{A}^1$-invariant sheaf of abelian groups. By [Mor12, Theorem 4.46], these are strictly $\mathbb{A}^1$-invariant when k is perfect. □

Definition 3.2.8. *For $n \geqslant 0$, a motivic space $\mathcal{X}$ is an n-Eilenberg-MacLane space if it is motivically n-truncated (see Definition 3.1.1) and motivically n-connective.*

Lemma 3.2.9. *For every $n \geqslant 0$ and every field k, there is an equality*

$$\mathcal{EM}_n(\mathcal{H}^{mot}(S)) = L_{\mathbb{A}^1}\mathcal{P}(S) \bigcap \mathcal{EM}_n(\mathcal{P}^{nis}). \tag{5}$$

Corollary 3.2.10. *Let $\mathcal{X} \in \mathcal{P}_{nis}(k)_{\geqslant 1}$ be a pointed object. Then the following conditions are equivalent:*

(1) $\mathcal{X} \in \mathcal{H}^{mot}(k)$.
(2) $\mathrm{K}_{nis}(\pi_i^{nis}\mathcal{X}, i) \in \mathcal{EM}_i(\mathcal{H}^{mot}(k))$ *for all* i.
(3) $\tau_{\leqslant i}^{nis}\mathcal{X} \in \mathcal{H}^{mot}(k)_{\leqslant i}$ *for all* i.

Proof. We identify the conditions here with those of Proposition 3.1.21.

For (2), use the identification in Lemma 3.2.9. To identify the condition (3) here with the condition (3) there, use Proposition 3.1.2. □

Remark 3.2.11 (On the postnicov convergence of the motivic connected motivic homotopy category)**.** The presentable subcategory of categorically connected motivic spaces $\tau_{\geqslant 1}^{mot}\mathcal{H}^{mot}$ is Postnikov convergent, at least over base fields. Indeed,

$$\tau_{\geqslant 1}^{mot}\mathcal{H}^{mot} = \tau_{\geqslant 1}^{nis}\mathcal{H}^{mot},$$

(Theorem 3.2.2), and $\tau_{\geqslant 1}^{nis}\mathcal{H}^{mot}$ is Postnikov convergent by Theorem 3.2.11. Thus, $\tau_{\geqslant 1}^{mot}\mathcal{H}^{mot}$ is Postnikov convergent.

Lemma 3.2.12. *Let k be a perfect field. Then there is an adjunction:*

$$\mathrm{B}_{\mathcal{H}^{mot}(k)} : Grp(\mathcal{H}^{mot}(k)) \leftrightarrows \mathcal{EM}_1(\mathcal{H}^{mot}) : \pi_1^{mot}$$

and for every $n \geqslant 2$ an adjunction:

$$\mathrm{B}^n_{\mathcal{H}^{mot}(k)} : Ab(\mathcal{H}^{mot}(k)) \leftrightarrows \mathcal{EM}_n(\mathcal{H}^{mot}) : \pi_n^{mot}$$

Proof. This follows from the facts that $\mathrm{B}_{\mathcal{H}^{mot}(k)} \simeq L_{mot}\mathrm{B}$ and that, for every $n \geqslant 1$, Proposition 3.1.20 gives $\pi_n^{mot} \simeq \pi_i^{nis}{}_{|\mathcal{H}^{mot}(k)}$. □

Theorem 3.2.13. *Let k be a perfect field. Then the adjunction in the above lemma induces equivalences of categories:*

$$Sh_{*,k}^{\mathbb{A}^1} \simeq \mathcal{EM}_0(\mathcal{H}^{mot}(k)) \text{ and } Grp_k^{\mathbb{A}^1} \simeq \mathcal{EM}_1(\mathcal{H}^{mot}(k)) : \pi_1^{mot}$$

and for all $n \geqslant 2$

$$Ab_k^{\mathbb{A}^1} \simeq \mathcal{EM}_n(\mathcal{H}^{mot}(k)) : \pi_n^{mot}.$$

Proof. Since L_{mot} is path-injective, this follows from Proposition 2.2.11 and Lemma 3.2.7. □

Such characterizations are largely unknown over arbitrary base schemes. Nevertheless, over an arbitrary base, we have the equivalence

$$\mathcal{EM}_0(\mathcal{H}^{mot}) \simeq Sh_k^{\mathbb{A}^1} = \mathrm{Disc}(\mathcal{H}^{mot}),$$

which is expected of an ∞-topos. Unfortunately,

Corollary 3.2.14 (SOsr12, Remark 3.5)**.** *Over any perfect field k, the motivic homotopy category $\mathcal{H}^{mot}(k)$ is not an ∞-topos.*

Remark 3.2.15. Before we prove this formally, let us note that the failure of the motivic homotopy category to be a topos was first brought to attention in [SOsr12, Remark 3.5]. However, their argument did not identify all the internal Eilenberg-MacLane objects of the motivic homotopy category, leaving it somewhat incomplete. This was addressed in model-categorical

language in [RS18, §4.3], where the authors gave a complete proof of the model-categorical claim that the "Motivic homotopy category is not a model topos" (see Proposition 4.12 in loc.cit.). I was unaware of the cited article while writing the first draft of this paper. While the statements are 'equivalent' in a suitable sense and, in fact, in the end require an example of an $\mathbb{A}^1$-invariant sheaf of groups that is not strongly/strictly $\mathbb{A}^1$-invariant, the two differ significantly in their presentation. In addition, instead of relying on the standard example of (which is abelian) the non-strongly $\mathbb{A}^1$-invariant sheaf of motivic sheaf of abelian groups, we provide a few non-commutative example.

$\mathbb{A}^1$-Invariant and Strongly $\mathbb{A}^1$-Invariant Group Sheaves

As remarked above, the key to the result lies in the fact that the inclusion $\mathrm{Grp}_k^{\mathbb{A}^1} \subsetneq L_{\mathbb{A}^1}\mathrm{Grp}_k$ is strict. We shall first list some examples of this sort before heading on to the formal proof. There is usually one example available in the literature, namely the one from [Cho14, §5]. We would like to provide some more to this line by abstracting out the methods of loc. cit. For this, we shall need a few preparatory lemmas.

Some exotic motivic spaces. In this subsection we shall provide a standard method for constructing new motivic spaces from old ones. The main problem, as is well known, is that Nisnevich sheafification often destroys $\mathbb{A}^1$ invariance. To remedy this, we shall derive a special condition under which Nisnevich sheafification does not destroy $\mathbb{A}^1$ invariance. We shall specialize to discrete motivic spaces for our purposes. To this end, we shall first recall the following result of Morel: every discrete motivic sheaf of groups is weakly unramified (i.e., induces an injection on dense open immersions) [Bac24, Corollary 2.4]. The converse is clearly false: weakly unramified sheaves (of Groups) need not be Nisnevich local. The aim of this section is to describe suitable conditions so that this converse is met, yielding the required generation of motivic sheaves. For this, we shall require the following stronger version of weak unramification:

Definition 3.2.16. *Let τ be a property of morphisms. We say that F is τ-weakly unramified if for a dominant τ-morphism $f : X \to Y$ the restriction $FY \to FX$ is injective.*

Remark 3.2.17. It is evident that the notion of weakly unramifiedness is equivalent to Zariski-weakly unramifiedness.

Example 3.2.18. The presheaf of global sections of the sheaf of Kähler differentials $X \mapsto \Omega_{X/k}(X)$ is étale-weakly unramified. Indeed, if $f : X \to Y$ is an étale morphism of smooth schemes over k then $f^*\Omega_{Y/k} \simeq \Omega_{X/k}$. On the other hand, since $\Omega_{Y/k}$ is a flat $\mathcal{O}_Y$ module and $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is injective (sicne f is dominant) it follows that the morphismm of $\mathcal{O}_Y$ modules

$$\Omega_{Y/k} \to f_*\mathcal{O}_X \otimes_{\mathcal{O}_Y} \Omega_{Y/k} \simeq f_*f^*\Omega_{Y/k} \simeq f_*\Omega_{X/k}$$

is injective $\mathcal{O}_Y$-module homomorphism. Taking global sections yields the injection:

$$\Gamma(Y, \Omega_{Y/k}) \to \Gamma(Y, f_*\Omega_{X/k}) \simeq \Gamma(X, \Omega_{X/k}).$$

Example 3.2.19. When $\tau = \varnothing$ is the empty property, we say that a τ-weakly unramified presheaf is fully-weakly unramified. This is equivalent to turning dominant maps into injections. A natural first candidate for a τ-weakly unramified presheaf is the assignment $X \mapsto K(X)$, where $K(X)$ denotes the field of rational functions on X. By definition, this assignment sends dominant morphisms to injective maps. However, it is not functorial with respect to arbitrary morphisms and hence does not define a presheaf on Sm_k. Nevertheless, this idea can be suitably generalized to construct presheaves with the desired property. Let G be a weakly unramified presheaf (induces injections on generic point inclusions) whose restriction to $\mathcal{F}_k$ takes values in Set^{inj} of sets and injections is fully weakly unramified. Indeed, for a dominant morphism $f : X \to Y$ the composition $GY \to GX \hookrightarrow G(K(X))$ is equal to the composition of injections $GY \hookrightarrow G(K(Y)) \hookrightarrow G(K(X))$.

Example 3.2.20. For every n, the Milnor K-theory presheaf K_n^M satisfies parts of the above condition. Indeed, K_n^M is injective on purely transcendental extensions [Wei13, Example 7.3.2]. Unfortunately, it is not injective on finite extensions. However, by Corollary 7.5.3 of *loc.cit.* it follows that $\mathbb{Q} \otimes K_n^M$ does have this property too. Since $\mathbb{Q}$ is a flat $\mathbb{Z}$-module and K_n^M is weakly unramified, $K_n^M \otimes \mathbb{Q}$ has all the required properties.

Remark 3.2.21. The version of the above example fails for the Milnor-witt K-thoery. For example, $GW(\mathbb{R}) \otimes \mathbb{Q} \simeq \mathbb{Q} \oplus \mathbb{Q}$ where as $GW(\mathbb{C}) \otimes \mathbb{Q} \simeq \mathbb{Q}$.

Remark 3.2.22. Using the general techniques of Example 3.2.19 is easy to show, extending the result of Example 3.2.18, that $\Gamma(\Omega_{-/k})$ is fully-weakly unramified when k is a field of characteristic 0.

Example 3.2.23 (Separated schemes)**.** Every separated scheme turns dominant maps into injections, i.e. fully-weakly unramified. Indeed, let Z be a separated scheme over k. Let $h : X \to Y$ be a dominant morphism of connected smooth schemes. Suppose $f, g : Y \to Z$ are two k morphisms such that $f \circ h = g \circ h$. It follows that f factors through the equalizer $E \hookrightarrow X$ (which is a closed immersion due to the separatedness of Z). The dominance of f then implies that $E \hookrightarrow X$ is dominant. Because all schemes are smooth, it follows that the equalizer is an isomorphism.

Here is the key result:

Lemma 3.2.24. *Suppose* F: Sets $\to$ Sets *preserves intersection squares and* G *is an étale-weakly unramified Nisnevich sheaf of sets on* Sm_k*. Then* $L_\Sigma FG \simeq L_{\mathrm{nis}} FG$*.*

Before we head on to prove the Lemma, we list some examples of endo functors of the type used in the statement above. *Functors preserving intersection squares.*

Example 3.2.25. The functors $F = F_{\mathrm{mon}}, F_{\mathrm{gr}}, \mathbb{Z}(-)$, i.e., the free monoid, free group, and free abelian group functors, and any composition of these functors, preserve intersection squares. So $L_{\mathrm{nis}} F_{\mathrm{mon}} \mathbb{G}_m$, $L_{\mathrm{nis}} F_{\mathrm{gr}} \mathbb{G}_m$, and $L_{\mathrm{nis}} \mathbb{Z}\mathbb{G}_m$ are motivic spaces.

Example 3.2.26. Among other functors with values in sets, here are a few classics:

(1) The cotensor $(-)^K$ with a fixed set K
(2) The (covariant) power set functor ${2^{(-)}}_+$: Given a set X this assigns the set of its non-empty subsets and a morphism $X \to Y$ to ${2^X}_+ \to {2^Y}_+$ given by images of subsets of sets. It is clear that this functor preserves injections and, moreover, intersections.

In order to prove Lemma 3.2.24, we shall need a crucial lemma that is very specific to the base being a field.

Lemma 3.2.27. *A presheaf* $\mathcal{F}$: $Sm_k^{op} \to Spc$ *satisfies Nisnevich excision if and only if* $\mathcal{F} \in \mathcal{P}_\Sigma(k)$ *and its restriction to the full subcategory* Sm_k^{conn} *of connected schemes satisfies Nisnevich excision.*

Proof. The forward implication is necessary by definition. We shall explain the sufficiency. For this, we first remind the reader of two important facts about the base being a field:

(1) A smooth scheme is irreducible iff it is connected. i.e., $Sm_k^{conn} = Sm_k^{irr}$.
(2) All closed subschemes of smooth k-schemes are Noetherian and hence have finitely many irreducible components.

Let the following be a Nisnevich square Q of non-empty schemes (the empty cases being trivial):

$$\begin{array}{ccc} U' & \xrightarrow{j'} & X' \\ \downarrow{\scriptstyle p} & & \downarrow \\ U & \xrightarrow{j} & X \end{array}$$

Decomposing X into its irreducible components X_i, it is evident that the above square is a disjoint union of Nisnevich squares Q_i of irreducible schemes. Since the product of pullback squares is a pullback square, we are reduced to the case where X is irreducible. But in that case, since j is an open immersion, U is irreducible as well. Unfortunately, since p is only étale, there is no reason for X' to be irreducible. Had p been an open immersion, we would clearly be done by the same reduction as before. In other words, F satisfies Zariski excision. For a general p, let $Z = X\backslash U$. The point is that the components of X' that do not intersect $p^{-1}Z$ induce an isomorphism on j'. So we may assume that $p^{-1}Z$ intersects every component of X'. If Z is irreducible, the homeomorphism $p^{-1}Z \to Z$ implies that $p^{-1}Z$ is irreducible and hence X' is irreducible by the previous line, and we would be done. Otherwise, let Z_i be the (finite) set of irreducible components of Z. Let $U_i := X\backslash(\sqcup_{j\neq i} Z_i)$. Since U_i is a Zariski cover of X (consisting of irreducible schemes since X itself is irreducible), by Zariski excision (which we have already obtained), we are reduced to checking the excision condition on U_i. But by construction, on U_i the required closed subset Z_i is irreducible. □

We now prove Lemma 3.2.24:

Proof. We want to show that $L_\Sigma FG$ is already a Nisnevich sheaf. Since points for the $\sqcup$-topology are precisely the connected schemes, by Theorem 3.2.27 we are reduced to checking the Nisnevich excision condition for FG along Nisnevich squares of connected schemes.

Observe that every morphism in such a Nisnevich cd-square of connected schemes is dominant. Since G is a Nisnevich-weakly unramified Nisnevich sheaf, it turns such squares into intersection squares of subsets. Since F is given to preserve such squares, we are done. □

Such 'intersection-preserving' transformations can therefore preserve étale-weakly unramified motivic sheaves of sets. This follows from the trivial but crucial observation that L_Σ preserves $\mathbb{A}^1$-invarince:

Lemma 3.2.28. *Over a field k, L_Σ preserves $\mathbb{A}^1$-local objects.*

Proof. Let $\mathcal{F}$ be an $\mathbb{A}^1$-invariant k-space. Since over a field every smooth scheme is a disjoint union of connected schemes, and $\mathbb{A}^1 \times_k -$ preserves disjoint unions, and (as mentioned earlier) points for the $\sqcup$-topology are precisely the connected schemes, we are reduced to checking the $\mathbb{A}^1$-invariance of $\mathcal{F}$ on connected schemes, which is given. □

Lemma 3.2.29. *Suppose F: Sets $\to$ Sets preserves intersection squares and G étale-weakly unramified motivic sheaf of k-sets. Then $L_{\mathrm{nis}}FG \simeq L_\Sigma FG$ is a motivic space.*

Proof. This is immediate from Lemma 3.2.28 and Lemma 3.2.24. □

Example 3.2.30. By Example 3.2.25 and Theorem 3.2.23, for an $\mathbb{A}^1$-rigid (separated) scheme A the sheaves of sets $L_{\mathrm{nis}}F_{\mathrm{mon}}A$, $L_{\mathrm{nis}}F_{\mathrm{gr}}A$, and $L_{\mathrm{nis}}\mathbb{Z}A$ are motivic . For example, $L_{\mathrm{nis}}F_{\mathrm{mon}}\mathbb{G}_m$, $L_{\mathrm{nis}}F_{\mathrm{gr}}\mathbb{G}_m$, and $L_{\mathrm{nis}}\mathbb{Z}\mathbb{G}_m$ are motivic spaces. Similarly, using Example 3.2.26 it follows that $L_{\mathrm{nis}}\mathbb{G}_m^K$ (where K is an arbitrary set) and $L_{\mathrm{nis}}2^{\mathbb{G}_m}{}_+$ are $\mathbb{A}^1$-local.

Example 3.2.31. Similarly, the methods of Example 3.2.20 can be combined with the proof of Lemma 3.2.24 to conclude that $\mathbb{Q}_{nis}(K^M_n)$ is a motivic sheaf of abelian groups.

Counterexample 3.2.32 (To show $\mathrm{Grp}^{\mathbb{A}^1}_k \subsetneq L_{\mathbb{A}^1}\mathrm{Grp}_k$)**.**

(1) *Abelian case:* The reduced subgroup sheaf $\mathbb{Z}(\mathbb{G})$ is $\mathbb{A}^1$-invariant (see Example 3.2.30 or [Cho14, the paragraph after Lemma 5.4]) but not strongly $\mathbb{A}^1$-invariant (see [Cho14, Lemma 5.6]).
(2) *Non-abelian case:* The free sheaf of groups generated by $\mathbb{G}_m$, denoted $F_{gr}(\mathbb{G}_m)$ is an $\mathbb{A}^1$-invariant sheaf of (non-abelain) groups (see Example 3.2.30). It follows that the reduced free group generated by the pointed sheaf $\mathbb{G} = (\mathbb{G}, 1)$, denoted $F_{gr}\mathbb{G}$, is again $\mathbb{A}^1$-invariant, being a retract of $F_{gr}(\mathbb{G}_m)$. Indeed, we know that $L_{nis}F_{gr}\mathbb{G} = \pi_1^{nis}\Sigma_{nis}\mathbb{G}$ while the free strongly $\mathbb{A}^1$-invariant sheaf of groups generated by $\mathbb{G}$ is
$$\pi_1^{mot}\Sigma_{mot}\mathbb{G} \simeq \pi_1^{\mathbb{A}^1}\mathbb{P}^1 = F_{\mathbb{A}^1}(1).$$
We claim that there is no isomorphism between the two. Indeed, we know that the center of $L_{nis}F_{gr}\mathbb{G}$ is trivial (being sectionwise (on connected schemes) a free group of a non-trivial sheaf) while that of $F_{\mathbb{A}^1}(1)$ is highly non-trivial (its center contains K^{MW}_2 [Mor12, Lemma 7.29(2)]).
(3) Let G be an étale weakly unramified motivic sheaf of sets that is not 1-brational local (i.e., there is a codimension at least 2 dense open immersion $U \hookrightarrow X$ (say of connected smooth schemes) such that $GX \to GU$ is not a bijection). Then $F_{gr}G$ is $\mathbb{A}^1$-invariant by Lemma 3.2.29, but it is not strongly $\mathbb{A}^1$-invariant. This happens because, by the celebrated theorem of Morel, strongly $\mathbb{A}^1$-invariant sheaves are necessarily 1-birational local ([Bac24, Lemma 2.8.]). It is not difficult to see (using the conservativity of F_{gr}) that 1-birationality of $F_{gr}G$ implies 1-birationality of G.
Examples of sheaves of this type are plenty. For example, let A be an $\mathbb{A}^1$-rigid variety of dimension > 1 and $Z \subset X$ a closed subset of codimension at least 2. Then $X\backslash Z$ is not 1-birational by construction, but is an étale weakly unramified (being a separated scheme) motivic sheaf (being an open subscheme of an $\mathbb{A}^1$-rigid scheme).

Remark 3.2.33. Clearly, because $M^{gp} \simeq \Omega \mathrm{B}_{nis}M$ and the loop functor Ω preserves $\mathbb{A}^1$-local spaces, strong $\mathbb{A}^1$-invariance of a monoid sheaf M would force that of M^{gp}. Hence the counterexamples (2) and (3) extend directly to free monoid sheaves via group completion: $F_{grp} \simeq (F_{mon})^{gp}$. Consequently, neither $F_{mon}(\mathbb{G})$ nor $F_{mon}(G)$ (for an étale weakly unramified motivic sheaf of sets G) is strongly $\mathbb{A}^1$-invariant, despite both being$\mathbb{A}^1$-invariant by Lemma 3.2.29.

Remark 3.2.34. The second example is not a special case of the third one. Indeed, since G_m is a strongly $\mathbb{A}^1$-invariant Nisnevich sheaf of groups, it is 1-birationally rigid (by [Bac24, Lemma 2.8.] again), so we can't conclude by (3) that $F_{gr}\mathbb{G}_m$ is strongly $\mathbb{A}^1$-invariant.

Remark 3.2.35. The analog of the second example is not true for arbitrary $\mathbb{A}^1$-rigid schemes. In fact, a similar claim is false for a large class of $\mathbb{A}^1$-invariant sheaves of sets, namely birational sheaves of sets. This happens because the free (abelian) group (or monoid) sheaf (even for the $\sqcup$-topology) generated by a birational local sheaf of sets is again birational local. And it is known that birational sheaves of monoids are strongly $\mathbb{A}^1$-invariant (Theorem 4.3.1) (see [Mai26a, Corollary 3.2.7.] for a complete explanation of this last fact). In fact, the following proposition is a stronger converse to this fact.

Proposition 3.2.36. *Let k be a perfect field. Let S be a motivic sheaf of pointed sets on Sm_k such that $L_\Sigma F_{gr}S$ is motivic. Then S is necessarily birational.*

Proof. We first derive an equivalent identification of the given condition and its conclusion.

(1) We have the following equivalences:

$$\mathrm{B}_{nis}F_{gr}S \simeq \mathrm{B}_{nis}\Omega\Sigma_{nis}S \simeq \Sigma_{nis}S.$$

From which it follows that $\mathrm{B}_{nis}F_{gr}S$ is $\mathbb{A}^1$-local (i.e. $L_{nis}F_{gr}S$ is strongly $\mathbb{A}^1$-invariant) iff $\Sigma_{nis}G$ is $\mathbb{A}^1$-invariant.

(2) Since F_{gr} is conservative, it is easy to see that S is birational local iff $L_\Sigma F_{gr}S$ is birational local.

Next we show that $\Sigma_{nis}S$ is $\mathbb{G}$ null. Since $\Sigma_{nis}S$ is a connected motivic (as deduced in the previous paragraph) space, by [ABH23, Lemma 3.1.15.(4)] it suffices to show that

$$\mathrm{Map}_*(\mathbb{G}, \Sigma_{nis}S) \simeq *.$$

But,

$$\mathrm{Map}_*(\mathbb{G}, \Sigma_{nis}S) \simeq \mathrm{Map}_{\mathcal{H}^{mot}(k)_\bullet}(\mathbb{P}, S) \simeq Shv_{k*}(\mathbb{P}, S),$$

where the last equivalence is by $\mathbb{A}^1$-invariance of S. Sicne S is discrete, its $\mathbb{A}^1$-invariance implies its $\mathbb{P}^1$-invariance [Mai26b, Example 2.2.7]. So we know that, indeed, $Shv_{k*}(\mathbb{P}, S) \simeq *$. Since $\Sigma_{nis}S$ is connected and k is a perfect field, by [Mai26a, Theorem 3.3.12.] we know that Σ_{nis} is birational local. Since Ω preserves birational local objects, it follows that $L_\Sigma F_{gr}S \simeq L_{nis}F_{gr}S \simeq \Omega\Sigma_{nis}S$ is birational local. □

We have thus obtained the following interesting result:

Proposition 3.2.37. *Let G be an étale weakly unramified sheaf of groups over a perfect field k. Then G is a birational sheaf iff it is a motivic sheaf such that the free group generated by the pointed sheaf G (at the unit element) is strongly $\mathbb{A}^1$-invariant.*

Proof. The only if part was discussed in Theorem 3.2.35 (note that this does not require étale unramifiedness). Using Lemma 3.2.29, the if part is a consequence of the Proposition above. □

We now prove Corollary 3.2.14.

Proof. Using the counterexamples above regarding the non-essential surjectivity of the inclusion $Grp_k^{\mathbb{A}^1} \subset L_{\mathbb{A}^1}Grp_k$, one may directly apply Corollary 2.2.12 to obtain the result; but we shall sketch the argument again in the present (motivic) context for the convenience of the motivic reader.

If $\mathcal{H}^{mot}(k)$ were to be an ∞-topos, one would have equivalences of categories given by the functor

$$\pi_1^{mot} \simeq \pi_1^{nis} : \mathcal{EM}_1(\mathcal{H}^{mot}) \to \mathrm{Grp}(\mathrm{Disc}(\mathcal{H}^{mot}))$$

[Lur09, Proposition 7.2.2.12]. However, we see that by the above theorem, the essential image of the same is the full subcategory $Grp_k^{\mathbb{A}^1}$ of $\mathrm{Grp}(\mathrm{Disc}(\mathcal{H}^{mot})) = L_{\mathbb{A}^1}Grp_k^{nis}$. But, as discussed in the first paragraph, this inclusion is not an equality. □

Remark 3.2.38. Since $\mathbb{Z}(\mathbb{G}_m, 1)$ is a sheaf of abelian groups, the proof above applies to all $n \geqslant 1$. In particular, there are $\mathbb{A}^1$-invariant sheaves of groups that are not strongly $\mathbb{A}^1$-invariant and hence not strictly $\mathbb{A}^1$-invariant. Hence, for any $n \geqslant 1$

$$\pi_n^{mot} \simeq \pi_n^{nis} : \mathcal{EM}_n(\mathcal{H}^{mot}) \to n\text{-}\mathrm{Grp}(\mathrm{Disc}(\mathcal{H}^{mot}))$$

is never an equivalence.

Having said that, we must explain how close/far the motivic homotopy category is from being an ∞-topos. More specifically, we shall try to see which defining property fails. To this end, we first recall Giraud's theorem on the characterization of an $infty$-topos from [Lur09, Proposition 6.1.5.3]:

Theorem 3.2.39 (∞-Categorical Giraud Theorem)**.** *An ∞-category is an ∞-topos if and only if it satisfies the following conditions:*

(1) $\mathcal{C}$ is presentable.
(2) Colimits in $\mathcal{C}$ are universal.
(3) Coproducts in $\mathcal{C}$ are disjoint.
(4) Every groupoid object in $\mathcal{C}$ is effective.

We have:

Theorem 3.2.40. *Over an arbitrary base scheme S, the motivic homotopy category $\mathcal{H}^{mot}(S)$ satisfies (1), (2). If $S = Speck$ is a field (or, more generally, 0-dimensional), then the motivic homotopy category $\mathcal{H}^{mot}(k)$ has disjoint coproducts, i.e., it satisfies (3).*

Proof. Being a localization at a small set of morphisms, the presentability of $\mathcal{H}^{mot}(S)$ is well known. Since L_{mot} is locally cartesian, and the inclusion $\mathcal{H}^{mot}(S) \subset \mathcal{P}(S)$ is closed under pullbacks (in fact, all limits), universality of colimits in $L\mathcal{C}$ follows immediately from that of $\mathcal{P}(S)$.

For the last statement, the point is that when k is a field, the Nisnevich local coproduct of two Nisnevich sheaves $\mathcal{X}$ and $\mathcal{Y}$ is given by $L_\Sigma(\mathcal{X} \sqcup_{pre} \mathcal{Y})$. Again, since the points of Sm_k are connected schemes, by Theorem 3.2.27, for this we are reduced to checking the Nisnevich excision condition on Sm_k^{conn}, which is clear.

Since $L_{\mathbb{A}^1}\mathcal{P}(k) \subset \mathcal{P}(k)$ is closed under coproducts, it follows from Lemma 3.2.28 that $L_\Sigma(\mathcal{X} \sqcup_{pre} \mathcal{Y})$ is $\mathbb{A}^1$-local whenever $\mathcal{X}$ and $\mathcal{Y}$ are. This way, we have shown that $\mathcal{X} \sqcup_\Sigma \mathcal{Y}$ is the motivic coproduct of two motivic objects $\mathcal{X}, \mathcal{Y}$. The disjoint coproduct condition for the motivic homotopy category can thus be obtained from the same condition in $\mathcal{P}_\sqcup(k)$ (which is obviously an ∞-topos). □

Remark 3.2.41. Since the inclusion $\mathcal{H}^{mot}(k) \subset \mathcal{P}_{nis}(k)$ is closed under filtered colimits, the above proof in fact shows that the inclusion is closed under all coproducts. The only missing piece in this colimit puzzle is therefore geometric realizations. In fact, the purpose of the counterexamples of Counterexample 3.2.32 is precisely this. The colimit of the simplicial diagram $\mathrm{B}_\bullet F_{gr}\mathbb{G}_m$ in the motivic homotopy category $\mathcal{H}^{mot}(k)$ (which is $\mathrm{B}_{mot}F_{gr}\mathbb{G}_m$) is not the colimit in $\mathcal{P}_{nis}(k)$. In section §4.5 we shall see that the birational motivic homotopy category recovers this gap by preserving even geometric realizations and thus is itself an ∞-topos.

3.3. **Motivic connective objects and grouplike objects.** In the previous section, we have seen that the motivic homotopy category fails to internally deloop arbitrary discrete motivic groups. In fact, we have seen that those that can be delooped are precisely those that arise as the internal homotopy groups of (internal) motivic Eilenberg-MacLane objects. This section is devoted to understanding the condition required for the internal delooping of arbitrary motivic grouplike monoids and thus to establishing a motivic version of the loop-deloop equivalence between the ∞-categories of internal motivic connected objects and of those special grouplike objects.

The failure of the equivalence of the functor $\pi_1^{mot} : \mathcal{EM}_1(\mathcal{H}^{mot}(k)) \to L_{\mathbb{A}^1} Grp_k^{nis}$, as explained in Corollary 3.2.14, extends to the deeper fact that the functor

$$\Omega : \mathcal{H}^{mot}(S)_{\geqslant 1} \to Grp(\mathcal{H}^{mot}(S))$$

is not an equivalence of infinity categories. The primary issue is that, as shown by Counterexample 3.2.32, this functor is not essentially surjective. It is easy to see that, over perfect fields, the spaces in the image must have strongly $\mathbb{A}^1$-invariant motivic π_0, since these are just motivic $\pi_1^{\mathbb{A}^1}$'s of the starting space [Mor12, Corollary 6.3]. In fact, this condition suffices to ensure a motivic delooping. Before we explain this, let us first restate the main definition and result of §2.3 applied to the motivic case:

Definition 3.3.1. *We say that a motivic monoid $\mathcal{M}$ is strongly L_{mot}-local if $\mathrm{B}_{mot}\mathcal{M} \simeq \mathrm{B}_{nis}\mathcal{M}$, i.e., if $\mathrm{B}_{nis}\mathcal{M}$ is motivic local. We denote the full subcategory of strongly L_{mot}-local spaces by $\mathrm{Mon}_{L_{mot}}(\mathcal{H}^{mot}(k))$. We denote the further full subcategory of grouplike objects by $\mathrm{Grp}_{L_{mot}}(\mathcal{H}^{mot}(k))$.*

Corollary 3.3.2. *There is an equivalence of ∞-categories:*

$$\mathrm{B}_{\mathcal{H}^{mot}(k)} : \mathrm{Grp}_{L_{mot}}(\mathcal{H}^{mot}(k)) \simeq \mathcal{H}^{mot}(k)_{\geqslant 1} : \Omega.$$

Proof. Due to Lemma 3.1.23, the categorical theorem Theorem 2.3.6 applies to the motivic localization L_{mot} of the Nisnevich topos. □

We first make a preliminary observation about the motivic counterparts of notons from §2.3:

Proposition 3.3.3. *A monoid motivic space is strongly L_{mot}-local iff it is strongly $\mathbb{A}^1$-invariant in the sense of* [EHK$^+$21, Definition 3.1.6(1)]. *If k is a perfect field, then this is iff $\pi_0^{nis}\mathcal{M}$ is strongly $\mathbb{A}^1$-invariant.*

Proof. The first line follows from comparing the definitions of the two terms. When k is a perfect field, the second line is a restatement of [EHK$^+$21, Proposition3.1.12(1)] under the first line. □

Lemma 3.3.4. *Let $\mathcal{G}$ be a grouplike motivic monoid space. Then $\pi_0^{mot}\mathcal{G} \simeq \pi_0^{nis}\mathcal{G}$.*

Proof. Since grouplike motivic monoid spaces are, in particular, motivic H-groups, this is a special case of Proposition 3.1.17. □

Corollary 3.3.5. *Over a perfect field, a grouplike monoid motivic space $\mathcal{G}$ is strongly L_{mot}-local if and only if $\pi_0^{mot}\mathcal{G}$ is strongly L_{mot}-local.*

Proof. This follows immediately from Proposition 3.3.3 and Lemma 3.3.4. □

Let us denote by $\mathcal{G}rp_{mot}(\mathcal{H}^{mot}(k))$ the full subcategory of $\mathrm{Grp}(\mathcal{H}^{mot}(k))$ consisting of motivic monoid spaces whose motivic path component π_0^{mot} is strongly L_{mot}-local.

Corollary 3.3.6. *Over a perfect field, there is an equivalence of categories*

$$\mathrm{B}^{nis} : \mathcal{G}rp_{mot}(\mathcal{H}^{mot}(k)) \simeq \mathcal{H}_\bullet^{mot}(k)_{\geqslant 1} : \Omega$$

where $Grp_{mot}(\mathcal{H}^{mot}(S))$ denotes the full subcategory of $\mathcal{G}rp(\mathcal{H}^{mot}(k))$ consisting of motivic monoids with strongly L_{mot}-local π_0^{mot}. This provides further evidence that $\mathcal{H}^{mot}$ is not an ∞-topos.

Proof. By Corollary 3.3.2, the claim follows from Corollary 3.3.5.

To justify the last line, it suffices to know that there exist grouplike motivic spaces whose motivic path components π_0^{mot} ($\simeq \pi_0^{\mathbb{A}^1}$) are not strongly $\mathbb{A}^1$-invariant. This is the content of [Cho14, the paragraph after Lemma 5.4], which we have used earlier in Corollary 3.2.14. □

3.4. **Motivic theory of covering spaces.**

Motivic theory of geometric coverings.

Definition 3.4.1. *A collection $\{\mathcal{U}_i \to \mathcal{X}\}$ of morphisms of Nisnevich local spaces is said to be a motivic Geometric covering if $L_{mot}\check{C}_\bullet p \to L_{mot}\mathcal{X}$ is a colimit diagram in $\mathcal{H}^{mot}(k)$, where $p : \bigsqcup \mathcal{U}_i \to \mathcal{X}$ is the disjoint union $\sqcup p_i$.*

Example 3.4.2. Every Nisnevich cover of a (even, not necessarily smooth) scheme is a Motivic geometric cover.

Corollary 3.4.3. *A k-space $\mathcal{X}$ is $\mathbb{A}^1$-connected if and only if there exists an $\mathbb{A}^1$-geometric covering (which is if and only if a Nisnevich-geometric covering) $\mathcal{U}_i \to \mathcal{X}$ such that each $\mathcal{U}_i$ is $\mathbb{A}^1$-connected and each $\mathcal{U}_{ij}$ admits a global point. Then $\mathcal{X}$ is $\mathbb{A}^1$-connected.*

Proof. If $\mathcal{X}$ is $\mathbb{A}^1$-connected then the identity $\mathcal{X} \to \mathcal{X}$ is a Nisnevich cover with this property, which in turn is an $\mathbb{A}^1$-grometric covering. Note that the existence of the rational point from connectivity follows from the effectivity of L_{mot}. The remaining implication follows from Corollary 2.4.7 in the presence of Lemma 3.1.23 and the fact that L_{mot} is effective, so that for any space $\mathcal{Y}$, existence of a global point of $\pi_0^{\mathbb{A}^1}\mathcal{Y}$ is equivalent to $\mathcal{Y}$ being pointed. □

Corollary 3.4.4. *A scheme X/k is $\mathbb{A}^1$-connected iff there is a Nisnevich covering $U_i \to X$ such that each U_i is $\mathbb{A}^1$-connected and for every pair i, j the fiber product $U_i \times_X U_j$ has a (k-)point.*

Corollary 3.4.5 (Motivic Van Kampen)**.** *Let $\mathcal{U}_i \to \mathcal{X}$ be an $\mathbb{A}^1$-geometric covering (for example, a Nisnevich local covering) such that $\mathcal{U}_{ijk}$ is $\mathbb{A}^1$-connected for every i, j, k (1-, 2-, and 3-fold). Then $\mathcal{X}$ is $\mathbb{A}^1$-connected, and the $\mathbb{A}^1$ homotopy group of $\mathcal{X}$ is given by the following coequalizer in $Grp_k^{\mathbb{A}^1}$:*

$$\underset{i,j}{*}\pi_1^{\mathbb{A}^1}(\mathcal{U}_i \times_{\mathcal{X}} \mathcal{U}_j) \rightrightarrows \underset{i}{*}\pi_1^{\mathbb{A}^1}(\mathcal{U}_i) \to \pi_1^{\mathbb{A}^1}(\mathcal{X}) \to *$$

Proof. This follows at once from Theorem 2.4.8 and Lemma 3.1.23, using the identification $\pi_1^{\mathbb{A}^1} = \pi_1^{mot}$ in Proposition 3.1.20. □

Corollary 3.4.6 (Generalized Morel's $\mathbb{A}^1$ Van Kampen, [Mor12, Theorem 7.12])**.** *Let $\{U_i \to X\}_{i\in I}$ be a Nisnevich covering of a smooth scheme X such that each U_{ijk} is pointed $\mathbb{A}^1$-connected for every $i, j, k \in \{I\} \cup I$ (where U_I is 1-, 2-, or 3-fold). Then X is $\mathbb{A}^1$-connected, and the $\mathbb{A}^1$ homotopy group of X is given by the following coequalizer in $Grp_k^{\mathbb{A}^1}$:*

$$\underset{i,j}{*}\pi_1^{\mathbb{A}^1}(U_i \times_X U_j) \rightrightarrows \underset{i}{*}\pi_1^{\mathbb{A}^1}(U_i) \to \pi_1^{\mathbb{A}^1}(X) \to *$$

Remark 3.4.7. While Morel establishes an $\mathbb{A}^1$-local Van Kampen theorem specifically for Zariski covers over perfect fields [Mor12, Theorem 7.12], the preceding result applies to more general covers. Furthermore, our statement includes a connectivity assumption on the triple

intersections to resolve a technical gap in *loc. cit.* The argument in *loc.cit.* relies on the exactness of the following diagram:

$$Hom_{\mathcal{G}r}(\pi_1^{\mathbb{A}^1}(\check{C}(\mathcal{U})), G) \to \prod_i Hom_{\mathcal{G}r}(\pi_1^{\mathbb{A}^1}(U_i), G) \rightrightarrows \prod_{i,j} Hom_{\mathcal{G}r}(\pi_1^{\mathbb{A}^1}(U_i \cap U_j), G).$$

Technically, this diagram represents the π_0 of the corresponding limit diagram of spaces:

$$\prod_i Map(U_i, \mathrm{B}G) \rightrightarrows \prod_{i,j} Map(U_i \cap U_j, \mathrm{B}G) \cdots$$

Since π_0 may fail to commute with limits, the required exactness is not automatic. Resolving the resulting cocycle condition on the double intersections inherently requires the connectivity of the triple intersections.

Motivic theory of spatial coverings

Definition 3.4.8 (Definition 2.4.9, Definition 2.4.17)**.** *(1) A morphism $f : \mathcal{Y} \to \mathcal{X}$ of motivic spaces is a motivic covering if $f \in \left(\mathcal{H}^{mot}_{/\mathcal{X}}\right)_{\leqslant 0}$. Denote the category of such coverings by $Cov_{\mathcal{H}^{mot}}(\mathcal{X})$.*

When $f : \mathcal{Y} \to \mathcal{X}$ is a morphism of Nisnevich local spaces, then:

(2) We say that it is an $\mathbb{A}^1$-fibration if it is a local object of $\mathcal{P}_{nis/\mathcal{X}}$ with respect to motivic equivalences contained in $\mathcal{P}_{nis/\mathcal{X}}$. We denote the category of $\mathbb{A}^1$-fibrations of $\mathcal{X}$ by $\mathcal{F}ib^{\mathbb{A}^1}_{\mathcal{X}}$.
(3) We call it an $\mathbb{A}^1$-covering if it is an $\mathbb{A}^1$-fibration and is a 0-truncated object in the ∞-category of $\mathbb{A}^1$-fibrations over $\mathcal{X}$. Denote the category of $\mathbb{A}^1$-coverings of $\mathcal{X}$ as $Cov_{\mathbb{A}^1}(\mathcal{X})$.

Proposition 3.4.9. *A morphism $\mathcal{X} \to \mathcal{Y}$ of motivic spaces is a motivic covering if and only if it is an $\mathbb{A}^1$ covering if and only if it is a Nisnevich covering.*

Proof. Follows immediately from Proposition 2.4.10 and Proposition 2.4.21. □

Remark 3.4.10. Because the motivic localization functor L_{mot} is effective (see Remark 2.4.11), we obtain, by Remark 2.4.14, a pullback square of 1-categories:

$$\begin{array}{ccc} \mathcal{C}ov_{\mathbb{A}^1}(\mathcal{X}) & \hookrightarrow & \mathcal{H}^{mot}_{/L_{mot}\mathcal{X}} \\ \downarrow & & \downarrow \\ \mathcal{C}ov_{nis}(\mathcal{X}) & \hookrightarrow & \mathcal{P}_{nis}(S)_{/X} \end{array}$$

Corollary 3.4.11. *Let $\mathcal{Z}, \mathcal{X}$ be Nisnevich-local spaces, with $\mathcal{X}$ pointed and $\mathbb{A}^1$-connected. Then a morphism $f : \mathcal{Z} \to L_{mot}\mathcal{X}$ is an $\mathbb{A}^1$-covering iff $fib(f) \in Sh^{\mathbb{A}^1}$.*

Proof. Follows immediately from Corollary 2.4.16 and from the fact that L_{mot} is closed under extensions over $\mathbb{A}^1$-connected objects [Mor12, Lemma 6.51]. □

Corollary 3.4.12. *Suppose $\mathcal{X}$ is a pointed, connected motivic space. Then $Cov_{\mathcal{H}^{mot}}(\mathcal{X})_*$ has an initial object, namely the ordinary simply connected cover.*

Proof. By [Mor12, Corollary 6.2], $\Pi^{\mathbb{A}^1}\mathcal{X}$ is a motivic space. Whence, the claim follows from Proposition 2.4.25. □

Corollary 3.4.13. *Suppose $\mathcal{X}$ is a Nisnevich sheaf of spaces. We then have an equivalence of categories*

$$\eta^*_{mot} : Cov_{\mathbb{A}^1}(L_{mot}\mathcal{X}) = Cov_{\mathcal{H}^{mot}}(L_{mot}\mathcal{X}) \leftrightarrows Cov_{\mathbb{A}^1}(\mathcal{X}) : L_{mot}$$

where η_{mot} is the localization unit $1 \to L_{mot}$. The category on the left-hand side is the category of covering spaces of $L_{mot}\mathcal{X}$ internal to the motivic homotopy category, while the right-hand side is the ∞-categorical analogue of Morel's category of $\mathbb{A}^1$-coverings of $\mathcal{X}$, as developed in [Mor12, §7].

Proof. Since L_{mot} is locally cartesian [Hoy17, Proposition 3.15] and effective (Remark 2.4.11), the result follows from Proposition 2.4.23. □

Theorem 3.4.14. *Let $\mathcal{X}$ be a pointed and motivically connected k-motivic space over a perfect field k.*

(1) Then the category $\mathcal{C}ov_{\mathcal{H}^{mot}}(\mathcal{X})_*$ has an initial object $\widetilde{\mathcal{X}}^{mot} \to \mathcal{X}$.
(2) This initial object is the unique one with a motivically simply connected source, i.e., it satisfies $\tau^{mot}_{\leqslant 1}\widetilde{\mathcal{X}}^{mot} \simeq *$. The inclusion

$$\mathcal{C}ov_{\mathcal{H}^{mot}(k)}(\mathcal{X})_* \subset \mathcal{C}ov_{\mathcal{P}_{nis}(k)}(\mathcal{X})_*$$

preserves this initial object, i.e., this initial object is the unique Nisnevich covering with a Nisnevich simply connected source.
(3) The canonical map $\pi^{mot}_1(\mathcal{X}) \to \underline{Aut}_{\mathcal{X}}(\widetilde{\mathcal{X}}^{mot})$ is an equivalence.

Proof. Since motivically connected spaces are $\mathbb{A}^1$-connected by Lemma 3.1.23, we know that $\mathcal{X}$ is $\mathbb{A}^1$-connected. By [Mor12, Corollary 6.2], it follows that $\Pi^{\mathbb{A}^1}\mathcal{X}$ is a motivic space. Whence Proposition 2.4.25 applies, and statement (1) follows.

Statement (2) is a formal consequence of the fact that, from the referred proposition, one has $(\widetilde{\mathcal{X}}^{mot}) \simeq \tau^{nis}_{>1}\mathcal{X}$, which is the unique simply connected (simplicial) cover of $\mathcal{X}$. Since this is motivic local and $\mathbb{A}^1$ connected, we have

$$\tau^{mot}_{\leqslant 1}\widetilde{\mathcal{X}}^{mot} \simeq \tau^{nis}_{\leqslant 1}\widetilde{\mathcal{X}}^{mot},$$

which is trivial, as obtained in the previous line.

The third statement follows from Lemma 2.4.31. □

Theorem 3.4.15 (compare the methods with [Mor12, Theorem 7.8])**.** *Let $\mathcal{X}$ be a pointed, $\mathbb{A}^1$-connected, Nisnevich-local space.*

(1) Then the category $\mathcal{C}ov_{\mathbb{A}^1}(\mathcal{X})_*$ has an initial object $\widetilde{\mathcal{X}}^{\mathbb{A}^1}$.
(2) The initial object is the unique $\mathbb{A}^1$-covering with $\mathbb{A}^1$ simply connected source.
(3) The canonical map $\pi^{\mathbb{A}^1}_1(\mathcal{X}) \to \underline{Aut}_{\mathcal{X}}(\widetilde{\mathcal{X}}^{\mathbb{A}^1})$ is an equivalence.

Proof. When $\mathcal{X}$ is pointed $\mathbb{A}^1$-connected, $L_{mot}\mathcal{X}$ is pointed and Nisnevich locally connected. Therefore, by the above corollary, Corollary 3.4.13, we may replace $\mathcal{X}$ with $L_{mot}\mathcal{X}$ and assume that $\mathcal{X}$ is $\mathbb{A}^1$-local. In that case, Theorem 3.4.14 applies. For the third statement to follow from the third statement of the previous theorem, it suffices to know that $\pi^{\mathbb{A}^1}_1\mathcal{X} \simeq \pi^{mot}_1\mathcal{X}$ (see Proposition 3.1.20). □

However, it might not be true that the ordinary simplicial covering of a space maps to the $\mathbb{A}^1$-simply connected covering under L_{mot}. This works for Nisnevich-locally connected spaces when $\tau^{nis}_{\leqslant 1}\mathcal{X}$ is $\mathbb{A}^1$-local, which happens precisely when $\pi^{nis}_1\mathcal{X}$ is strongly $\mathbb{A}^1$-invariant.

Proposition 3.4.16. *Suppose $\mathcal{X}$ is a Nisnevich-local, pointed, connected space with a strongly $\mathbb{A}^1$-invariant fundamental group. Then L_{mot} takes the universal Nisnevich-local cover of $\mathcal{X}$ to the universal $\mathbb{A}^1$-simply connected cover.*

Proof. This follows immediately from [Mor12, Theorem 6.59] for $n = 1$ and from the identification of the $\mathbb{A}^1$-simply connected covering with $(\widetilde{\mathcal{X}}^{\mathbb{A}^1}) \simeq \tau^{nis}_{>1} L_{mot}\mathcal{X}$. □

Corollary 3.4.17 (Mor12, Lemma 7.5(1))**.** *Let G be a strongly $\mathbb{A}^1$-invariant Nisnevich sheaf of groups. Then any G-torsor is an $\mathbb{A}^1$-covering.*

Proof. This follows at once from Corollary 2.4.22 and the definition of strong $\mathbb{A}^1$ invariance. □

Proposition 3.4.18 (This was stated in [Mor12, Remark 7.10])**.** *Suppose, $\mathcal{Y} \in \mathcal{P}_{nis}(k)$ and $y : * \to \mathcal{Y}$ a base point. Then pull back along y induces a functor*

$$y^* : Cov_{\mathbb{A}^1}(\mathcal{Y}) \to Cov_{\mathbb{A}^1}(*) = Sh_k^{\mathbb{A}^1}.$$

This functor factors through the category of $\mathbb{A}^1$-invariant Nisnevich sheaves (of sets) having an action by $\pi_1^{\mathbb{A}^1}(\mathcal{Y}, y)$ (by the canonnical functor forgetting the action):

$$\Gamma_y^{\mathbb{A}^1} : Cov_{\mathbb{A}^1}(\mathcal{Y}) \longrightarrow \pi_1^{\mathbb{A}^1}(\mathcal{Y}, y)\text{-}Sh_k^{\mathbb{A}^1}.$$

When $\mathcal{Y}$ is $\mathbb{A}^1$ connected, $\Gamma_y^{\mathbb{A}^1}$ is an equivalence of categories.

Proof. The existence of the pullback functor follows from the pullback stability of $(\mathbb{A}^1)$ coverings, Lemma 2.4.13 (3). That it factors through $\pi_1^{\mathbb{A}^1}(Y, y)\text{-}Sh_k^{\mathbb{A}^1}$ is a consequence of Lemma 2.4.29, in view of the fact that when $\mathcal{Y}$ is $\mathbb{A}^1$ connected, there is an equivalence

$$\Pi^{\mathbb{A}^1}\mathcal{Y} \simeq \Pi^{mot} L_{mot}\mathcal{Y},$$

or, in other words, that $\Pi^{\mathbb{A}^1}\mathcal{Y}$ is $\mathbb{A}^1$ local (see Proposition 3.1.21).

Since $\mathcal{H}^{mot}$ is closed under extensions among connected objects (see [Mor12, Lemma 6.51]), the claim follows immediately from Proposition 2.4.30. □

Motivic Galois correspondence

Theorem 3.4.19. *Over a perfect field k, the motivic homotopy category satisfies 1-Galois correspondence: given a pointed motivically connected motivic space $\mathcal{X}$, there is an equivalence*

$$Cov_{\mathcal{H}^{mot}(k)}(\mathcal{X})^{>0}_* \simeq Sub_{Grp_{mot}(k)}(\pi_1^{mot}\mathcal{X})$$

Proof. This follows from Theorem 2.4.32. □

Theorem 3.4.20. *Let $\mathcal{X}$ be a pointed $\mathbb{A}^1$-connected k-space over a perfect field k. Then there is an equivalence of categories between the 1-category $Cov_{\mathbb{A}^1}(\mathcal{X})^{>_{\mathbb{A}^0}}$ of $\mathbb{A}^1$-coverings of $\mathcal{X}$ with $\mathbb{A}^1$-connected total spaces and the poset of strongly $\mathbb{A}^1$-invariant subgroup sheaves of $\pi_1^{\mathbb{A}^1}\mathcal{X}$.*

Proof. Apply Theorem 2.4.33. □

Remark 3.4.21. Although one can formulate a theorem concerning the "normal" part of the Galois correspondence, its statement is less satisfactory at present because it is unclear whether group-theoretic normality in the Nisnevich-local setting coincides with motivic local normality in the category $\mathrm{Grp}_k^{\mathbb{A}^1}$. See [ABH25, Remark 3] for a discussion of this issue, and Lemma 2.7

therein for related results. However, for abelian groups, Morel's celebrated *strongly implies strictly* theorem implies that motivic and ordinary normality coincide over perfect fields; see [ABH25, Proposition 2.8]. Consequently, the normal part of the Galois correspondence can be formulated for spaces with abelian $\mathbb{A}^1$-fundamental group.

4. The internal homotopy theory of $\mathcal{H}^{bir}$

The main observation of the first three subsections of the previous section is that the motivic homotopy category, constructed out of smooth schemes, even over a field, is not an ∞-topos, in contrast to the ordinary homotopy theory of smooth manifolds. To elaborate, let us have a quick recap of how these categories are born from universality:

Proposition 4.0.1. *Let S be a Qcqs scheme. Then the functor*

$$h_S^{mot} := L_{mot} h_S : Sm_S \to \mathcal{H}^{mot}(S)$$

is universal among functors to a presentable ∞-category that contracts the affine line and sends the Nisnevich squares to pushout squares. In other words, the large co-presheaf

$$Cat_\infty^{\mathbb{A}^1,nis}(Sm_S, -) : Pr^L \to \mathcal{G}_1\text{-}Spc$$

sending a presentable ∞-category $\mathcal{D}$ to the full subcategory of $Cat_\infty(Sm_S, \mathcal{D})$ satisfying the two properties mentioned above, is representable by $h_S^{mot} : Sm_S \to \mathcal{H}^{mot}(S)$.

Proof. See [Rob12, Theorem 5.2]. □

In the classical case of smooth manifolds, the answer is well known and justifies the construction of the motivic homotopy category:

Proposition 4.0.2. *Let* SmMan *be the category of smooth manifolds. Then the co-presheaf*

$$Cat_\infty^{\mathbb{R}^1,open}(\mathrm{SmMan}, -) : Pr^L \to \mathcal{G}_1\text{-}Spc$$

sending a presentable ∞-category $\mathcal{D}$ to the full subcategory of $Cat_\infty(SmMan, \mathcal{D})$ consisting of functors that contract the real line $\mathbb{R}^1$ and map twofold open cover squares to pushout squares is representable by the embedding SmMan $\hookrightarrow$ *Spc.*

Proof. This is [Dug99, Remark 3.4.10]. □

Evidently, the resulting category in the classical case is not just a presentable ∞-category but, in fact, an ∞-topos. However, the schematic case is not as beautiful. Specifically, in Theorem 3.2.13 we have seen that the motivic homotopy category need not be an ∞-topos. The problem is that, in the case of manifolds, one can always construct a 'good' cover, in the sense that, 'open cover locally,' every manifold is homeomorphic to some $\mathbb{R}^n$; i.e., given a manifold M, one can always find an open cover $\{U_i \subset M\}$ such that each simplex of its Čech nerve is homeomorphic to a union of copies of $\mathbb{R}^{dimM}$. Even though one knows that (say over a field) etale locally every smooth scheme X is isomorphic to $\mathbb{A}^{dimX}$, this does not pass to the corresponding Čech nerve, so that even etale locally schemes might have $\mathbb{A}^1$-locally non-zero homotopy dimension. Thus, even the etale local motivic homotopy category is not an ∞-topos.

Instead of seeking a suitable topology in which such a dimension reduction (from pure homotopy dimension to $\mathbb{A}^n$'s) is possible, algebraic geometry offers an interesting way to brute-force this reduction. Namely, because fields have homotopy dimension zero, the desired ∞-topos might arise by identifying every variety with its function field, or equivalently, by identifying every open subscheme of a given scheme in Sm_k. This is precisely the birational motivic homotopy

category of [Mai26a] (where it is denoted $\mathcal{H}^{b\mathbb{A}^1}(S)$), [BE21] (where it is denoted $L^0_{bir}\mathcal{H}^{\mathbb{A}^1}(S)$), and [Pel14], which we now recall below.

Given a Qcqs scheme S, we define the birational motivic homotopy category, denoted $\mathcal{H}^{bir}(S)$, as the localization of $\mathcal{H}^{mot}(S)$ at the set of all dense open immersions in Sm_S. The key observation of [Mai26a] is what makes the contents of this section work and thus suggests that the birational motivic homotopy category is close to being an ∞-topos. Let us recall it here for the convenience of the reader:

Theorem 4.0.3. *There is an equivalence of ∞-categories:*

$$\mathcal{H}^{bir}(S) = \mathcal{H}^{b}(S) := L_{dense}\mathcal{P}_{\Sigma}(S).$$

Here, $\mathcal{P}_{\Sigma}(S)$ is the full subcategory of $\mathcal{P}(S)$ consisting of presheaves that turn coproducts of schemes into products of spaces. Being (identically) the category of sheaves $\mathcal{P}_{\sqcup}(S)$ for the disjoint cover topology $\sqcup$, this is also an ∞-topos [BH21, Lemma 2.4], just like the Nisnevich one $\mathcal{P}_{nis}(S)$. But this new topos is quite well-behaved compared to the Nisnevich one. To see this, let $\tau^{\sigma}_{\leqslant n}$ denote the truncation functors for the various topologies we deal with. The ordinary truncation $\tau_{\leqslant n}$ of $\mathcal{P}(S)$ preserves products and hence takes $\mathcal{P}_{\Sigma}(S)$ to itself. Since $\mathcal{P}_{\Sigma}(S) = \mathcal{P}_{\sqcup}(S)$ is a left exact localization of $\mathcal{P}(S)$, the previous sentence, together with [Lur09, 5.5.6.28], implies that $\tau^{\sqcup}_{\leqslant n} \simeq \tau_{\leqslant n|\mathcal{P}_{\Sigma}(S)}$. This implies, in particular, that $\pi^{\sqcup}_i \simeq \pi_{i|\mathcal{P}_{\Sigma}(S)}$. (Such a statement is not true Nisnevich locally, and the correct relation there is only $\tau^{nis}_{\leqslant n} \simeq L_{nis}\tau_{\leqslant n}$.)

4.1. **Internal Birational motivic truncations and Birational motivic homotopy group.** We now discuss the birational local analogue of §2.1. We won't follow the lines of §3.1 to restate the definitions for the birational category; we recommend that the reader look at §2.1 or §3.1. Let us denote the internal truncations and the internal homotopy groups of the presentable ∞-category $\mathcal{H}^{bir}(S)$ by $\tau^{bir}_{\leqslant n}$ and π^{bir}_i, respectively.

Lemma 4.1.1. *An S-space $\mathcal{X}$ is a birational motivic space if and only if, for every $i \geqslant 0$, $\pi_i\mathcal{X}$ is birational motivic.*

Proof. By Theorem 4.0.3, this follows from the observation that $\pi_{i|\mathcal{P}_{\Sigma}(S)}$ is automatically a Σ-presheaf (i.e., a $\sqcup$-sheaf). □

Therefore, we see that $\tau_{\leqslant n}$ maps $\mathcal{H}^{bir}(S)$ to itself:

Theorem 4.1.2. *There are identifications (of truncations restricted to the the birational homotopy category)*

$$\tau^{bir}_{\leqslant n} = \tau_{\leqslant n|\mathcal{H}^{bir}(S)} = \tau^{\sqcup}_{\leqslant n|\mathcal{H}^{bir}(S)} = \tau^{nis}_{\leqslant n\,|\mathcal{H}^{bir}(S)}$$

and of homotopy groups

$$\pi^{bir}_i = \pi_{i|\mathcal{H}^{bir}(S)}.$$

Proof. This follows directly from the fact that $\tau_{\leqslant n}$ preserves birational local objects, say by [Lemma 4.1.1]. Indeed, for

$$\mathcal{X} \in \mathcal{H}^{bir}(S) \text{ and } \mathcal{Y} \in \mathcal{H}^{bir}(S)_{\leqslant n} = \mathcal{H}^{bir}(S) \bigcap \mathcal{P}$$

one computes

$$\begin{aligned} Map_{\mathcal{H}^{bir}(S)_{\leqslant n}}(\tau^{b}_{\leqslant n}\mathcal{X}, \mathcal{Y}) \simeq Map_{\mathcal{H}^{bir}(S)}(\mathcal{X}, \mathcal{Y}) &= Map_{\mathcal{P}(S)}(\mathcal{X}, \mathcal{Y}) \\ &\simeq Map_{\mathcal{P}(S)_{\leqslant n}}(\tau_{\leqslant n}\mathcal{X}, \mathcal{Y}) \\ &= Map_{\mathcal{H}^{bir}(S)_{\leqslant n}}(\tau_{\leqslant n}\mathcal{X}, \mathcal{Y}) \end{aligned}$$

(here the last equality follows from the fact that $\tau_{\leqslant n}\mathcal{X} \in \mathcal{P}(S)_{\leqslant n}$ is birational and hence belongs to $\mathcal{H}^{bir}_{\leqslant n}$ as well, while it was given that $\mathcal{Y} \in \mathcal{H}^{bir}(S)_{\leqslant n}$. The result then follows by the Yoneda lemma. Moreover, because L_{nis} of a birational local object is the object itself [Mai26a, Theorem 2.1.1], we have $\tau^{bir}_{\leqslant n} = \tau^{nis}_{\leqslant n\,|\mathcal{H}^{bir}(S)}$.

It then follows from definition that

$$\pi^{bir}_i := \tau^{bir}_{\leqslant 0}\Omega^i = \tau_{\leqslant 0}\Omega^i = \pi_{i|\mathcal{H}^{bir}(S)}.$$

□

Theorem 4.1.3. *$L_{\leqslant 0} = \pi^b_0$, and the restriction*

$$\tau^{bir}_{\leqslant 0} : \tau_{\leqslant 0}\mathcal{H}^{bir}(S) \to \mathrm{Disc}\mathcal{P}_{nis}(S) = Sh^{nis}_S$$

is simply the inclusion of birational sheaves.

Proof. This follows immediately from the observation that π_0 of birational motivic spaces is birational (Lemma 4.1.1). □

Theorem 4.1.4. *The birational motivic homotopy category is postnikov complete.*

Proof. By construction, $\mathcal{H}^{bir}(S) \subset \mathcal{P}(S)$ is closed under limits and, by Theorem 4.1.2, under truncations. Since $\mathcal{P}(S)$ is Postnikov complete, so is $\mathcal{H}^{bir}(S)$. □

4.2. **Birational Eilenberg Maclane spaces.** In Section §3.2, we saw that the ordinary motivic homotopy category fails to be an ∞-topos. The argument stemmed from the observation that, in the ordinary motivic case, not every discrete motivic group admits an internal delooping. In this section, we shall show that this issue disappears after birational localization. The key algebraic reason is that birational sheaves are flasque.

We denote by Sh^b_S the category of sheaves of sets that are birational local. This is precisely the full subcategory of discrete/0-truncated objects of $\mathcal{H}^{bir}(S)$. Similarly, Grp^b_S and Ab^b_S denote the full subcategories of birational local sheaves of groups and abelian groups, respectively. Regarding Definition 2.2.10, we say that,

Definition 4.2.1. *A presheaf of (abelian) groups G is (n-strictly) strongly L_{bir}-local if ($\mathrm{B}^n_{nis}G$) $\mathrm{B}_{nis}G$ is L_{bir}-local. We denote the full subcategory of n-strictly L_{bir}-local groups by n-$Grp(Disc(\mathcal{H}^{bir}(S)))$.*

Lemma 4.2.2. *A presheaf of groups G is L_{bir}-local if and only if it is strongly L_{bir}-local. In other words, there is an equivalence $Grp^b_S = Grp_{L_{bir}}(Disc(\mathcal{H}^{bir}(S)))$.*

Proof. This is [Mai26a, Corollary 3.2.9]. □

Similarly,

Lemma 4.2.3. *Let $n \geqslant 2$. A presheaf of groups G is n-strictly L_{bir}-local if and only if G is L_{bir}-local.*

Definition 4.2.4. *A pointed birational local space $x : * \to \mathcal{X}$ is said to be:*

(1) birationally connected (1-connective) if $\tau^{bir}_{\leqslant 0}\mathcal{X} \simeq *$.

(2) birationally globally $(n-1)$-connected (or, n-connective) if it is connected and for all $0 < i < n$ $\pi_i^{bir}(\mathcal{X}, x) \simeq *$.

Definition 4.2.5. *We say that a pointed birational local space is birationally n-Eilenberg-MacLane if it is birationally globally n-connective and birationally n-truncated.*

Theorem 4.2.6. *Let S be a QCQS scheme. Then, similar to that of an ∞-topos* [Lur09, Lemma 7.2.2.11 (1)]*, there are equivalences of categories:*

$$\begin{aligned} Sh^b_{*,S} \simeq \mathrm{Disc}(\mathcal{H}^{bir})_\bullet \simeq \mathcal{EM}_0(\mathcal{H}^{bir}) : \pi_0^{bir} \\ Grp^b_S \simeq Grp(\mathrm{Disc}(\mathcal{H}^{bir})) \simeq \mathcal{EM}_1(\mathcal{H}^{bir}) : \pi_1^{bir} \\ Ab^b_S \simeq Ab(\mathrm{Disc}(\mathcal{H}^{bir})) \simeq \mathcal{EM}_n(\mathcal{H}^{bir}) : \pi_n^{bir} \text{ for all } n \geqslant 2 \end{aligned}$$

Proof. Since L_{bir} is path-injective (Theorem 4.1.3), this follows from Proposition 2.2.11 and Lemma 4.2.3. □

Remark 4.2.7. Here is a rather straightforward argument. The induced presheaf Eilenberg-MacLane functors take a birational group object to birational Eilenberg-MacLane spaces, with inverse given by presheaf homotopy groups. Explicitly, there are equivalences

(1) $\mathrm{K}(-,0) : Shv^b_S \simeq \mathcal{EM}_0(\mathcal{H}^{bir}(S)) : \pi_0$.
(2) $\mathrm{K}(-,1) : Grp^b_S \simeq \mathcal{EM}_1(\mathcal{H}^{bir}(S)) : \pi_1$.
(3) $\mathrm{K}(-,i) : Ab^b_S \simeq \mathcal{EM}_i(\mathcal{H}^{bir}(S)) : \pi_i$ for all $i \geqslant 2$.

In short, we have equivalences of categories $\mathrm{K}(-,n) : n\text{-}Grp^b_S \simeq \mathcal{EM}_n(\mathcal{H}^{bir}_S)$ for each $n \geqslant 0$. The identification of the theorem above with this statement follows from two facts (compare with the motivic case). First, π_i of a birational local space is birational. Second, $\mathrm{K}(A,n)$ is birational local iff A is a birational presheaf. Indeed, these can be checked on the presheaf π_i's by Lemma 4.1.1.

Now the equivalence

$$\mathrm{K}(-,n) : n\text{-}Grp_S \leftrightarrows \mathcal{EM}_n(\mathcal{P}(S)) : \pi_n$$

restricts to the desired equivalence by restricting both sides to $\mathcal{H}^{bir}$ and noting that

$$\mathcal{EM}_n(\mathcal{H}^{bir}(S)) = \mathcal{EM}_n(\mathcal{P}(S)) \bigcap \mathcal{H}^{bir}(S)$$

[Proposition 2.2.7] and that $\pi_i^{bir} = \pi_{i|\mathcal{H}^{bir}(S)}$ Theorem 4.1.2.

Remark 4.2.8. An algebraic reason this is not surprising is that birational sheaves are flasque over arbitrary *qcqs* base schemes. (To see this, let $U \hookrightarrow X$ be an open immersion (not necessarily dense). Obviously, $(X \backslash \bar{U}) \sqcup U \hookrightarrow X$ is a dense open immersion. Thus, if F is birational,

$$F(X) \to F(X \backslash \bar{U} \sqcup U) \cong F(X \backslash \bar{U}) \times F(U)$$

is an isomorphism. By composing with the (surjective) projection $F(X \backslash \bar{U}) \times F(U) \twoheadrightarrow F(U)$, we see that the canonical map $F(X) \to F(U)$ is indeed surjective.) But flasque sheaves of abelian groups cannot have higher Nisnevich cohomology groups. (Indeed, this follows from [Lemma 1.39, 1.40 Rio02] over noetherian bases and thus over all qcqs bases by noetherian approximation.) To follow the above theorem for $n \geqslant 2$, it suffices to verify the same on generators of $\mathcal{H}^{bir}_S$, namely

$h^b(X)$ for smooth X/S. But,

$$\pi_i Map(h^b(X), \mathbf{K}(A-,n)) = \pi_i \mathcal{P}_{nis}(Sm_S)\big(X, \mathbf{K}(A-,n)\big) \tag{6}$$

$$= \begin{cases} A \text{ for } 0 = i \\ H^{n-i}(X;A) = 0 \text{ for } 0 < i \leqslant n \\ \pi_{i-n} Map_{\mathcal{P}_{nis}(S)}\big(X, A\big) = 0 \text{ for } i > n \text{ [as } A \text{ is discrete]} \end{cases} \tag{7}$$

The case $n = 1$ is a little delicate, though it is also straightforward, since we are expecting $H^1_{nis}(-;G)$ to be birational and hence a Nisnevich sheaf in particular.

Remark 4.2.9. This equivalence is an ∞-topos-theoretic property (Remark 2.3.2) that fails in the standard motivic context (Corollary 3.2.14). The theorem thus shows that a further birational localization forces the motivic homotopy category to exhibit properties of an ∞-topos.

Theorem 4.2.10. *Over a qcqs scheme, every object $\mathcal{E} \in \mathcal{H}^{bir}(S)$ has 0-cohomological dimension. Hence, in particular, $\mathcal{H}^{bir}(S)$ has 0-cohomological dimension.*

Proof. By definition, for any $n > 0$ and X smooth over S,

$$\mathbf{H}^n_{\mathcal{H}^{bir}(S)}(A)(X/S) := \pi_0\Big(\mathrm{Map}_{\mathcal{H}^{bir}(S)}\big(L_{bir}X, \mathbf{K}(A-,n)\big)\Big) = \pi_0(\mathbf{K}(AX,n))$$

vanishes.

Since $\mathcal{H}^{bir}(S)$ is generated under colimits by $L_{bir}h_S(X)$ (for smooth X/S), we conclude that $\mathbf{H}^n(\mathcal{H}^{bir}(S)(\mathcal{E})$ is 0 for all $n > 0$. □

Remark 4.2.11. These properties do not already hold in any other higher birational motivic categories (see [Mai26c] for the definition). In fact, the same example as in [Cho14] shows that the 1-birational motivic homotopy category is not an ∞-topos either. Indeed, his $\mathbb{A}^1$-local example $\mathbb{Z}^{nis}(\mathbb{G}_m)$ is not strongly $\mathbb{A}^1$-invariant. From his proof, or otherwise, it is clear that $\mathbb{Z}^{nis}(\mathbb{G}_m)$ is a summand of $\mathbb{Z}^{\sqcup}(\mathbb{G}_m)$, and is thus 1-birational local since $\mathbb{G}_m$ is so. But $\mathcal{EM}_1(\mathcal{H}^1(k))$ consists of strongly $\mathbb{A}^1$-invariant, strongly 1-birational-invariant sheaves of groups. In fact, this failure is not just to the $\mathbb{A}^1$-local aspect; it is of birational nature as well. There are n-birational, strongly $\mathbb{A}^1$-invariant abelian groups that are not strictly n-birational. One quick example is K^M_{n+1}.

4.3. **Birational groups/monoids.** We can generalize the results of the previous section about birational sheaves of groups to arbitrary birational monoid spaces. Recall that the birational homotopy category is presentable, and hence there is an internal bar construction. We shall denote this by $\mathrm{B}_{bir} : \mathrm{Mon}(\mathcal{H}^{bir}) \to \mathcal{H}^{bir}$. As in §2.3, one may define a notion of strongly L_{bir}-local group objects, but the following theorem shows, as in Lemma 4.2.3, that such a concept is unnecessary.

Theorem 4.3.1. *There is an equivalence $\mathbf{B}_{bir} \simeq \mathbf{B}_{|Mon(\mathcal{H}^{bir}(S))}$.*

Proof. It follows from Lemma 2.3.3 that $\mathrm{B}_{bir} \simeq L_{bir}\mathrm{B}$, where B is the presheaf-level bar construction. Thus, it suffices to know that B preserves birational local objects. This is [Mai26a, Lemma 3.2.4 (3)]. □

We caution that $\mathbf{B}_{mot} \not\simeq \mathbf{B}_{|Mon(\mathcal{H}^{mot}(k))}$.

Theorem 4.3.2. *L_{bir} commutes with the (Nisnevich) group completion of commutative monoids in $\mathcal{P}(S)$ (as well as in $\mathcal{P}_{nis}(S)$).*

Proof. Since $\mathcal{M}^+ \simeq \Omega\mathbf{B}\mathcal{M}$ and L_{bir} commutes with Ω on connected objects and with $\mathbf{B}$ on all objects, the result follows immediately. □

Recall that the following is a property of an ∞-topos:

Proposition 4.3.3. *The functor*

$$\Omega : \mathcal{H}^{bir}(S)_{\geqslant 1} \to Grp(\mathcal{H}^{bir}_{\bullet}(S)) \simeq Grp(\mathcal{H}^{bir}(S))$$

is an equivalence of infinity categories, with inverse given by the bar construction $\mathbf{B}_{|bir}$.

Proof. Since $\mathbf{B}_{bir} \simeq \mathbf{B}_{|Mon(\mathcal{H}^{bir}(S))}$, the co-unit

$$\mathbf{B}_{bir}\Omega \to 1_{\mathcal{H}^{bir}(S)_{\geqslant 1}}$$

is an equivalence, as it is equivalent to the restriction of the equivalence $B\Omega \to 1_{\mathcal{P}(S)_{\geqslant 1}}$.

Because grouplike objects are group-complete, this is just a restatement of [Mai26d, Theorem 3.3.7]. □

Note that this is a property of an ∞-topos that fails to hold in the ordinary motivic homotopy category Corollary 3.3.2. We conclude this section with one more topos-theoretic property of the birational motivic homotopy category:

Proposition 4.3.4. *Suppose S is a geometrically unibranch scheme with finitely many generic points. Then the birational homotopy category of S has a subobject classifier.*

Proof. It suffices to show that the subobject classifier of $\mathcal{P}_\Sigma(Sm_S)$ is birational local. Indeed, since a localization is closed under limits, the subobject functor is simply the restriction of the subobject functor from the parent category to the localized one.

In general, the subobject classifier of $\mathcal{P}_\Sigma(Sm_S) = \mathcal{P}_\sqcup(Sm_S)$ is given by the sifted sheaf:

$$X/S \mapsto \Omega^{\sqcup}_S(X) := \{W/S \big| W \text{ is a clopen subset of } X\}$$

Since X has a finite decomposition by its irreducible components (which are thus both open and closed) (see the lemma below),

$$\Omega^{\sqcup}_S(X) := \{W/S \big| W \text{ is a component of } X\}.$$

It is obvious that $\Omega^{\sqcup}_S$ is birational local. □

Lemma 4.3.5. *Suppose S is a Qcqs scheme. And $f : X \to S$ a smooth (qc and separated) morphism.*

(1) If S is geometrically unibranch then the irreducible components of X are disjoint.
(2) If S has finitely many generic points, then so does X.

Proof. (1). We first show that X has the same set of properties. Since being geometrically unibranch is a local condition, we reduce to the following algebra problem: if $A \to B$ is an essentially smooth local homomorphism of local rings such that A is geometrically unibranch, then so is B. This follows from [Sta26, Lemma 0DQ1 (1)] and [Sta26, Lemma 0C37 (3)]. Therefore, it suffices to show that the irreducible components of a geometrically unibranch scheme are disjoint. This is, in fact, true for all unibranch schemes: if x is a point that belongs to two irreducible components, then $\mathcal{O}_{X,x}$ will have at least two minimal primes.

(2). This follows from the fact that f maps generic points to generic points and smooth schemes over fields have this property. □

Remark 4.3.6. The motivic homotopy category most likely does not possess a subobject classifier. In fact, the subobject classifier Ω_S^{nis} of Shv_S^{nis} cannot be $\mathbb{A}^1$-local, since there are Nisnevich covers of X does not extend (along the 0 section) uniquely to a Nisnevich cover of $\mathbb{A}^1 \times X$.

Theorem 4.3.7. *The adjunction $\mathcal{H}^{bir}(S) \leftrightarrows \mathcal{P}(S)$ is finitary-monadic. In other words, the idempotent monads induced by the adjunctions identify $\mathcal{H}^{bir}$ as their Eilenberg-Moore category of finitary algebras. In detail, this means that the induced monad preserves sifted colimits.*

Proof. In light of the theorem above, this is a consequence of Beck's Monadicity Theorem [Lur17, Theorem 4.7.0.3]. In fact, this holds for any localization closed under geometric realizations. Indeed, the conservative condition is redundant for fully faithful functors. □

Remark 4.3.8. It turns out that any Bousfield localization is finitary monadic. Consequently,

$$\mathcal{H}^{bir}(S) \subset \mathcal{P}_\Sigma(S), \mathcal{H}^{bir}(S) \leftrightarrows \mathcal{P}_{nis}(S), \mathcal{H}^{bir}(S) \leftrightarrows \mathcal{H}^{mot}(S), \text{ and } \mathcal{H}^{mot}(S) \leftrightarrows \mathcal{P}_{nis}(S)$$

are all monadic. However, localization via descent is not finitary-monadic in general. The only finitary monadic localization here that is obtained by imposing descent is $\mathcal{P}_\Sigma(S) \subset \mathcal{P}(S)$. Even the localization $\mathcal{P}_{nis}(S) \subset \mathcal{P}(S)$ is not monadic. To see this, observe that in a monadic localization, an object X is local if and only if the localization map $X \to LX$ admits a retract. Consider the separated presheaf $\underline{\mathbb{Z}}$ generated by $\mathbb{Z}$ (or any other constant set, for that matter):

$$U \mapsto \underline{\mathbb{Z}}(U) = \begin{cases} \mathbb{Z} \text{ if } U \neq \varnothing \\ 0 \text{ otherwise} \end{cases}$$

Evidently, the sheafification $\underline{\mathbb{Z}} \to \mathbb{Z}^{nis}$ is not an equivalence. However, it possesses a retraction induced by the "identification of generic points map" $(-)^{(0)} \to *$. The same example shows that this issue persists for the motivic adjunction $\mathcal{H}^{mot}(S) \subset \mathcal{P}(S)$.

4.4. **Birational coverings. Birational Geometric coverings**

Definition 4.4.1. *A collection $\{\mathcal{U}_i \to \mathcal{X}\}$ of morphisms of presheaves of spaces is said to be a birational geometric covering if $L_{bir}\check{C}p \to L_{bir}\mathcal{X}$ is a colimit diagram in $\mathcal{H}^{bir}(k)$, where $p : \bigsqcup \mathcal{U}_i \to \mathcal{X}$.*

Lemma 4.4.2. *An $\mathbb{A}^1$-geometric covering is a birational geometric covering.*

Proof. Since $L_{bir} \simeq L_{bir}L_{mot}$, this is obvious. □

So,

$$\text{Nis-geom-cover} \implies \mathbb{A}^1\text{-geom-cover} \implies \text{bir-geom-cover}$$

Corollary 4.4.3. *An S-space $\mathcal{X}$ is birationally connected if and only if there exists a birational (which is if and only if a Nisnevich if and only if an $\mathbb{A}^1$-) geometric covering $\mathcal{U}_i \to \mathcal{X}$ such that each $\mathcal{U}_i$ is birationally connected and, for every pair i, j, the sheaf $\pi_0^b(U_i \times_X U_j)$ has an S-rational point.*

Proof. If $\mathcal{X}$ is birationally connected, then the identity map $\mathcal{X} \to \mathcal{X}$ is a Nisnevich cover with the given property. Using the above chain of implications, we reduce to the case of a birational geometric cover. But this follows from Corollary 2.4.7 in the presence of Theorem 4.1.3. □

Corollary 4.4.4. *A scheme X is birationally-connected iff there is a $U_i \to X$ be a Nisnevich covering of a scheme X/S and such that for every pair i, j the $\pi_0^b(U_i \times_X U_j)$ has an S-rational point. If U_i is birationally-connected for all i, then*

Remark 4.4.5. Compare with the ordinary motivic counterparts Corollary 3.4.3, Corollary 3.4.4. The condition there is slightly stronger due to the effectiveness of L_{mot}.

Corollary 4.4.6 (Birational Motivic Van Kampen)**.** *Let $\mathcal{U}_i \to \mathcal{X}$ be a birational-geometric covering (for example, a Nisnevich local covering or a motivic geometric covering) such that $\mathcal{U}_{ijk}$ is birationally connected for every i, j, k (1-, 2-, or 3-fold). Then $\mathcal{X}$ is birationally connected, and the birational motivic homotopy group of $\mathcal{X}$ is given by the following coequalizer in Grp_S^b:*

$$\underset{i,j}{*}\pi_1^b(\mathcal{U}_i \times_{\mathcal{X}} \mathcal{U}_j) \rightrightarrows \underset{i}{*}\pi_1^b(\mathcal{U}_i) \to \pi_1^b(\mathcal{X}) \to *$$

Proof. This follows immediately from Theorem 2.4.8 and Theorem 4.1.3. □

Theorem 4.4.7 (Birational version of Morel's Van Kampen, [Mor12, Theorem 7.12])**.** *Assume S is qcqs. Let $\mathcal{U} := \{U_i \to X\}$ be a basic Nisnevich cover in Sm_S such that the 1-, 2-, and 3-fold fiber products are birationally connected and compatibly pointed across all fiber products of all finite pairs. Then there is a coequalizer diagram in Grp_S^b:*

$$\underset{i,j}{*}\pi_1^b(U_i \times_X U_j) \rightrightarrows \underset{i}{*}\pi_1^b(U_i) \to \pi_1^b(X) \to *$$

Remark 4.4.8. When all the connectivity conditions are replaced by $\mathbb{A}^1$-connectivity, we can derive our birational Van Kampen theorem Theorem 4.4.7 directly from the $\mathbb{A}^1$ motivic Van Kampen theorem. Indeed, if all the relevant schemes are $\mathbb{A}^1$-connected, then [Mai26a, Proposition 3.6.7] tells us that the sequence involving π_1^b is simply the application of π_0^b to Morel's Van Kampen sequence. The result then follows from the fact that, being a localization, $\pi_0^b : Grp^{\mathbb{A}^1} \to Grp^b$ preserves colimits.

Birational Spatial coverings

Definition 4.4.9 (Definition 2.4.9, Definition 2.4.17)**.** *(1) A morphism $f : \mathcal{Y} \to \mathcal{X}$ of birational motivic spaces is a birational covering if $f \in \left(\mathcal{H}^{bir}_{/\mathcal{X}}\right)_{\leqslant 0}$. Denote the category of such coverings by $Cov_{\mathcal{H}^{bir}}(\mathcal{X})$.*

When $f : \mathcal{Y} \to \mathcal{X}$ is a morphism of S-spaces, then:

(2) We say that it is a birational fibration if it is a local object of $\mathcal{P}_{/\mathcal{X}}$ with respect to birational motivic equivalences contained in $\mathcal{P}_{/\mathcal{X}}$. We denote the category of birational fibrations of $\mathcal{X}$ by $\mathcal{F}ib^b_{\mathcal{X}}$.
(3) We call it an $\mathbb{A}^1$ covering if it is a birational fibration and is a 0-truncated object in the ∞-category of birational fibrations of $\mathcal{X}$. Denote the category of birational coverings of $\mathcal{X}$ by $Cov_b(\mathcal{X})$.

Proposition 4.4.10. *A morphism $\mathcal{X} \to \mathcal{Y}$ of birational motivic spaces is a birational covering if and only if it is a (Nisnevich) covering if and only if it is an $\mathbb{A}^1$-covering.*

Proof. Follows immediately from Proposition 2.4.10. □

Proposition 4.4.11. *Let G be a birational sheaf of groups. Then any Nisnevich G-torsor $\mathcal{Y} \to \mathcal{X}$ is a birational cover.*

Proof. Since $Grp_{Lbir}(\mathcal{H}^{bir}) \simeq Grp(\mathcal{H}^{bir})$ (Theorem 4.3.1), this follows from Corollary 2.4.22. □

Remark 4.4.12. Using the model categorical definition of Morel, one may use the original idea from [Mor12, 7.5 (1)] to prove the above theorem. Let us repeat it for the convenience of the reader. Suppose we are given a commutative square on the left, where $\mathcal{A} \to \mathcal{B}$ is a birational weak equivalence

$$\begin{array}{ccc} \mathcal{A} & \longrightarrow & \mathcal{Y} \\ \downarrow & & \downarrow \\ \mathcal{B} & \longrightarrow & \mathcal{X} \end{array} \qquad\qquad \begin{array}{ccccc} \mathcal{A} & \longrightarrow & \mathcal{Y} \times_{\mathcal{X}} \mathcal{B} & \longrightarrow & \mathcal{Y} \\ & \searrow & \uparrow a & & \downarrow \\ & & \mathcal{B} & \longrightarrow & \mathcal{X} \end{array}$$

Consider the diagram on the right, obtained by pulling the torsor to $\mathcal{B}$. It follows that the projection map $\mathcal{Y} \times_{\mathcal{X}} \mathcal{B} \xrightarrow{\phi} \mathcal{B}$ is a G torsor. The commutativity of the starting diagram yields that this torsor is trivial on $\mathcal{A}$. Since $\mathrm{B}\mathcal{G}$ is birational local by Theorem 4.3.1, and $\mathcal{A} \to \mathcal{B}$ is given to be a birational weak equivalence, it follows that the map $[\mathcal{B}, \mathrm{B}G] \to [\mathcal{A}, \mathrm{B}G]$ is an isomorphism. So the triviality of the torsor on $\mathcal{A}$ implies that it trivializes on $\mathcal{B}$ as well. Let $s : \mathcal{B} \to \mathcal{Y} \times_{\mathcal{X}} \mathcal{B}$ be the corresponding trivialization. Then the composite

$$\mathcal{B} \xrightarrow{s} \mathcal{Y} \times_{\mathcal{X}} \mathcal{B} \xrightarrow{a} \mathcal{Y}$$

makes the bottom triangle commute. Obviously, the upper triangle may not commute. However, since the torsor $\mathcal{Y} \to \mathcal{X}$ is a $\mathcal{G}$-torsor trivializing on $\mathcal{A}$, any two such trivializations are $\mathcal{G}$ automorphic. Therefore, there is a section $g : \mathcal{A} \to \mathcal{G}$ such that $g(a \circ s)$ works.

Theorem 4.4.13. *Let $\mathcal{X}$ be a pointed birational motivic space (not necessarily connected).*

(1) Then the category $\mathcal{C}ov_{\mathcal{H}^{bir}}(\mathcal{X})_*$ always has an initial object, say denoted by $\widetilde{\mathcal{X}}^{bir}$.
(2) This initial object is the unique one with a birationally simply connected source (i.e., $\tau^{bir}_{\leqslant 1}\widetilde{\mathcal{X}}^{bir} \simeq *$). Each of the inclusions

$$\mathcal{C}ov_{\mathcal{H}^{bir}}(\mathcal{X})_* \subset \mathcal{C}ov_{\mathcal{P}_{nis}}(\mathcal{X})_* \subset \mathcal{C}ov_{\mathcal{P}}(\mathcal{X})_*$$

preserves this initial object. In other words, this initial object is the unique (Nisnevich) covering with a (Nisnevich) simply connected source.
(3) The canonical map

$$\pi_1^{bir}(\mathcal{X}) \to \underline{Aut}_{\mathcal{X}}(\widetilde{\mathcal{X}}^{bir})$$

is an equivalence.

Proof. (The first one is mentioned at [Mai26a, Corollary 3.6.2], whose proof is similar to Proposition 2.4.10. We shall, however, prove this using the categorical techniques developed in §2 just to keep up with the style of the paper.)

From Theorem 4.1.2, it follows that $\Pi^b\mathcal{X}$ is a birational motivic space. Thus, Proposition 2.4.25 applies, and statement (1) follows.

Statement (2) is a formal consequence of the fact that, from the referred proposition, one has $(\widetilde{\mathcal{X}}^{bir}) \simeq \tau_{>1}\mathcal{X}$, where this last term is the unique (Nisnevich) simply connected (simplicial) cover of $\mathcal{X}$. Since this is already birational local, we have

$$\tau^{bir}_{\leqslant 1}\widetilde{\mathcal{X}}^{bir} \simeq \tau_{\leqslant 1}\widetilde{\mathcal{X}}^{bir} \simeq \tau_{\leqslant 1}\tau_{>1}\mathcal{X} \simeq *.$$

The third statement follows from Lemma 2.4.31 because L_{bir} is fundamental on every object. □

Remark 4.4.14. Compare the above theorem with its ordinary motivic counterpart, namely Theorem 3.4.14. There, a similar statement was proven for internally connected motivic spaces. Here, in the birational case, however, we did not require the base to be connected. Thus, we again see that the birational motivic homotopy category, $\mathcal{H}^{bir}$, behaves like an ∞-topos (see Remark 2.4.24). Though in the motivic case, we did not really prove that an initial internal motivic covering may not exist when the base is not internally (motivically) connected. It is interesting in its own right to know whether this has a positive answer.

Since L_{bir} is neither locally cartesian (see [Mai26a, Counterexample 3.2.13]) nor effective (see [Mai26a, Counterexample 3.5.7]), proving the following by pulling back the (birational) universal cover of $L_{bir}\mathcal{E}$ to $\mathcal{E}$ (similar to Morel [Mor12, $\mathbb{A}^1$-coverings]) seems difficult.

Corollary 4.4.15 (compare with the $\mathbb{A}^1$ analogue, Corollary 3.4.13, no connectivity condition is required there)**.** *Let $\mathcal{X}$ be an $\mathbb{A}^1$-connected (equivalently, motivically connected when the base is a field, Lemma 3.1.23) space. Then we have an equivalence of categories*

$$\eta_{bir}^* : Cov_b(L_{bir}\mathcal{X}) = Cov_{\mathcal{H}^{bir}}(L_{bir}\mathcal{X}) \leftrightarrows Cov_b(\mathcal{X}) : L_{bir}$$

where η_{bir} is the localization unit $1 \to L_{bir}$.

Proof. Recall that L_{bir} is locally cartesian on motivically connected objects [Mai26a, Corollary 3.2.16]. Moreover, since $\mathcal{X}$ is an $\mathbb{A}^1$-connected space, [Mai26a, Corollary 3.5.2] implies that $L_{bir}\mathcal{X}$ is connected. Therefore, the localization map $\mathcal{X} \to L_{bir}\mathcal{X}$ is an effective epimorphism in $\mathcal{P}_{nis}(S)$ [Lur09, Proposition 7.2.1.14]. With these observations, the claim follows from Proposition 2.4.23. □

Theorem 4.4.16. *Let $\mathcal{X} \in \mathcal{P}(S)$ be an $\mathbb{A}^1$ connected space. Then*

(1) $\mathcal{X}$ has a birationally simply connected birational cover $\widetilde{\mathcal{X}}^b$.
(2) Choice of a base point makes this birationally simply connected cover into the initial object of $Cov_b\mathcal{X}_*$.
(3) The canonical map $\pi_1^b\mathcal{X} \to Aut_{\mathcal{X}_*}(\widetilde{\mathcal{X}}^b)$ is an isomorphism.

Proof. Since $\mathcal{X}$ is $\mathbb{A}^1$-connected, so is $L_{bir}\mathcal{X}$ by [Mai26a, Corollary 3.5.2]. Therefore, by the above corollary, replacing $\mathcal{X}$ with $L_{bir}\mathcal{X}$ we may assume that $\mathcal{X}$ is birational local. By [Mor12, Corollary 6.2], it follows that $\Pi^b\mathcal{X}$ is a birational motivic space. So Proposition 2.4.25 applies, and statement (1) follows.

Statement (2) is a formal consequence of the fact that, from the referred proposition, one has $\tilde{\mathcal{X}} \simeq \tau_{>1}\mathcal{X}$, which is the unique simply connected (simplicial) cover of $\mathcal{X}$.

The third statement follows from Lemma 2.4.31. □

Remark 4.4.17. The main hurdle when $\mathcal{X}$ is not motivically connected, but only birationally connected, is that

$$\mathcal{Z} \simeq \mathcal{X} \times_{L_{bir}\mathcal{X}} \mathcal{Y} \to \mathcal{Y}$$

need not be a birational weak equivalence, since our category is not locally cartesian. Thus, the above theorem is weaker than its $\mathbb{A}^1$-counterpart [Mor12, Theorem 7.8].

Remark 4.4.18. When the base space is actually Nisnevich connected, the result is a little stronger. In that case, the simply connected cover maps to the birational simply connected cover under L_{bir} (see [Mai26a, Corollary 3.6.4]).

Proposition 4.4.19. *Suppose $\mathcal{Y} \in \mathcal{P}(S)$ and $y : * \to \mathcal{Y}$ is a base point. Then pullback along y induces a functor*

$$y^* : Cov_b(\mathcal{Y}) \to Cov_b(*) = Sh_k^b.$$

This functor factors through the category of birational sheaves (of sets) with an action by $\pi_1^b(\mathcal{Y}, y)$ (via the canonical functor forgetting the action):

$$\Gamma_y^b : Cov_b(\mathcal{Y}) \longrightarrow \pi_1^b(Y, y)\text{-}Sh_k^b$$

which is an equivalence of categories when $\mathcal{Y}$ is birationally connected.

Proof. The existence of the pullback functor follows from the pullback stability of (birational) coverings, Lemma 2.4.13 (3). That it factors through $\pi_1^b(\mathcal{Y}, y)\text{-}Sh_k^b$ is a consequence of Lemma 2.4.29, in view of the fact that

$$\Pi^b\mathcal{Y} \simeq \Pi^{bir} L_{bir}\mathcal{Y}$$

(in other words, $\Pi^b\mathcal{Y}$ is birational local) (see Theorem 4.1.2).

Since $\mathcal{H}^b$ is closed under extensions over connected objects [Mai26a, Lemma 3.3.1], the claim follows immediately from Proposition 2.4.30. □

4.5. **On the birational motivic homotopy category being an ∞-topos.** At the beginning of §4, we explained how the transition from ordinary motivic homotopy to birational motivic homotopy suggests a near 'classical topological' behavior, allowing us to brute-force various topos-theoretic properties. In fact, through §4.1, §4.2, and §4.3, we have proven many of those topos-theoretic properties. However, the birational localization functor $L_{bir} : \mathcal{P}_{nis}(S) \to \mathcal{P}_{nis}(S)$ is not even locally cartesian [Mai26a, Counterexample 3.2.13], let alone left exact. At this juncture, we are thus unable to answer the following question over an arbitrary scheme:

Question 4.5.1. *For a Qcqs scheme S, is the presentable ∞-category*

$$\mathcal{H}^{bir}(S) := L_{bir}\mathcal{H}^{mot}(S) \simeq L_{dense}\mathcal{P}_\Sigma(S)$$

(where the last equivalence is due to [Mai26a, Corollary 2.2.12(2)]*) an ∞-topos?*

In fact, we have only been able to show the existence of a subobject classifier for geometrically unibranch Qcqs schemes with finitely many irreducible components; see Proposition 4.3.4. In the result below, we will show that, under this condition on the base scheme, Question 4.5.1 actually has a positive answer:

Theorem 4.5.2. *Suppose S is a Qcqs, geometrically unibranch scheme with finitely many generic points (for example, the spectrum of a field). Then $\mathcal{H}^{bir}(S)$ is an ∞-topos.*

Proof. The proof follows from the abstract lemma below (Lemma 4.5.3) and the fact that, under the conditions of the theorem, $\mathcal{H}^{bir}$ is closed under colimits in $\mathcal{P}_{nis}(S)$ (as well as in $\mathcal{P}_\Sigma(S)$; see Proposition 4.5.4 below). □

Lemma 4.5.3. *Suppose $\mathcal{C}$ is an ∞-topos and $L\mathcal{C} \subset \mathcal{C}$ is an accessible localization which is closed under all colimits. Then $L\mathcal{C}$ is an ∞-topos.*

Proof. Since $L\mathcal{C}$ is a presentable ∞-category, it suffices to check conditions (3) (ii), (iii), and (iv) from [Lur09, Theorem 6.1.0.6 (3)]. Each of these conditions concerns interactions between certain limits and colimits. Since these conditions hold in $\mathcal{C}$, and since $L\mathcal{C}$ is closed under both limits and colimits in $\mathcal{C}$, the result follows. □

Proposition 4.5.4. *Let S be a Qcqs, geometrically unibranch scheme with finitely many generic points. Then $L_{bir} : \mathcal{P}_\Sigma(S) \to \mathcal{P}_\Sigma(S)$ and $L_{bir} : \mathcal{P}_{nis}(S) \to \mathcal{P}_{nis}(S)$ preserve all colimits. In other words, both $\mathcal{H}^b(S) \subset \mathcal{P}_\Sigma(S)$ and $\mathcal{H}^b(S) \subset \mathcal{P}_{nis}(S)$ are closed under all colimits.*

Proof. Using [Mai26a, Corollary 3.1.6], it suffices to show that L_{bir} preserves coproducts. Suppose $j : U \hookrightarrow X$ is a dense open immersion in Sm_S. Since X is a finite disjoint union of its irreducible connected components (Lemma 4.3.5), we can assume that j is a finite disjoint union of dense open immersions of connected schemes. Since connected schemes are points in the $\sqcup$-topology, it follows that coproducts (computed in $\mathcal{P}_\Sigma(S)$) of birational local objects are birational local.

Since $\mathcal{P}_{nis}(S)$ is closed under coproducts in $\mathcal{P}_\Sigma(S)$, the claims about the Nisnevich topology follow from the one proven above. □

Corollary 4.5.5. *Under the above condition on S, there are essential geometric morphisms of ∞-toposes*

$$\mathcal{P}_{nis}(S), \mathcal{P}_\Sigma(S) \to \mathcal{H}^{bir}(S).$$

Proof. Owing to Proposition 4.5.4, the inclusions $\mathcal{H}^{bir}(S) \subset \mathcal{P}_\Sigma(S)$ and $\mathcal{H}^{bir}(S) \subset \mathcal{P}_{nis}(S)$ admit further right adjoints by the adjoint functor theorem. □

Actually, using the main result of [Mai26c], we can easily get rid of the geometrically unibranch condition:

Theorem 4.5.6. *Let S be a Qcqs scheme with finitely many generic points. Then $\mathcal{H}^{bir}(S)$ is an ∞-topos.*

Proof. By [Mai26c, Theorem 4.3.4] we know that there is an equivalence of ∞-categories:

$$\mathcal{H}^{bir}(S) \to \prod_{\eta \in S^{(0)}} \mathcal{H}^{bir}(k(\eta))$$

where the product on the right-hand side is computed in Cat_∞. By Theorem 4.5.2, we know that each category $\mathcal{H}^{bir}(k(\eta))$ in the product above is an ∞-topos. So it suffices to know that the Cartesian product of ∞-toposes is an ∞-topos. This is [Lur09, Proposition 6.3.2.1]. □

To compare with Proposition 4.0.1, we summarize the above result in the following manner:

Corollary 4.5.7. *Let S be a Qcqs scheme with finitely many generic points. Then*

$$h^b_S(-) : Sm_S \to \mathcal{H}^{bir}(S)$$

contracts the affine line, sends the Nisnevich (and in fact, cdh) squares to pushout squares, and has as its target an ∞-topos of cohomological dimension 0, which is Postnikov complete whenever $\dim(S) < \infty$.

Proof. The $\mathbb{A}^1$ and Nisnevich parts follow from the definition: $\mathcal{H}^{bir}(S) := L_{bir} L_{\mathbb{A}^1} L_{nis} \mathcal{P}(S)$. The cdh descent case follows from [Mai26a, Remark 2.1.5]. That the target is an ∞-topos is the theorem above. Its Postnikov completeness is proven in Theorem 4.1.4, and 0-cohomological dimension in Theorem 4.2.10. □

Conclusion and Open questions

We conclude this paper with a set of open questions that the author plans to return to in the future:

(1) Although $\mathcal{H}^{bir}(S)$ exhibits several topos-theoretic properties over an arbitrary base scheme S (as shown in §4.1–4.3), its status as an ∞-topos in full generality (for schemes with infinitely many generic points) remains open.

(2) It also remains an open question whether the Birational Yoneda functor of Corollary 4.5.7 has a universal property in $\mathcal{T}op_{\infty}$ (similar to the one in Proposition 4.0.1).

(3). Finally, since in *pro*-$\mathcal{H}^{bir}$ every scheme is identified with its discrete set of generic points (which has Nisnevich homotopy dimension 0), it is reasonable to hope that the birational ∞-topos is locally of homotopy dimension 0.

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